中南大学

博士学位论文

湘西沃西金矿综合地电方法研究

姓名：IRAJ VOSOUGHI NIRI

申请学位级别：博士

专业：APPLIED GEOPHYSICS AND INFORMATION TECHNOLOGY

指导教师：何继善

20031201

ABSTRACT

INTEGRATED GEOELECTRICAL METHODS OVER WOXI GOLD DEPOSIT

BY

Iraj Vosoughi Niri

The study area, Woxi gold-antimony, tungsten deposit, is located in Woxi town of Yuanling County of West Hunan, at 110° 54' E and 28° 32' N and with an area of 12 square kilometers,114 km to the southwest of Changde City and 79 km to the east of Yuanling. Structures in the area can be divided into arc, WE-trending and NE-trending structures. The mining area is situated at the northeast wing of Xian'ebaodan Anticline. The outcropped strata are mainly the Proterozoic Lengjiaxi Group, Banxi Group and Sinian system as well as the Cambrian system. Under the role of early Wuling Movement, commonly known as the Dong扎n Movement, the Lengjiaxi and Banxi Groups contacted in an incomplete angle.

To study geology of the ore deposit, mapping geological structures and looking for new prosperous mineralized zones, geoelectrical methods including CSAMT, MT, and IP were employed. Scalar controlled-source audio magnetotellurics (CSAMT) survey was carried out over the Woxi deposit in two individual stages (A and B). For the first time in China, Tensor CSAMT was employed in the third stage (stage C). The CSAMT data provided important information about the mineralized layers and structures in the study area. Follow-

up detailed geological mapping and drilling works verified most of the achieved results. In addition, RRI inversion was used as an effective method for producing 2-D resistivity images from CSAMT measurements made along surface traverses.

Near-ore IP profiling results along three survey lines subjected to intensive drilling and geological studies show efficiency and reliability of the implemented CSAMT method.

Pseudo-random multi-frequency IP sounding shows high conformity with the produced 2-D CSAMT models. The drilled boreholes based on the achieved results demonstrate the effectiveness of the combined use of both methods

MT data was collected in 11 stations on 20 km line with 2 km spacing to study deep structures of the region. The result shows a relatively deep concealed extension fault under the Woxi deposit.

As indicated by employed integrated geoelectrical methods outcome, field geology and borehole data, some new faults were traced and mapped and two important large-scale reserves were explored in the area. The results of geophysical work along with fieldwork and borehole data directed us to new idea about geology of the deposit. In the offered model the concealed main fault in the area, named Wuhe (武赫) along with other main faults in the area such as Woxi Fault and other newly mapped, F2, F3, F4, and Ganziping faults are all were classified as normal faults in an extensional basin.

Based on the proposed model, exploration should be carried out towards north of Ganziping and northwest of Xintianwan Faults. Meanwhile, prospecting in the east and west should also be paid close attention.

ACKNOWLEDGMENT

With the feeling of deepest and special gratitude, I thank those I can never express my real sentiments of appreciation. Those who helped me so much, knowingly and unknowingly.

I am deeply grateful to my advisor Professor J. He for valuable instructions, comments and constructive suggestions through this study and contributing sincerely from the beginning of the work. A very sincere expression of thankfulness and acknowledgment is extended to Professor J. Tang who contributed sincerely supplying informative data, preparing necessary research facilities and for his constructive instructions. I would not hesitate to record my hearty and everlasting valuation and thankfulness to Dr. Lu, Dr. Xi, Dr. Cheng, Mr. Liu, Mr. Xi, Mr. Yang, and Mr. Zhang, for their efforts and helps in fieldwork and assistance in data processing. I would like also to express my gratitude to Professor D. Yang, former director of Int. Exch. Division for his sincere collaboration.

This work would not have been possible without the cooperation of many individuals most of all Ms. Guo for her valuable help in literature review, former and current directors of Int. Exch. Division, hospitable people of Xiangxi and last but not the least my Chinese and foreign friends who did not deny me of their kind assistance wherever and whenever it was needed.

Chapter One
Geology of the Study area

I. Geography and geology of Hunan Province

A Geographical position

Hunan Province is situated on the middle reaches of the Yangtze River of South China to the south of Dongting lake, at 108° 47'-114° 15' E and 24° 39'-30° 08' N, and has a total area of 210000 square kilometers with a population of 55 million. Changsha City, the provincial capital, is located at 113° 01' E and 28° 07' N (Fig. 1-1). This region is characterized by developed strata, various sedimentation types, frequent magmatic activities, complicated geological structures, excellent metallogenic conditions and rich ore resources.

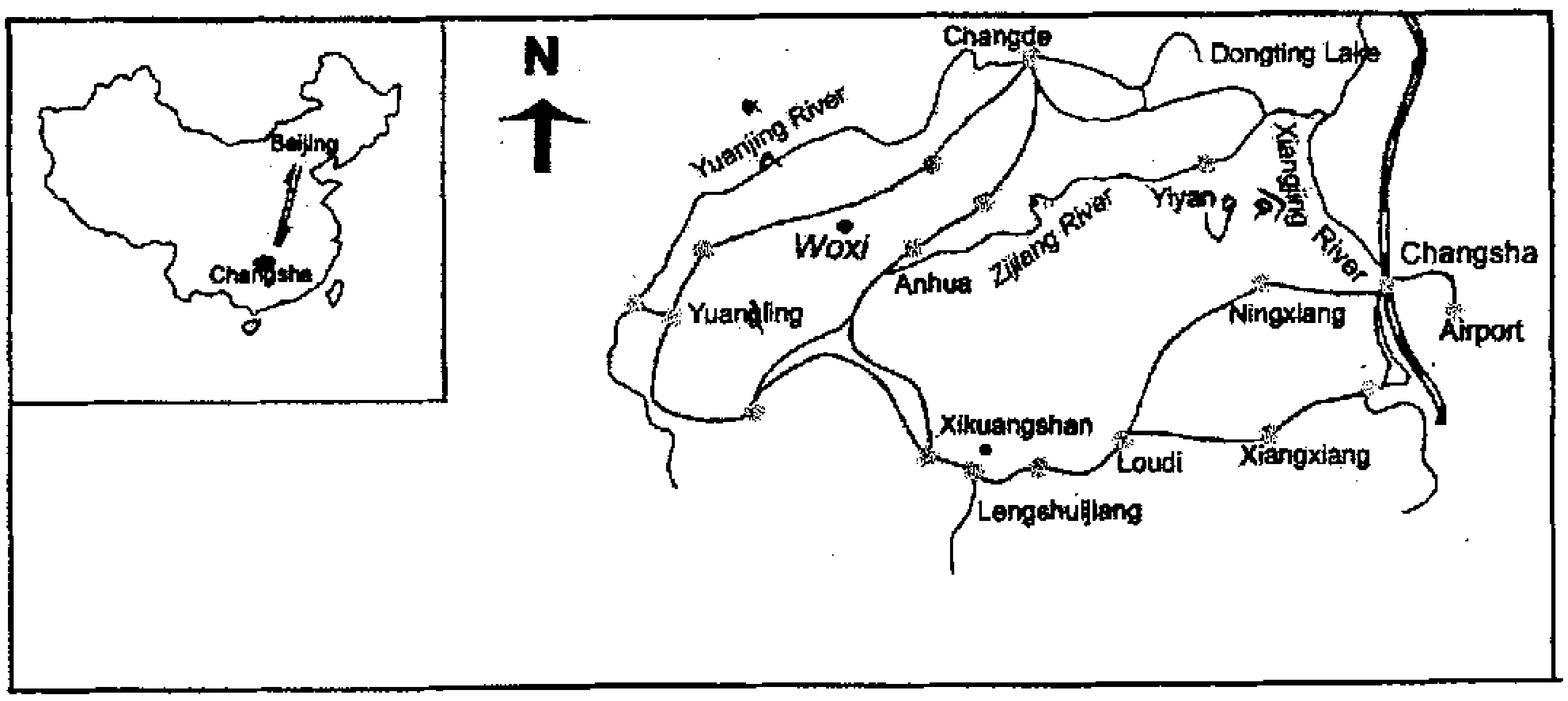


Figure 1-1: Schematic map showing the geographical location of the Woxi deposit.

B Transportation

Hunan has good transport facilities. Changsha's Huanghua International Airport has flights to all parts of the country including Hong Kong. Railway routes cover all parts of the province and link it to all other major cities of the country. A network of highways links up all cities and towns of the province. The four rivers of Xiangjiang, Zijiang, Yuanjiang and Lijiang flow northward into the Dongting Lake, and then connect the Yangtse River, so the water routes are convenient.

C Landforms

Hunan is mountainous in the east, west and south. The middle part of the province is hilly land and the northern part is the Dongting Lake plains. The highest peak is 2099 meters above sea level, the lowest land 23 meters above sea level. Xuefeng Mountain is located in the north. Its south section is about 1500 meters to 1000 meters. Xuefeng Mountain is the watershed between the Zi jiang River and the Yuan jiang River. The Woxi ore deposit is situated in the north section of Xuefeng Mountain.

D Climate

Hunan has a subtropical monsoon moist climate. Its annual average temperature is 16~18°C. In January the lowest temperature averages is 4~7°C; and in July the highest temperature is 27~30°C on month average. The low and high records are 18°C and 43°C respectively.

E Geologic structures

Hunan is located on the south bank of the middle reaches of Changjiang and at the northern margin of Nanling mountain range. In topography the east, west and south sides are surrounded by mountain ranges while the north is open, forming a horseshoe basin. This relief reflected the basic features of the geological structures. On the west side, the Wuling and Xuefeng mountain ranges trend in a general NE-NNE direction which is the third composite upwarding zone of the Neocathaysian System. On the south, the different trends of mountain ranges reflected the basic feature of the compounding of the Nanling latitudinal structural system, the Cathaysian structural system and the Meridinal structural system.

Based on the stratigraphic development, the genesis and development of structural system, and the stratigraphic unconformities and disconformities, the main crustal movements in Hunan province are classified as: Wuling, Xuefeng, Caledonian, Anyuan, Ningzhen and Himalayan movements. Among them Xuefeng movement has an important relation with mineralized zones of the Woxi deposit

Xuefeng movement occurred during the late Proterozoic, it was an isostatic crustal movement that caused the disconformity-unconformity relation between Wuqiangxi and Dongshanfeng formations of the Sinian periods. Disconformity to low angle unconformity relation prevailed in the west Hunan and southwest Hunan zone, in local places showing medium angle unconformity. It has maintained trend of Wuling movement as the activities are strong in the northwest and weak in the southeast Structures formed during this movement have strong contrast thereby provided favorable condition for formation of glaciers, deposition in early Sinian and the changes of lithofacies.

F Sedimentary rocks

Sedimentary rocks are well developed and widely distributed in Hunan province with numerous rock types and abundant sedimentary and stratabound type deposits. The sedimentation process displays a regular pattern of variation.

Sedimentary rocks of the area are mostly classified based on the plan recommended by the Chengdu Institute of Geology with reference to other related sources that emphasizes on their genesis. Moreover, according to the source of the sediments they are classified into sedimentary rock of terrestrial origin, volcanic origin and of intrabasin origin, and according to the composition-genesis and texture-genesis classification principle it is further led to subdivisions and small divisions.

Main types of the sedimentary rocks in the area are: Normal terrigenous clastic rocks, mainly including conglomerate, sandstone and siltstone, Pyroclastic rocks , Argillaceous rocks, Carbonate rocks, and other rock types such as siliceous, ferruginous, manganese and phosphatic rocks that are also developed.

G Stratigraphy

Hunan province provides an extensive distribution of strata, covering about 91.7 of the province's surface area. The stratigraphic units are well developed, ranging from the Middle Proterozoic to the Quaternary. Most of the stratigraphic units in each system are complete with good outcrops and bountiful fossils. It is especially noted that Hunan province extends across the two Paleozoic seas of Yangtze and South China, showing obvious lateral transition and many sedimentary types, therefore is one of the critical regions for the stratigraphical studies in southern China.

Based on its lateral changes, the stratigraphy in Hunan can be divided into two stratigraphic zones, the northwest Hunan zone and the south and central Hunan zone. The former possesses the characters of the Yangtze zone while the latter is a part of the South China zone. In some of the systems that are located between the two zones a transitional zone can be recognized. In a vertical section there are four major sedimentation stages. The Lengjiaxi group is a flysch formation of huge thickness indicating the features of an active type zone and it is the first stage of sedimentation development in the province. This group bottom is not seen in this district but exposure metamorphosed stratum clearly. Development of the Banxi group, the Sinian and the Lower Paleozoic is basically the same, forming another sedimentation stage. Its general feature is that in the southern part the active type sedimentation still preserves while in the northern part it is transformed into stable type sedimentation. Hence, a gradual change in lithology from mainly carbonates pelites and siliceous rocks in the north to mainly sandstones and pelites in the south can be observed, their thickness also becomes larger accordingly. From the upper Paleozoic to the Triassic a stable sedimentation took over in the whole province. The sedimentation features and variation patterns are different from the former stage. In this stage the Devonian and the Carboniferous systems are not well developed in the northwest Hunan zone but they are widely distributed in the south and central Hunan zone with great thickness and showing the pattern of increasing argillaceous and arenaceous ingredients and decreasing carbonate rocks from south to north. This is an inverse sedimentation feature to the previous stage, obviously of different genetic mechanism. In the lower and middle series of the Permian and Triassic system, especially in the Permian, carbonate rocks predominate in the northern part while sandstones, pelites increase towards the south. The Indosinian movement whose activity reached its peak after the Middle Triassic epoch brought the end to the marine sedimentation that had existed in the Hunan province for a very long period. Taking its place were the scattered deposits of terrestrial lake basins comprising the fourth sedimentation stage.

Stratigraphic units of different periods in the province are classified to the formation, some divided into members while a few are classified into groups or series. Sedimentary strata in Hunan is rich in mineral resources, the major ones are coal, iron, manganese copper, phosphorous, salts of calcium, sulfate, barite, gold, tungsten, antimony, uranium, vanadium, lead, and zinc etc.

H Magmatic rocks and ore resources

Magmatic rocks are quite well developed in Hunan, their outcrop area amount reaches to 8.3% of the total area of the province. They are mainly intrusions of few volcanics, including acidic, intermediate, basic and ultrabasic rocks. A lot of igneous rocks in different age occur in the Xiangzhong region. The statistics shows that there are 89 igneous rocks (or dike group) with the outcropping area larger than 0.3 square kilometer.

Proportion of acidic rock is about 99%, mainly as granitoids and hypabyssal facies granite porphyry and quartz-porphyry. Formation time of the magmatic rocks include Wuling, Xuefeng, Caledonian, Indosinian, Yanshanian and Himalayan periods. Intermediate and acidic intrusions are best developed in the early Yanshanian period and Indosinian period, Caledonian period is next, while basic, ultrabasic intrusions and volcanics are better developed in the Waling and Xuefeng periods. Lamprophyres are mainly formed in the Indosinian and Yanshanian periods. Most of the intermediate and acidic intrusions form composite bodies and polystage multi-intrusions of the same period. In general, large magmatic activities occur in the uplifting areas around the Xiangzhong Basin, most of them being the intermediate-acidic rocks of the Indosinian-Yanshanian periods; within the basin are some small dikes.

The intermediate, acidic intrusive rocks in Hunan are intimately related to ore resources. Ore resources related to magmatic rocks (especially to intermediate and acidic rocks) are very abundant, the main ones include tungsten, tin, molybdenum, bismuth, copper, lead, zinc and rare metal and rare earth etc. Most of them are distributed near the outer contact zone and within the intrusive body. They are controlled by the occurrence of the magmatic body and its size, shape, emplacement level, erosion grade, country rock, composition of the magma, rock forming process, and some other factors. The antimony mineralization and deposits are always far from the magmatites and have no direct relationship with them. Those related to volcanic rocks are also very few, the only mineralization seen are Fe and Cu etc.

I Gold deposits distribution in Hunan

There are dozens of big and small ore deposits in the region, which are controlled by north-east-east (NEE) structural zone. They form gold, antimony and tungsten ores in West Hunan. They can be divided into: Woxi-Lengjiaxi gold, antimony and tungsten area, Liulincha Gold area in the north and Xian-Taoan scheelite area (in which gold is associated) in the south. Among them, Woxi gold-antimony-tungsten Deposit (Woxi deposit) is a large deposit and Xi'an scheelite Deposit is a medium-sized deposit, and others are small deposits or mineralization spots.

The Woxi tungsten-antimony-gold deposit along with Xikuangshan antimony deposit occur tectonically in the transitional zone between the

Yangtze plate and the Hunan-Guangxi Hercynian fold belt of South China mobile zone (Figure 1-2). The Woxi antimony-tungsten-gold deposit lies in the northwest border of the Xiangzhong Basin and just at the northwest-toward protruding part of the pre-Cambrian Xuefeng uplifting area. Antimony deposits occur always near deep-large faults. The Xikuangshan antimony deposit, for example, is at the cross position of the Chengbu-Taojiang deep-large fault and the buried, northwest-striking Shuangfeng-Xikuangshan basement fault. The Chengbu-Taojiang fault may be a lithosphere fault and is considered as an old subduction-collision belt because of its thick lithosphere of 300 kilometers and the "V"-shaped profile. On the both sides of this fault many antimony mineralizations in different sizes occur. and form the Xiangzhong Central Hunan antimony

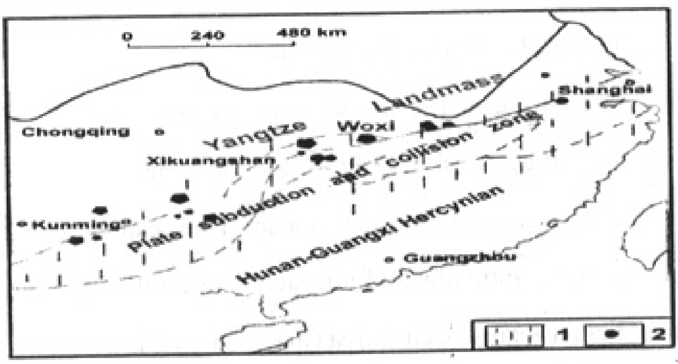


Figure 1-2. Sketch map showing the distribution of major antimony deposits in south China. 1. Antimony ore deposits distribution zones, 2. antimony deposits. Woxi deposit is located nearly the Xikuangshan antimony deposit in the transmitted zone between Yangtze plate and Hunan-Guangxi Hercynian fold belt of south China mobile zone.

metallogenic belt. The Woxi ore deposit is located on the lower block of the Huangtudian-Luxi east-striking, deep large fault along which are distributed a number of tungsten, antimony and gold deposits, making up the well-known Xiangxi antimony-tungsten-gold metallogenic belt (Figure 1-3).

The Layer controlling of Gold Deposit of Precambrian System in Hunan is very noticeable, not only because 95% of gold deposits in the whole province are situated in the Precambrian System, but also a series of characteristics such as gold-containing of the stratums and trace elements distribution pattern verify this point.

Figure 1-3. A sketch of the tectonic system and antimony deposits in Xikuangzhong (central Hunan) region. 1: Sb ore deposits (spot); 2: anticline and synclines; 3: Fault; 4: compresso-shear fault; 5: Cretaceous-Tertiary Basalt; 6: Hercy-Indosinian Fold belt; 7:Caledonian Fold belt; 8: Xuefeng uplifting. Woxi deposit is shown on the map.

The Proterozoic strata in Xiangxi area are distributed by stratabound gold deposits represented by Xichong, Woxi and Mobin, which constitute 72% of gold deposits in Hunan. These deposits are mostly layered ores and mostly believed that are closely related to structural expanding space. Distribution of Precambrian gold deposit in Hunan and their distribution is shown in figure 1-4. Table 1-1 shows the Precambrian

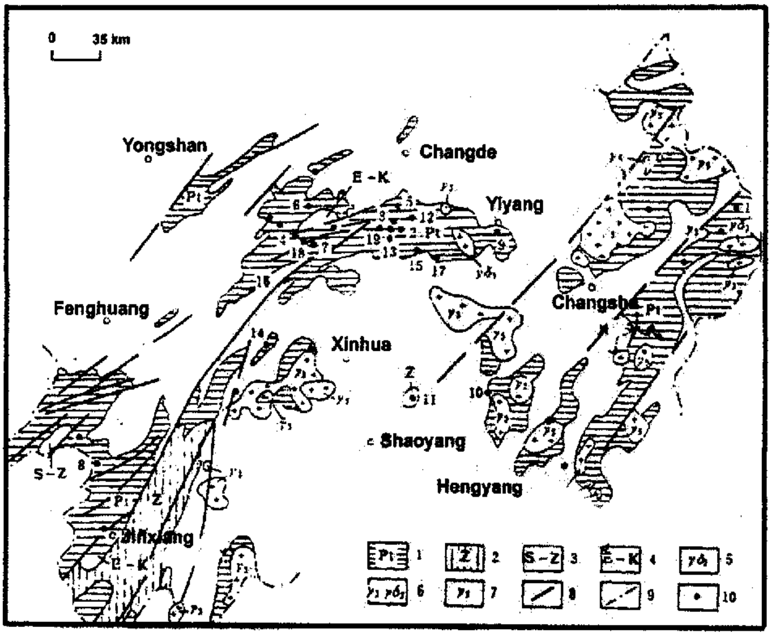


Figure 1-4. The distribution map of Precambrian gold deposits (occurrences) in Hunan Province. 1- Lengjiaxi Group and Banxi Group; 2- Sinian System; 3- Sinian System to Silurian System; 4- the Cretaceous period to the Tertiary period; 5- Granites of Xuefeng period; 6- Granite and granodiorites of Gallatin period; 7- Granite of the Yinzhi-Yanshan period; 8- Fractures; 9- provincial Boundary; 10- Gold deposits and their serial numbers: 1- Huangjindong Gold Mine; 2- Xichong Gold-Antimony Mine; 3 - Canglangping Gold Mine; 4 - Woxi Gold-Antimony-Tungsten Mine; 5 - Huang-tudian Gold Mine; 6 - Liulincha Gold Mine; 7 - Xi'an Tungsten (Gold) Mine; 8 - Mobin Gold Mine; 9 - Yiyang's southern suburbs Gold Mine; 10 - Jinkengchong Gold Mine; 11 - Longshan Gold-Antimony Mine; 12-Hexinqiao Gold-Antimony Mine; 13-Fuzhuxi Gold-Antimony Mine; 14-Longwangjiang Gold-Antimony-arsenic Mine; 15-Wangjiacun antimony-gold mine; 16- Yangpinmao gold-antimony mine; 17- Banxi Xiaogang Gold-Antimony Mine; 18 - Tanghuping Gold (Tungsten) Mine; 19 - Huayanchong Gold Mine.

gold abundance in Hunan. It indicates that background value of gold in Precambrian System in Hunan is generally higher than Clark value. Quartz-vein gold rich layers are ore minerals enrichment zones. Table 1-2 shows degree of gold enrichment (about 4~20 times) comparing to the background.

Table 1-1. Gold abundance statistics of the Precambrian System in Hunan.

Regions	section number	Section length (km)	Host rock	Number Of Samples	Au(ppb)
Longshan Xinshao			Z1	77	7.1
			Ptbnw		9.2
Woxi Gold-Antimony-Tungsten deposit	I, II, III	16	Ptbnw	32	4.0
			Ptbnw	75	4.7
			Ptlj	22	4.0
Xiangxi Xi'an Tungsten (Gold) deposit			slates	60	4.7
			carbonate rocks	11	8.2
			sandstones	20	6.2
Xiangdong Huangjindong Gold deposit	A-A1	5	$Ptlj_2^a$	13	4.7
			$Ptlj_2^a$	50	1.8
			$Ptlj_2^a$	65	7
Section from Taziao to Anhua Yantou Chong (crossing canglangping Gold Mine)		6.2	Ptbnw	11	3
			Ptbnm	28	4.5
			Ptlj	8	3.8
Section from Xiangxi Taoyuan Sandushui to Anhua Baiguoyan Region	IV, V	27	Ptbnw	23	6.8
			Ptbnm	67	5.8
			Ptlj	63	4.1

Index: Z1: Lower Tongjiangkou Group of Sinian System; Ptbnw: Wuqiangxi formation (Banxi Group); Ptbnm: Madiyi formation (Banxi Group); Ptly: Lengjiaxi group.

Table 1-2. Gold Abundance of major ore-hosting strata of the Precambrian System in Hunan.

Regions	Section number	(m)	Host rock	Number of Samples	Au(ppb)
Huangjindong gold mine	28 0 15	250 270 500	Ptlj2	11 11 9	42.7 62.9 19.3
Yiyan southern suburb gold mine		3000	Ptlj	144	74.0
Woxi deposit		100	Ptbnm	7	21.0
Canglanping gold deposit		100	Ptbnm	13	17.4
Mobin gold deposit			Ptbnw	22	21.7

II. Geology of Woxi deposit

A Introduction

Hunan Province has a long gold-producing history and has exploited placer gold, vein gold and associated gold successively. Woxi antimony-tungsten-gold deposit is located in Woxi town of Yuanling County of West Hunan, at 110° 54' E and 28° 32' N and with an area of 12 square kilometers, 114 km to the southwest of Changde City and 79 km to the east of Yuanling. The highway traffic from Changsha to Woxi includes Changsha-Changde section of 189km of the No.319 national highway (cement surface) and Changde-Woxi section of 114km of the provincial-level highway (asphalt surface).

The economy in this area is mainly agriculture, forestry. Labor is ample. So many people shifted to side occupation. Village town and individual economy grow fast. In south-east, the number of people pan for gold is very high, and the economy is active. Woxi deposit is the main mine in the area.

The Woxi Au-Sb-W deposit (Xiangxi gold deposit) was discovered in 1875, which was found as placer gold in stream at first, and then traced back to the primary vein gold deposit and exploited by hand. In 1895, stibnite was discovered in Tianxiangwan in this district and exploited by a private company in 1898. In 1903, Hunan Mining General Company was set up. The exploitation of antimony was very prosperous from 1908 to 1918. In 1946, the exploitation of scheelite turned to tungsten and antimony, but gold still was a byproduct.

Since 1946, the characteristics of associated antimony-tungsten-gold deposit has been recognized. Exploration work has proven that antimony and gold are in large scale and tungsten is in medium scale. (1) Finally as part of my research work, detailed geophysical work using integrated geoelectrical methods were carried out to get detailed information about ore deposit, mineral veins location and possible mineralized zones in the area.

B Igneous rocks

Magmatic rocks are sparsely distributed in the area. They are well developed in the eastern sector of Xuefeng arcuated mole track, minor in the central sector and less in the southwestern sector. The outcropped rock bodies in the eastern sector are mainly Yanshanian Dashenshan granite stock and Cangshuipu granite stock as well as Caledonian Yanbaqiao and Taojiang granodiorite rock bodies. Rocks mainly consist of biotite-granite, biotite-adamellite, granodiorite, biotite-plagiogranite and boitite-monzonite-granodiorite. Yanshanian gabbro-diabase dike, granite-porphyry and quartz porphyry as well as Caledonian basic-alkalic, lamprophyre, diorite and granite-porphyry dikes occur in the central sector of the mole track.

Igneous activity was not very strong in the ore district. Basaltic andesite - andesite is observed at 800m above sea level of the Xian'ebaodan

Mountain on the southern margin of the ore-field. In addition in the directions of north and east outside the mining area, small numbers of Caledonian diabase veins and lamprophyre veins are distributed. According to research, this is not closely connected with mineralization.

C Stratigraphy

The outcropped regional strata can be divided into four structural layers. From down to up, they are: (1) Lower Proterozoic Lengjiaxi Group

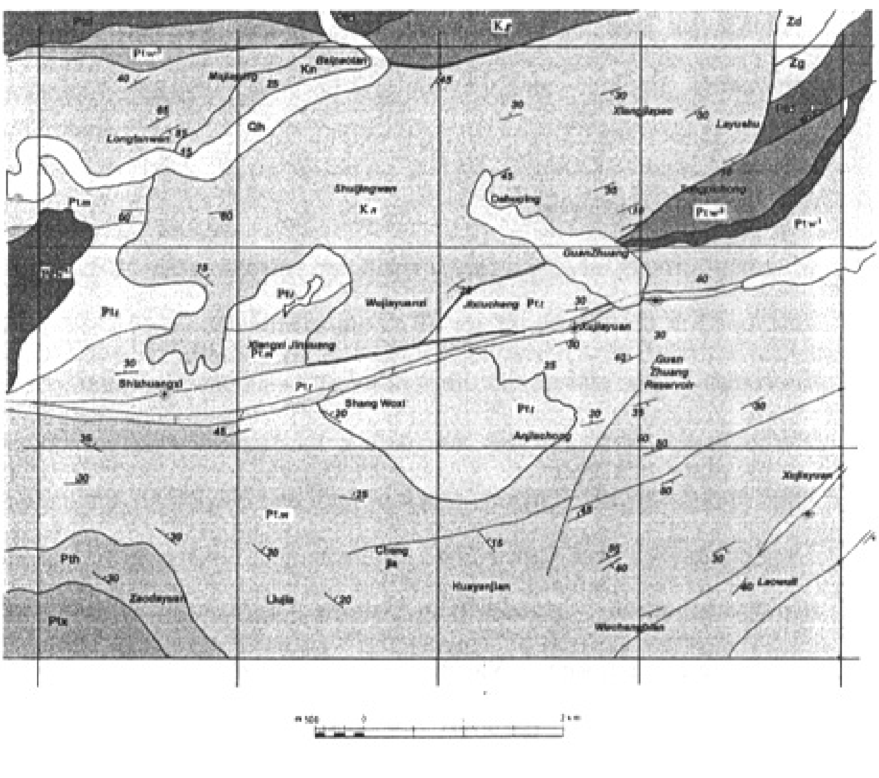


Figure 1-5. Geological map of Woxi area. Structures and stratigraphy of the area are shown clearly. Woxi and Ganziping Faults are considered as boundaries of the mineralization area.

(Pt, lj); (2) Upper Proterozoic Banxi Group including Madiyi (Pt_3bnm) and Wuqiangxi(Pt_3bnw) Formations; (3) Sinian, Cambrian, and Ordovician periods (4) Upper Cretaceous period (Fig.1-5). Xian'ebaodan anticline and Tuomao Mountain Duplicated anticline are located in the east and west sides of the south of the mining area respectively, where the oldest strata, Lengjiaxi Group is shown. In the middle is Woxi- Tanghuping reverse "S" structure where Madiyi and Wuqiangxi Formations are shown. In the north, the Woxi Fault being the boundary, basins of Wuqiangxi Formation, Sinian and the Cambrian systems are distributed. (Figure 1-6)

The outcropped strata are mainly Proterozoic Lengjiaxi Group, Banxi Group and Sinian system as well as the Cambrian Period. Under role of early Wuling Movement also called Dong'an Movement, they two contacted in an incomplete angle. The rocks are a set of layered rock strata of near-shore and shallow-sea rock , such as slightly metamorphic slates, sandy slates and sandstones, occasionally, tuffy sandstones, tuffy slates, dolomitite gray rocks and dolomitite marls can be seen as well. Sinian and Cambrian sandstones and shales are distributed at the north-west and southeast wings of the Xuefeng arcuated mole track. Cretaceous system rocks are a set of red thick-layered conglomerates, which are distributed in the north of this region.

The ore district lies in the northeast limb of an EW-trending dome and by the south side of the NEE-trending Woxi large fault. The structure of the ore district is generally a monoclinal strata pitching to the northeast. However, its occurrence changes gently both in strike and dip. As a result, the Madiyi Formation's strata form six large **roll-folds** from the west to the east of the ore district. These roll-fold are the best positions of mineralization. Consequently from west to east in the range of 7.5 km^2 they compose six following ore blocks: Hongyanxi, Yuershan, Sujiaxi, Shiliupenggong, Shangwoxi and Jingchangwan.

The strata exposed in the ore district include slate or sandy slate of Lengjiaxi Group of the Middle Proterozoic, epimetamorphic limestone of Banxi Group of the Upper Proterozoic and red clastic rocks of Cretaceous System. The Banxi Group is the main ore-bearing stratum, including Madiyi (lower) and Wuqiangxi Formations (upper).

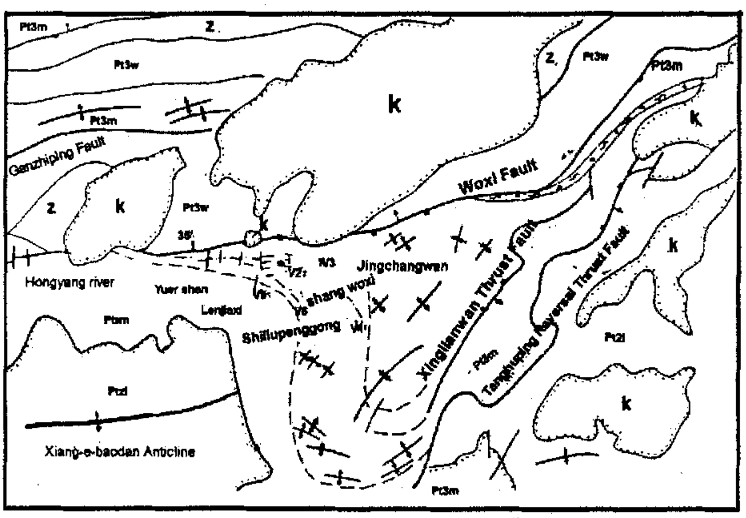


Figure 1-6. A geological sketch map of Woxi deposit. 1: Cretaceous; 2: Sinian; 3: Wuqianxi Formation (Banxi Group); 4: Madiyi Formation (Banxi Group); 5: Lengjiaxi Group; 6: ore vein; 7: anticline; 8: syncline; 9: fault: 10: reversal S-type structure; 11: unconformable interface; 12: conformable interface; 13: Compressive shear zone

1 Lengjiaxi Group of the Middle Proterozoic

Lengjiaxi group is a part of the middle Proterozoic and is the oldest outcropped strata in the province. It is a flysch formation consisted of an assemblage of mainly light grayish weakly metamorphosed fine grained clastic rocks, clay rocks and fine grained tuffaceous clastic rocks with intercalations of dolomite, limestone lumps in the bottom part of the formation and sandstones in the top part, locally with basic and intermediate-acidic lava interlayers. Maximum thickness of the Lengjiaxi group is over 25000m, the bottom has not been observed. It is best developed in east and northeast Hunan and can be further divided into 5 formations and a certain number of series, all conformably related. In other regions, it is either not outcropped or sparsely distributed, hence stratigraphic division is not made.

2 Banxi Group of the Upper Proterozoic

Banxi group belongs to the lower part of Upper Proterozoic, distributed mostly in the Jiangnan old plate but not exposed in the south Hunan zone. The type locality of the Banxi Group is situated at Banxi, Yuanling, where it includes two formations, Madiyi down and Wuqiangxi up. Combination of rocks is purple, mignonette slate with epimetamorphic sandstone, feldspar SiO_2 sandstones and quartz lenticular

Madiyi Formation, 637-937 meters thick, consists of purple slate, sandy slate and metamorphic siltstone. Its middle-upper part, characterized by calcareous slate with calcareous bandings and nodules, is the host rock of orebodies. This formation is one of the good symbolic strata in Banxi Group. Most of the principal minerals in the area exist in this formation, which is an important mineral-bearing rock system in the area (2). The higher layers are usually alternately red and green and become predominantly green upwards. It has obvious layer-controlling role to gold in the area and therefore is the important ore-hosting stratum.

The Wuqiangxi Formation begins with pebbly coarse- to medium-grained sandstone, but consists mostly of green and grey banded slates, often tuffaceous and silicified. Graded bedding and flute marks are frequent in the slates, which may represent flysch beds of genuine turbidity character, especially to the south of Yuanling and Chenxi. Further to the south, in south-eastern Guizhou and adjacent parts of Hunan, the Banxi Group becomes much thicker and entirely green in color. Thus in eastern Guizhou and western Hunan the Banxi group comprises two different facies, a red and a green, probably representing different water depths during deposition. Banxi Group can be divided into north and Central Hunan Zones:

North Hunan zone can be subdivided into the Linxiang-shimen and Yueyang-Huaihua subzones. Woxi deposit is located in the Yueyang-Huaihua subzone's exposure area. Yueyang-Huaihua subzone composed of grayish green, purplish red weakly metamorphosed conglomerates, sandstones, slates, tuffites etc. Interlayers of carbonates and carbonaceous slates are found in the lower part, occasionally with copper mineralization or interlayers of copper-bearing slates, locally with basic and intermediate acidic volcanics. The lower cycle is the Madiyi formation which is in unconformable relation with the underlying Lengjiaxi group (1). The higher cycle is the Wuqiangxi formation, resting conformably over the Madiyi formation(1). Wuqiangxi Formation, 520 meters thick, has the lithology of gray or gray-green feldspar-quartz sandstone interbedded with sandy slate and slate. Wuqiangxi Formation contacts Madiyi Formation. by a fault. The property and thickness of rocks of this group are stable from down to up and it is divided into three segments (2) Table 1-3 shows the minor elements' contents in the regional strata.

Table 1-3. The average contents of minor elements in Proterozoic strata of Xiangxi

lithology	V(10^{-6})	Co(10^{-6})	Ni(10^{-6})	W(10^{-6})	Zn(10^{-6})	Au(10^{-9})	Ag(10^{-6})	As	Hg(10^{-9})	Sb(10^{-6})	Sr(10^{-6})	Ba(10^{-6})
Wuqingxi Fm. sandstone(20)	60.3	12.8	9.6	3.2	102.7	2.4	7.8	4.5	43	1.7	100	859.2
Madiyi Fm. slate(10)	119.8	22.3	25.5	5.2	137.1	3.4	5.0	2.3	47	2.1	45.9	441.8
Lengjiaxi Gr. Slate(7)	145.5	21.7	27.7	4.5	156.5	3.6	5.0	9.5	52	2.4	27.6	109.6
Upper crust*	**60**	**10**	**20**	**2.0**	**71**	**1.8**	**5.0**	**1.5**	**80**	**0.2**	**350**	550

Data from No. 418 Geological Team of Huadong Bureau of Geology and Exploration for Nonferrous Metals

Based on table 1-3 and exploitation features of ore minerals in the study area, Lengjiaxi and Banxi Groups are major strata source of ore, specially Lengjiaxi Group that is considered as a stratiform rich in Au, Sb, and W. Analysis of host rocks in Banxi Group rocks shows its relative richness in Au, W and Sb. Moreover Au content is generally high in claystones than sandstones because claystones have strong absorbing capacity which increases content of Au. Therefore claystones are good targets for mining. Distribution of Banxi Group is wide, about 80% of the total area, so there are abundant sources of mineral-forming materials and then good condition for mineralization.

D Structural geology

1 Introduction

Woxi deposit situated in the central sector of Xuefeng arc-structural mole track located on the southern margin of Jiangnan Earth axis (or called Tailong) in Yangtze paraplatform. Eighty percent of gold deposits in Hunan occur in this mole track.

In middle and late periods of the Proterozoic Era, this region was a near-shore, shallow-sea basin, where huge geosynclinal clastic rocks with near-shore, shallow-sea mud and sands as main parts, and occasionally tuffs, were sedimented. In the Paleozoic Era, this region gradually rose. In the Caledonian Period, huge arcuated mole tracks were formed. Later, except that in some parts sedimentation and depression of basins took place, bulging areas remain till now. Under long-time role of extrusion and deformation in Wuling, Xuefeng, Caledonian, Woxi and Yanshar Movements, strata in the Xuefeng arcuated mole track experienced strong metamorphism and deformation.[a5] The regional geological structures are well developed, Xuefeng arcuated mole track, Dongting depressed area and depressed area in the centre of Hunan are the first-order structures. Second-order structures are eastwest, northwest and NNE-trending structural systems, shear structure, Yuanma basin and Changtao basin. Gufushan anticline and Jingzhuxi reversed fault as the third-order structures controls directly the distribution of auriferous ore belts, fourth-order structures are Woxi major fault and Xian'ebaodan box plunging anticline and have obviously controlling role on gold deposits.

Regional structure is a north-east-east duplicate anticlinal structure made up of Xiangshuidong Anticline, Xian'ebaodan Anticline and Mingyueshan (Bright Moon Mountain) Anticline, called Gufoshan Anticline. Meanwhile, north-east-east Woxi, Lengjiaxi, Tanghuping and Tangjiayuan Reverse Faults are developed. These anticlines and big Faults form the north-east-east structural zone in this region, which is Grade I structure controlling gold, antimony and tungsten zones in west Hunan.

According to mechanical property and formation relationship, structures in the area can be divided into arc, WE-trending and NE-trending structures. The mining area is situated in the northeast limb of Xian'ebaodan Anticline. It is an inclining monoclinal structure rising like an arc in the direction of northeast. Strata of Madiyi Formation in the area were under the continuous role of strong extrusion and deformation and displacement in Xuefeng, Caledonian and following other structural movements. During Caledonian Movement and later structural movements, folds, fractures, foliation, joints and splitting grains are developed, which play a multistage mineral controlling role on mineralization in the area. Distribution of gold deposits is directly controlled by the large-scale Woxi, Tanghuping and Xintianwan Faults.

2 Woxi Fault

(i) Characteristics

Structures always have an important role in mineralization and therefore in exploration. Any prospecting and exploration work in the area needs to study the structures. In recent years, detailed study of the Woxi Fault appears to be very useful in ore genesis and prospecting. Nature and characteristic of the Woxi Fault is an important basic geologic problem of Woxi deposit. Woxi fault is the main structure in the area and should be studied in detail and very carefully. After combining the regional and local research results on the Fault with the formation and evolution of regional and ore-bearing structures, some new facts about the nature of the Woxi Fault and its geologic significance were discovered. To judge about its type, some characteristics of the fault should be considered:

The Woxi Fault (F_1) is the longest east west fault throughout the whole ore district. It strikes east-west and dips north to north-north-west with a dipping angle of 25°~ 45° degrees. Fault planes in the west is relatively moderate, less than 30°; that in the east is relatively steep, more than 40° (Figures 1-5, 1-6 and 1-7).

Its outcropped length is up to 20 kilometers, developed between stratum of Madiyi and Wuqiangxi Formations, and crosses with strata of Madiyi Formation with an acute angle. Its fracture zone's width varies between 20~130 meters. Fault shear zone in quartz arkoses of Wuqiangxi Formation is wider. This zone in the west is narrower than east. Based on current drilling data, hanging-wall (or northern part of Woxi Fault) is Wuqiangxi Formation, with the cover of the Cretaceous red strata, in which no orebody has been found; its lower block (or southern part) is the Madiyi Formation which hosts the orebodies. It is Contact between lower block of the fault and purplish-red slates in Madiyi Formation is clear and its fault gouge has 0.1~1.48m thick. The fault has the characteristic and sign of hydrothermal metamorphic solution activity in many periods. However, some researchers it is not an ore-conducting structure in the area and it has played a late destructive role to veins at the west wing of the mining area.

Because of those characteristics, it is mostly thought that Woxi Fault develops between strata in Madiyi Formation and Wuqiangxi Formation, and is an along-the- layer fault.

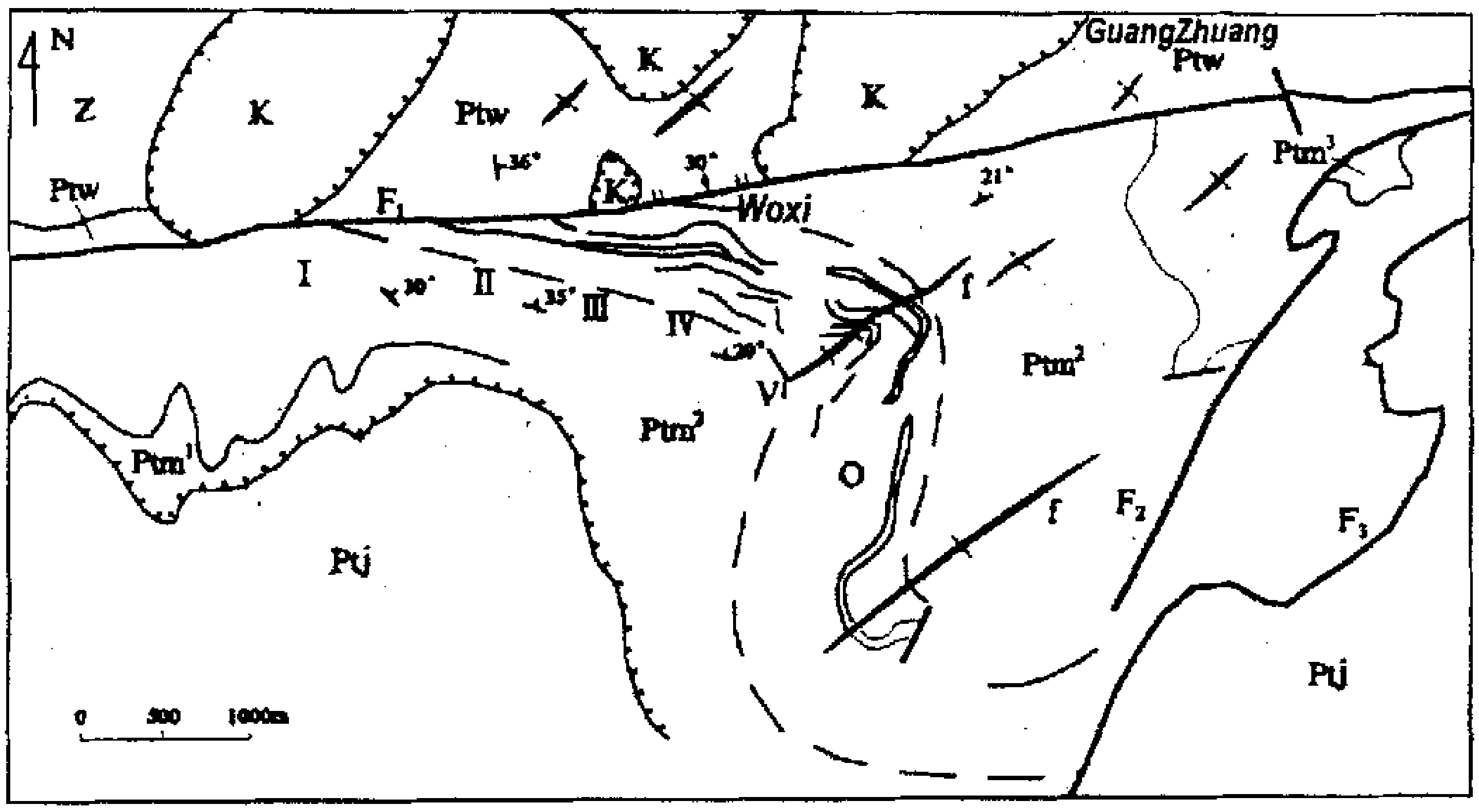


Figure 1-7. Sketch map of regional structural of Xiangxi gold mine. K: Cretaceous; Z: Sinian; Ptm; Madiyi Formation Banxi Group (Proterozoic) (divided into upper, middle and lower segments); Ptj; Lengjiaxi Group (Proterozoic); f: fold (anticline and syncline); F1: Woxi Fault; F2:Xintianwan Fault; F3: Tanghuping Fault;
O: Layered-vein and altered slate zone in ore beds; I: Hongyangxi ore block; II: Yu'ershan ore block; III. Sujiaxi Mineral Z.; IV: Shiliupenggong (Woxi) ore block; V: ShanWoxi ore block

However, this is open to question. Because: influenced in the zone of Woxi Fault:

(1) The brittle fractures of Wuqiangxi Formation strata influenced in the zone of Woxi Fault broken zone can be up to 9 meter.

(2) In the west of Woxi deposit, such as Yu'ershan ore block, the producing form of strata in Woxi Fault and Madiyi Formation is gradually similar to layered-vein and alteration zones with identical strata, and the strata cross each other. Besides, in the south of Guanzhuang, cutting of Woxi Fault through middle and upper segments of Madiyi Formation can be seen (Fig. 1-7).

(3) A few geological cross-section shows that the Woxi Fault is an non along the layer fault with strata in Madiyi and Wuqiangxi Formations (Fig. 1-8).

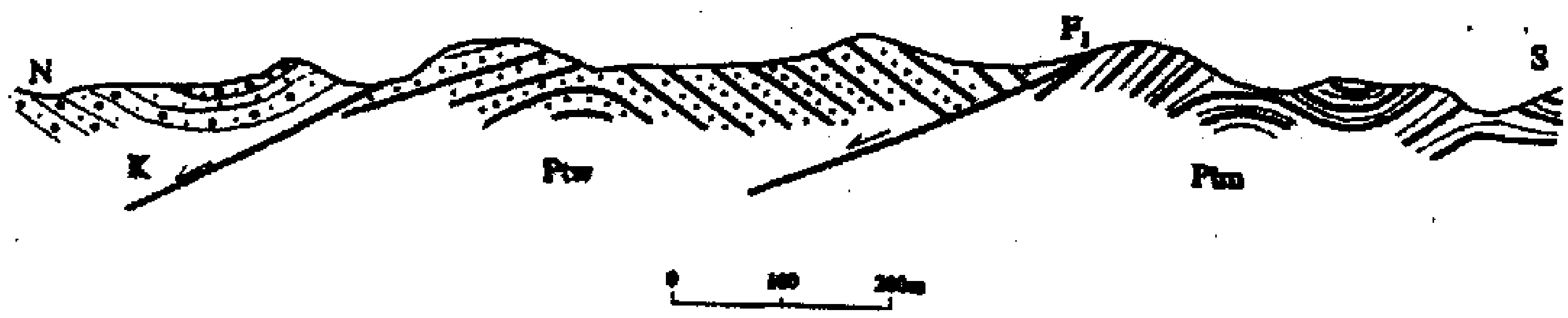


Figure 1-8. Sketch showing relation between Woxi Fault and the strata.

(4) In Yu'ershan ore block, main veins exist directly in the Woxi fault zone. While the branch vein existing in fractures between layers of purplish-red slates and Woxi Fault form a "人" shape, and the angle between them is about 15°~50° (Fig. 1-9).

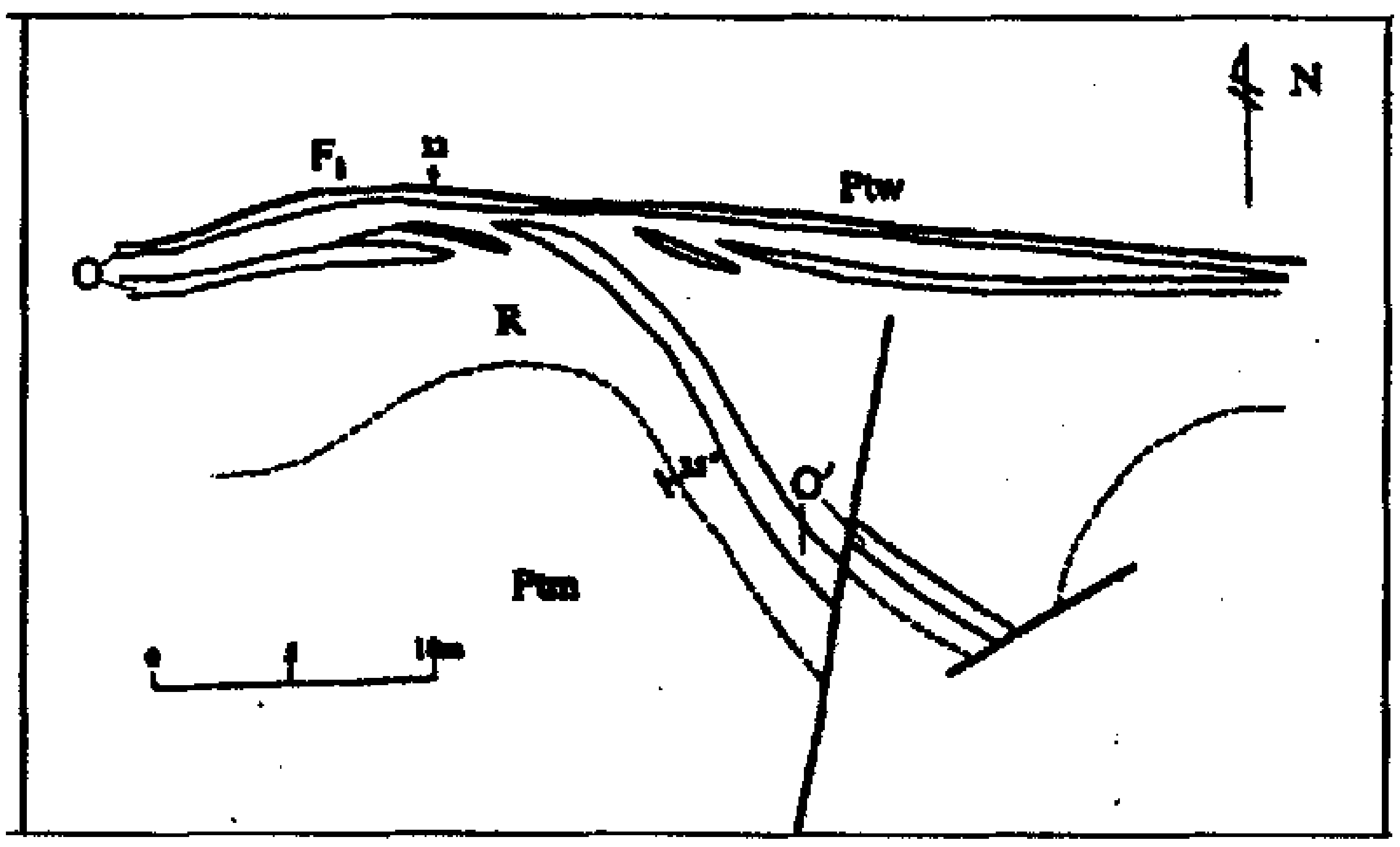


Figure 1-9. Sketch showing relation between faults, strata and ore layers in Yu'ershan ore block. R: host rocks; O: main vein; O': branch vein; other symbols the same as figure 1-7.

All above shows that Woxi Fault is a non-along the layer non along the layer fault. On the strike, it slantingly crosses with strata in Madiyi

Formation with an angle of 15°~50°. On the dip, they cross with an angle of 5°~10°, the strata are smooth and the fault is steep, but both have slightly slanting producing forms.

(ii) Normal or reverse?

Some researches believe that Woxi Fault is an expanding shearing fault. This is important, because it is connected with the problem of mining in the upper block of the Fault. It is not along the layer, and it is characterized by alteration, mineralization and activation in many stages. The dip angle of Woxi tensile-shear (listric extension) normal fault is steeper than bedded vein systems. It may cut the orebody in deep position so that there is the foreground of looking for orebody in upper block at depth. In addition, there are foregrounds to look for ore deposits in Yu'ershan and Hongyanxi districts, etc., in western sector of Woxi fault, as well as in Dongzichong-Xiaotaoyuan zone of northern part well.

There are two famous different ideas about the Woxi fault: 1) A reversed fault or a covered thrust fault or reversed thrust with the property of sinistral compression and twist; 2) A normal fault with the property of sinistral expansion and shearing. Moreover there are two important geological role of Woxi fault: ① position of a series of layered veins in upper parts of orebodies in Yu'ershan is shifted and the shifted and lost orebodies are on the west of the Yu'ershan; ② some researchers believe that Woxi Fault is both an ore-transmitting and ore-hosting structure. Through fieldwork geology, it can also be deduced that Woxi Fault has horizontal shearing activities of the northern block moving west and the southern block moving east.

The earliest Woxi fault was a listric extension fault, which formed in Xuefeng Movement (at 800 Ma or more). In Caledonian movement it was of right-lateral thrust and cut-off the bedded veins above known orebodies in Yu'ershan. From the displaced bedded orebodies lie on the east of Yu'ershan. From Indo-China movement to Yenshan movement, it was sinistral normal fault. It was neither a structure as passageway for ore fluid nor a host structures displayed normal fault.

Although Woxi fault was considered as reversed fault that has over thrust and nappe characteristics but other researchers believe that it is not a reverse fault on the whole, Analyzed on the structures of the plane and the section, the upper block of Woxi Fault is new strata, the low part is old strata and in the middle some strata within the layers are lost. Therefore according to the principle of fault effect, it is no doubt that it is a normal fault.

(iii) Relation with mineralization

The question is this: does Woxi Fault formed before or after mineralization? Some researchers think that the Fault played a destructive role to orebodies. However, many evidences show that there are hydrothermal alterations and mineralization in the Woxi Fault zone.

(1) Alteration: In the fractured zone, quartz veins, silicification and coarse-grain pyritization in the zone of Woxi Fault.

(2) The contents of geochemical ore-bearing elements in the crack zone are high:11 samples of breccia in the fault being analyzed, the content of gold is $0.33\times10^{-}$~$0.5\times10^{-}$, that of Sb is 0.01%~0.1%, WO_3 0.01%~0.17%. The

average content of gold in fault clay (0.05~1.02mm thick; the average thickness is 0.33m) in the lower block of Woxi Fault is $1.09\times10^{-}$. The increase of contents of ore-bearing elements in Woxi Fault in Yu'ershan ore blocks are more obvious (Table 1-4).

Table1-4 Analysis of ore elements in the fault zone, Yu'ershan ore block

ore blocks	broken zone in the upper block			fault clay		
	ώ (Au)/10^{-6}	ώ(Sb) /10^{-2}	ώ(WO_3)/10^{-2}	ώ(Au) /10^{-6}	ώ(Sb) /10^{-2}	ώ(WO_3) /10^{-2}
east ore block	1.035	0.021	0.031	0.94	0.025	0.016
west ore block	1.027	0.057	0.015	1.27	0.054	0.183
Majiayuan ore block	0.75	0.015	nd	1.00	0.025	nd

Ore bodies existence in the Woxi Fault zone: For example, Woxi Fault controlls the shape and scale of main orebodies in Yu'ershan ore block. Main veins in this ore block exist in the broken zone in the lower block of Woxi Fault and are on top of purplish-red slates in Madiyi Formation (Fig. 1-9), that is, ore bearing quarts veins in the main vein are big regular veins produced close to Woxi Fault. The strike of the quartz veins is east west, they slope towards north with dip angle of 20°~30°.

The existing orebodies were get rich by transformation (reconstruction process) of Woxi Fault. For example, orebodies of quartz veins in the west ore block are produced along Woxi Fault, so they are thicker than those in other ore blocks and their grades are high (Table 1-4). The east ore block is thinner and worse than the west one and there is an altered slate layer between it and Woxi.

The gallery along V_1 of Yu'ershan has revealed that Woxi Fault is the roof of the vein and the relation between Woxi Fault and V_1 also reflects transformation and enriching. For instance, veins of Mineral Block 908 and Mineral Block 907 are directly produced at the bottom of Woxi Fault, the veins are thick and their grades are high. There is a layer of alteration rocks (0.1~0.5m thick) between veins of Mineral Block 904 and Woxi Fault, the veins are thin and their grades are low (Table 1-5).

The transformation and enriching of Woxi Fault can be called structural mineralization or fracture structural mineralization.

Table 1-5 Relation of Woxi Fault to thickness and grade of V_1 vein, Yu'ershan ore block

Ore blocks' number	Thickness and grade of orebodies			
	Thickness of veins (m)	ω (Au)/10^{-6}	ω(Sb) /10^{-2}	ω(WO_3)/10^{-2}
904	1	1	1	1
907	1.88	1.35	1.59	1.88
908	1.38	1.77	0.51	6.25

Evidences provided by materials about age of lead isotope for mineralized activities shows the age of V_3 in Woxi ore block is 808 Ma, and V_4 is 636.18. Age of K-Ar isotope in purplish-red sericite slates is 281.30Ma. These statistics of age at least can prove that in structural movement of the Yinzhi period (Yanshan Period), Woxi Fault played an active piling and transformational role to mineralization.

3 Other faults

(i) Fault in the west:

In addition, there is also an induced, secondary fault (F_7) at Shiliupenggong area. It is in the pit near no.1 prospecting line. It crosses with Woxi Fault in the east with an acute angle and extends to line 39 in the west. The strike is over 1000 m long. Its slope extends to 400 m in depth. Strata of the F_1's lower block in the Madiyi Formation, develops eight interlayer faults. Their occurrence generally consists with sloping to north or northeast at 25~35 degrees, extending 650~5300 meters along its strike. These interlayer faults are the most principal ore-controlling structures, and beside them the joints and cracks well developed and were later filled with veins and veinlets, forming the net-vein and joint-vein orebodies. They are often cut or dislocated by the post-metallogenic faults (Fig. 1-7).

Because it is covered by surficial sedimentary layer, it doesn't appear on the surface. The strike of the fault is N 50°~70°E, the dip is NW and the dip angle is 30°~70°. Along the strike and the dip, it rises and falls like smooth waves, but its upper block is steep and lower block smooth. The fault plane is smooth and there is 5~20 mm fault gouge. The producing form of the fault plane roughly moves N~W and S~E. V1~V4 are jagged and the distance between faults is above 130 m. It is biggest post-mineralization fault. Veins in the area are controlled by the lower block of the fault, and a few ore shoots are developed in V4 (1~10 middle segments) near the fault. While in the upper block of the fault, only a few small orebodies and mineralization are seen. According to this, it can be speculated that F7 was formed after mineralization. Due to north-south extrusion, the fault was shut, which was separated from mineralized hydrothermal solution from northeast, so that in the lower block minerals gathered and sedimented to form ore shoots while in the upper block minerals were relatively lacking. It is a fault, which was mainly interlocked after mineralization and has activities in many periods.

(ii) Interlayer faults:

Developed from calcium-rich purplish-red sericitic slates in middle and upper parts of Madiyi Formation in lower block of Woxi Fault. Under the role of extrusion and deformation by Grade I structure in the area, Xian'ebaodan anticline and inferior crossing folds, they were formed by sliding, breaking and stripping structures produced on stratification planes of weak strata. The producing form and lithology are basically identical. Occasionally, slantingly cutting strata with a small angle are seen. In the strike, the fault rises and falls like smooth waves. Eight major interlayer faults are found in the area. Their strike is 650~5300 m long, depth of extension of inclination is up to 2500 m. Each of four interlayer faults has 1~2 branch faults which control the shapes and scales of branch veins.

(iii) Post-mineralization faults:

Fault after mineralization in this area is well developed. Group with the strike of NE~SW and dip of NW~N is the most developed. Dip angles are 30°~80°. Belonging to an expanding twisting normal fault, it often cuts veins into stairs. The distances of 2~5 m and 20~50 m are common. Next, the distance of fault with the strike of NW and the dip of NE or SW is generally 1~5 m, which to some extent destroys orebodies

4 Folds:

Strike of strata in ore-bearing Madiyi Formation is centered on No.16 shed, in the west east-west and in the east southeast. The inclination is north to northeast and dip angle is 25^0~35^0. Along the strike and inclination, the strata rise and fall smoothly and wave likely so that a series of inclined skirt-edge crossing pleats centering on No.16 shed are formed. From west to east, there are Hongyanxi anticline and syncline, Yuershan anticline and east syncline, ShanWoxi and Jinchangwan anticlines sucessively which are Grade II mineral-controlling structures. Along weak stratifications, these rock strata at the axis of the folds form sliding, fracturing and stripping structures for many times, which become good spaces for concentration, flowing and filling of mineral fluids to control shape and distribution of ore bodies. The size of crossing pleats arcs, the degree of opening and closing and the direction of inclination strictly controls the scale and the direction of extension and expansion. The wider its arc, the longer strike of the orebody and vice versa, and greater the depth of leaning extension, the greater of orebody size..

E Mineralogy and mineralography

Woxi gold deposit is a multistage mineralization deposit. Mineralogy of this area is simple, mainly native gold, stibnite, scheelite, Wolframite, ferberite, aurostibnite and pyrite, and next small amounts of wustite, as well as chalcopyrite, sphalerite, galena, arsenopyrite and chalcopyrite. Secondary minerals are limonite, antimony, tungsten, etc. Gangue minerals are mainly quartz (above 80%), next are calcite, chlorite, sericite, ankerite and small amounts of pyrophyllite, Illite, kaolinite etc.

F Orebodies shape

Woxi Gold-Antimony-Tungsten Deposit, Woxi deposit, is a unique large deposit in which gold, antimony and tungsten (scheelite) intergrowth. The three elements have their individual industrial and economic values, which are rare in the same kind of deposits in other parts of the world. The deposit is located in the west of Woxi-Lengjiaxi Formation gold zone. From west to east Woxi deposit includes six following ore blocks: Hongyanxi, Yu'ershan, Sujiaxi, Shiliupenggong, ShanWoxi and

Jinchangwan. The Shiliupenggong ore block is the largest, then Yu'ershan and finally Lijiaxi ore blocks. All the six ore blocks have identical geological features on the whole. Size of ore bodies in other mineral blocks is relatively small. The deposit is produced from purplish-red sericite slates containing calcium in middle and upper parts of Madiyi Formation, Banxi Group. Ore-hosting quartz veins fill interlayer, branching and inferior expanding and shearing fractures. Based on producing forms, the quartz veins can be divided into layered veins, stockwork ores (reticulated thin veins) and layer-cutting joint veins. The scale of layered veins is the largest and reticulated veins are the second.

Prospecting in the deposit in the begining was mainly done in Shiliupenggong ore block- about 210 meters below sea level. V_1, V_3 and V_4 are ore veins where prospecting reached to mineralized zones. Each vein parallels and is distributed with upper and lower parts overlapping (Fig. 1-10). Level distance between V_3 and V_4 is 110~140 m or more, and between V_1 and V_3 is about 10~30 m. Each vein is a layered smoothly slanting thin vein. Their thickness is 0.1~3.2 m, average being 1.06 m, and dip angle is 20°~35°. Layered veins are combinations of many "lath-like ore columns". The columns are 50~350 m long on the run, slantingly extend 800~2300 m, incline towards northeast with angle of 40°. Between the columns, there are zones with 50~200 meter width without mineralization or with low-grade ore minerals. Ore bodies expand and narrow down on the strike, branching and compounding are obvious, and the coefficient of change of thickness is 71%. Generally, there are mineralization of gold, antimony and tungsten in ore bodies, but degrees of mineralization are different, and degrees of regularity of distribution of three kinds of useful components are different. Mineralization of gold is the most stable, next is antimony but tungsten is unstable. They are 20~120 m long on the strike, extend slantingly 50~300 m. Thickness of the ore bodies is 2~7 m, average is 4.5 m. Stockwork ore bodies are attached to layered veins but are distributed independently. Their producing form is similar to layered veins, and dip angle is 25°~35°.

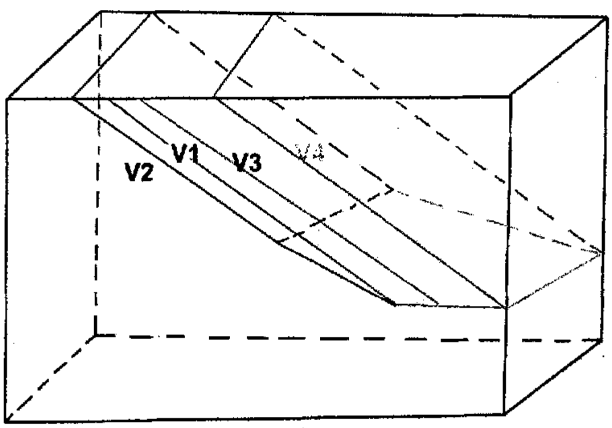


Figure 1-10. Diagram of the shape of long cylinder of the ore vein

Joints are like thin veins, the walls of veins are flat and straight. Mineralization of joint veins is strong, their grade is high, and scale is small. Joint veins have 20~40 m length on the strike, and 10~60 m length on the run. They are attached to and controlled by layered veins. Dip angle is 40°~80°. Post-mineralization Faulting did great damage to ore bodies.

1 Interlayered vein

Ore-hosting quartz veins fill interlayers fractures . The producing form of veins is basically the same as host rock and parts of veins slantingly cross with host rock with a small angle. In the west of mining area, strike is nearly east west and dip is north. From Shiliupenggong ore block to the east, strike turns to northwest, incline is northeast, dip angle is generally 25° ~35° and in some parts, it can be up to 40°~50°. Seven layered veins have been explored in the mining area. Emplaced along interlayer faults, they extend parallel to each other. Their striking length is 300~3500 meters, and dipping extension 500~2300 meters (controlled by drilling); on the strike section, they show lenticular shape (fluctuated with the host strata); on dip section, they have tabular shape along the fold-axis (Figures 1-10 and 1-11). Interlayered veins are main orebodies and account for two thirds of total reserves of the mine.

2 Stockwork

They fill in joints and fissures which developed mainly beside the interlayered veins and sometimes in the stratification plane. They consist of three-dimensional network of planar, irregular, to dendriform veinlets closely enough spaced that the whole mass can be mined. Individual veinlet has 0.5~5 centimeters width and 0.4~5 meters length.

3 Joint veins

Joint veins or structure filling veins are also called layer-cutting (joint) vein fill in fractures beside the interlayered veins. Its individual vein's length is 10~40 meters in strike line, 10~16 meters length in dip line with 0.2~1.5 meters thickness and 40~70 degrees dip angle. Despite their small size, they have high grade.

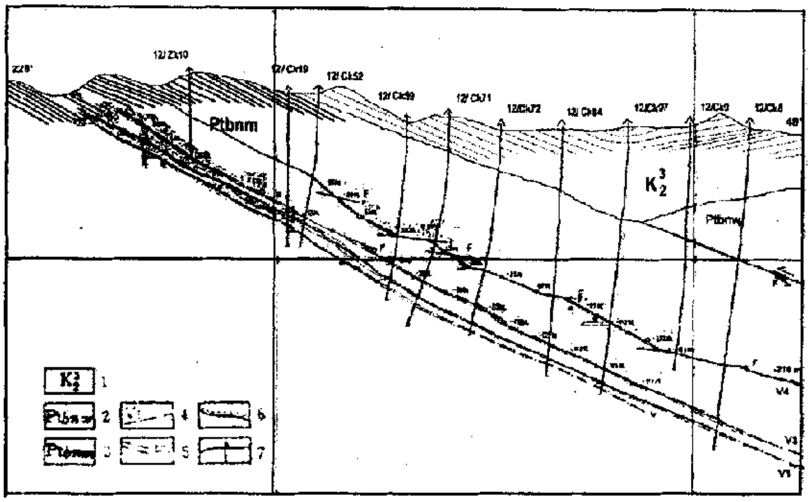


Figure 1-11. geological section along no. 4 exploration line of the Woxi ore deposit. 1: Cretaceous; 2: Wuqianxi Formation; 3: Madiyi Formation; 4: Ore vein and its serial number; 5: Fault and its serial number; 6: Unconformity; 7: Drill hole

G Wallrock alterations

Alteration of wall rocks in the deposit is well developed. Wallrock alterations of hanging wall and footwall of orebody are widespread and strong. Main alterations produced by Middle–low–temperature hydrothermal alteration are silicification, pyritization, carbonatization, chloritization, illitization, kaolinization, pyrophyllitization and sericitization which are developed to less or more in the foot and upper blocks of veins. First three alterations are in closest relationship with gold mineralization. Decolorization and pyritization always exist in mineralized area but chloritization can be seen in the poor mineralized area. Various alterations can occur in the same place, and they overlap each other to form the alteration zones. Wallrock alteration near the veins show obvious zoning, from inside of ore vein to outside being illitizaiton, sericitization-pyritization, quartz-pyritization, sericite-quartz-chlorite zones as well as iron and titanium oxides zone.

Strong alteration along with wide distribution of purplish-red slates in Madiyi Group, and faulted structures development are most favorable geological condition for mineralization in the region. All types of alteration often overlap on faded altered wall rocks, are developed on both sides of veins and host rocks. They are distributed belt-like and thus called alteration belts. Their width is generally 1~3 m to maximum 20 m. Alteration belts in lower blocks of veins are wider than upper blocks, and they are asymmetrically distributed. Alteration of wall rocks is controlled by development degree of structural fractures, wall rocks properties and mineralization strength.

Alterations beside ore vein V_3 are obvious on the ground. Silicification and sericitization (bleaching) are dominated and form the quartz lump and decolorized belt. Various alterations are spatially overlapped by each other, but still show the zoning (table 1-6).

Table 1-6. The alteration zoning in the deposit

zoning	type	width	**Alteration assemblage**
altered slate	purple sericitization	20~30m	**sericite, quartz**
weak alteration zone	bleaching, carbonatization sericitizaton, chloritizaton	1~2m	**calcite, sericite chlorite**
strong alteration zone	silicification, pyritization, carbonatization, bleaching	0.6~1m	**quartz, pyrite, sericite, pyrophyllite**
clay zone	illitization	0.01m	**illite, ore minerals**
ore vein			

H Structure and mineralization

Many factors including geologic conditions play role in gold mineralization in the Woxi deposit. But among the factors, structures play the most obvious and important role.

1 Control of Fracture Zone on Mineralization

In Guanzhuang Fracture Zone taking Panxiangpu-Xiaotaoyuan Fault (F_{15}) and Guanzhuang-Huangtupu Fault (F_{20}) as boundaries, a series of gold (antimony, tungsten) deposits are distributed, which is unusually identical with combination of ore minerals such as gold, antimony and tungsten (Fig 1-5).

2 Secondary faults and fractures role on mineralization

Besides major faults, interlayer fractures, feather-like fractures and secondary faults on the both sides of ore controlling faults control the scale, shape and distribution of veins. Among the six gold spots in the area, there are three spots in which veins were controlled by interlayer fractures, and other three were controlled by feather-like fractures and secondary faults. This fully explains that secondary faults, feather-like fractures and interlayer fractures provided good reserving spaces for ore-bearing hydrothermal solution, and were act as ore hosting structures.

The Structural features of layered veins reflect layered control and role of metamorphic hydrothermal are major mineralization factors of the deposit. Strength of mineralization and alteration is closely related to degree of development of faulted structure.

Chapter 2

Controlled Source Audio-frequency Magnetotellurics (CSAMT)

I. Introduction

Controlled source audio-frequency magnetotellurics (CSAMT) is a frequency-domain electromagnetic sounding technique used to establish a three dimensional (3-D) apparent resistivity map of the earth. The technique uses a fixed, grounded dipole or horizontal loop as an artificial signal source. The CSAMT method was introduced by Goldstein (1971) and Goldstein and Strangway (1975) to overcome problems encountered by the audio magnetotelluric (AMT) and Magnetotelluric (MT) methods. CSAMT is similar to the natural-source magnetotellurics (MT) and audio-frequency magnetotelluric (AMT) techniques; the differences is using of the artificial CSAMT signal source at a finite distance. The source provides a stable, dependable signal, resulting in higher precision and more economical measurements than are usually obtainable with natural-source measurements in the same spectral bands. However, the controlled source can also complicate interpretation by adding source effects, and by placing certain logistical restrictions on the survey. CSAMT method has proven particularly effective in mapping the top 2 to 3 km of the earth's crust.

The CSAMT source usually consists of a grounded electric dipole about 1 to 3 km in length. Ideally located at least four skin depths ($4\delta=$ $2012 \sqrt{p/f}$; ρ =resistivity, f = signal frequency) from the area where soundings are to be made. Measurements are made within the 0.1 Hz to 10 kHz frequency band. Orthogonal electric and magnetic field components are measured, ideally in the plane-wave portion of the field far from the source (Figure 2-1). The ratio of orthogonal, horizontal electric and magnetic field magnitudes yields the apparent resistivity. Magnitude and phase are measured for two to five electric and magnetic field components (Ex, Ey, Hx, Hy, Hz), using either one or two sources. Grounded dipoles detect the electric field and magnetic antennas sense

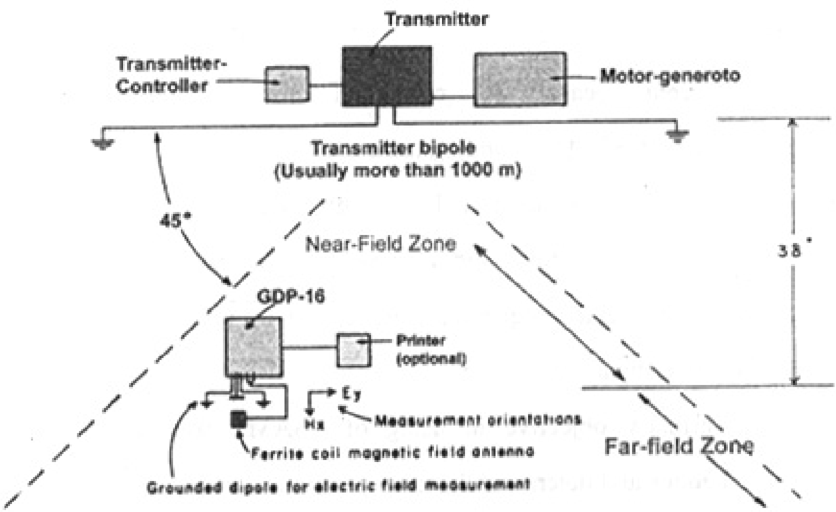


Figure 2-1. Explanation of standard CSAMT field set-up. Measurments can be made anywhere inside the cone delimited by the dotted lines. For far-field (plane wave) measurements, receiver lines should be no closer than three δ, for the lowest frequency to be used, to the transmitter dipole. Measurements may be made closer than three δ (near field) but interpretation is complicated. Receiver dipole size depends on survey requirements: desired resolution, signal strength

needed to overcome local noise. Transmitter dipole size can be increased to provide more signal.

the magnetic field. The difference between the electric and magnetic field phase angles provides impendence phase. In tensor measurements these quantities may be treated by standard MT processing techniques.

Controlled-source audio-frequency magnetotelluric (CSAMT) measurements have been used in mineral exploration since 1974 with significant success. However, data interpretation is often difficult and complex when the transmitter and receiver are separated by the lowest frequency used. Field experiments and numerical modeling have helped in the development of techniques to remove or mitigate near-field effects and transmitter-site overprint contamination.

II. Objectives

In recent years the controlled source audio-frequency magnetotellurics (CSAMT) sounding technique has gained acceptance as a viable geophysical exploration tool. Its high data quality, good lateral resolution, cost-effectiveness, and relative ease of interpretation made CSAMT an effective technique for a variety of energy-exploration and geotechnical investigations.

Our primary objective in using of CSAMT was to look for mineralized zones and determine the most prosperous zones in about 1000 m depth. Moreover, tensor CSAMT was employed to study the mineralization and especially structures that are related to mineralization, in very reliable way.

Surveying and modeling procedure will be described using resistivity characteristics of different geological stratigraphical units in the specific area and some theories about the stratigraphy, major faulting and folding structures, and locating possible mineralized zones in the area will be presented. Behavior of the source fields, the surveying and interpretation methods will be discussed as well.

As CSAMT method is quite sensitive to the conductive targets, it becomes feasible to test and compare the results of IP and CSAMT methods for the detection of conductive ore bodies in the study area. In order to get results that are more exact some selected test sites were subjected to extensive drilling and detailed geological investigations. The core samples had been taken to measure their electrical properties and the geophysical responses of the weathered strata, rock units, and target mineralization.

The pseudosections derived from the IP and CSAMT data are compared in view of the lateral and vertical resolutions that will be discussed in detail.

Success of the geophysical survey depends on the careful examination of the physical response of an individual method using the results of the core sample measurements or forward modeling and depends on understanding the relation between these factors and the geological situation. Considering all the mentioned parameters, Comparison of all the employed techniques will be done to find out their efficiency, reliability and accuracy in study of such typical gold deposit.

III. CSAMT technique description

The advantageous of Controlled source Audio-Magnetotellurics (CSAMT) are several: i) Signals are stronger, therefore the sending equipment doesn't need to be as sensitive as that for MT or AMT. ii)

Because of the coherent signal, the usual signal processing and enhancement techniques are far more effective. iii) Thus CSAMT surveyed can be much faster than AMT surveys.

However, there are some disadvantageous as well. One disadvantageous with respect to the natural field mode is the nearness of the signal source. In the natural field methods, the signal source is, in effect, infinitely distant. This "plane wave" assumption simplifies both the mathematics of the technique and the interpretation of AMT/MT data. When a controlled source is used, the "plane wave" assumption is no longer true (at least not close to the transmitter), and the calculated apparent resistivity must be corrected for the "near-field" effect.

In standard operation, six E-field magnitudes and one H-field magnitude are simultaneously measured at 16 binary related frequencies (0.25 to 8192 Hz). The transmitter bipole is located several kilometers from the area to be surveyed. A large trapezoidal survey area, on either side of the transmitter bipole, can be covered from a single transmitter setting. This makes the CSAMT very cost-effective for mapping. In the study area, in Scalar CSAMT (stags A and B), 10 binary related frequencies (from 8 to 4092) were measured. The tensor CSAMT measurements (stage C) at each station were made at 11 frequencies from 4 to 4096 Hz with binary steps. In those selected sites for measuring very low frequency, this frequency ranges began from 0.25 HZ.

Apparent resistivities are calculated in real time for the six E-dipole sounding locations, at each of the 10 frequencies. Phase, E-field, H-field and the standard deviations are also measured and stored. Later in a field-based camp, the data are dumped into a field computer and processed automatically to generate an apparent resistivity pseudosection plot. A near-field correction is also applied.

A. CSAMT parameters measured and calculated

CSAMT directly measures four basic field quantities: (I) electric field magnitude, (2) electric field phase, (3) magnetic-field magnitude and (4) magnetic field phase. These quantities were measured as scalar, and tensor surveys. In the study area, apparent resistivity and phase difference were calculated directly from these measured quantities. Electric field magnitude (E, MKS units of volts per meter or CGS units of millivolts per kilometer) was obtained from the potential difference measured across a grounded dipole. For ease of interpretation, all data was normalized for current. E-field data was analyzed as a separate parameter in order to determine sources of noise or peculiar responses and to identify the incursion of near-field source effects. E-field data was adversely affected by "static" warping due to edge effects from small 2-D and 3-D bodies. They also can be affected by source overprint and the electrode contact resistance (ECR) effect.

Magnetic field magnitude, H (MKS units of ampere per meter, or CGS units of milligammas, where 1 = 10 Aim = I nT) was obtained from the voltage measured in a high-gain antenna and is normalized for current and antenna characteristics. H is affected by source effects, but not by static effects. Hence, H-field inversions are sometimes useful for producing static-free resistivity data. Vertical H was more sensitive to lateral structure. Like E, H is useful for identifying the incursion of near-field effects. Both the electric field (E-field) and the magnetic field (H-field) were measured, at the frequency transmitted from the remote transmitter bipole source. The magnitudes of both the E-field and H-field are determined, as well as their relative phase. The measured data are digitally stacked, filtered and processed in real-time; the apparent resistivity is also calculated at each frequency. Apparent resistivity (ρ_a, units of ohmmeters) is the quantity most used in CSAMT work. The parameter is calculated directly from E and H by use of equation

$$Z = |E/H| \qquad (1)$$

Various methods can be used to calculate apparent resistivity (Spies and Eggers, 1986), but the simplified Cagniard relation was used for the calculation.

$$\rho_a = 1/5f\ |Ex/Hy|^2 \quad \text{ohm-m} \qquad (2)$$

Where ρ_a is apparent resistivity in ohm-m; f is frequency in Hz; Ex is the E-field magnitude, in mv/km, parallel to the transmitter dipole; Hy is H-field magnitude perpendicular to the E-dipole in gammas. However, note that the Cagniard relation holds only for the far-field, or plane- wave zone. In the transition zone and near-field zones, the Cagniard relation is invalid and yields unrealistic resistivity values. To calculate the exact apparent resistivity in the near-field and transition zone, the exact geometry of the receiver-transmitter locations is drawn in all three stages a, b and c (Fig 2-5, 2-6, and 2-7).

Phase difference (φ, units of milliradians) is the phase of the impedance, and is calculated from the difference between φ_E and φ_H). The phase difference in a homogeneous earth is $\pi/4$ rad (785 mrad) in the far-field, and zero in the near-field. Phase difference is proportional to the slope of the log of the resistivity; values higher than pi/4 indicate high-over-low resistivity layering, while values lower than r/4 indicate low-over-high layering.

$$\Phi = \Phi E - \Phi H \qquad (3)$$

Where Φ is phase differences in radians; ΦE is E-field phase and ΦH is H-field phase. In addition to the parameters mentioned above, the standard deviation of each parameter, Ex, Hy, p_a, ΦE, and ΦH, are calculated for signal verification. Phase difference is a very useful and helpful quantity in evaluating noise and is also useful for identifying the location of the transition zone, where phase contours in a pseudosection take on an artificial "layered" appearance.

Apparent resistivities were calculated in real time for the all E-dipole sounding locations, at each of the acquired frequencies. All the measured and calculated data were transferred and stored in the Mass storage device at the measuring site. Later in a field base camp, the data are dumped into a field computer and processed automatically to generate an apparent resistivity pseudosection plot. A near-field correction is also applied. A 1-D inversion program has also been developed for the field computer.

The main problem of the CSAMT method is the near-field effect that limits the depth of investigation. In addition, the transition-field notch generates elliptical contours of low resistivity that can be misinterpreted easily with the anomaly of the ore body.

A schematic of the basic equipment configuration is shown in Figure 2-2. The large transmitter dipole is located as far away from the receiver dipole as is practical - usually three skin depths at the lowest frequency being used; where skin depth is defined as:

$$\delta = 503 \sqrt{p/f} \text{ m} \qquad (4)$$

where δ= skin depth in meter, ρ = resistivity in ohm-m and f = frequency in Hz.

The receiver setup was comprised of a grounded dipole oriented parallel to the transmitting dipole for maximum electric field pickup, and a high-gain coil oriented at right angles to the dipole for magnetic field detection. The E-field dipole is usually between 15 and 150 m in length but in the study area varies between 50 (stages A&B) to 100 and more (stage C), depending upon the survey objectives.

B. CSAMT signals and parameters

1 Field Strength

At a given receiver location, the field strength depends upon various factors: location of measuring point (relative to Tx, bipole), transmitter bipole length, current in the bipole, earth resistivity and measuring frequency. It is important to know the approximate field strength to be expected, in order to plan the survey grid and to locate the transmitter properly, to give the optimum survey results. This transmitter location was on of difficult tasks, in one hand to get strong signal transmitter should be not far from the receivers' site but on the other hand it should be set far away to get signal in far-field zone. The best way is to prepare a very powerful generator (Like GGT-30) that is able to supply strong signals in far distances. However, topography and logistics are also important. It is always not easy to get a proper almost plan view place to spread 2000

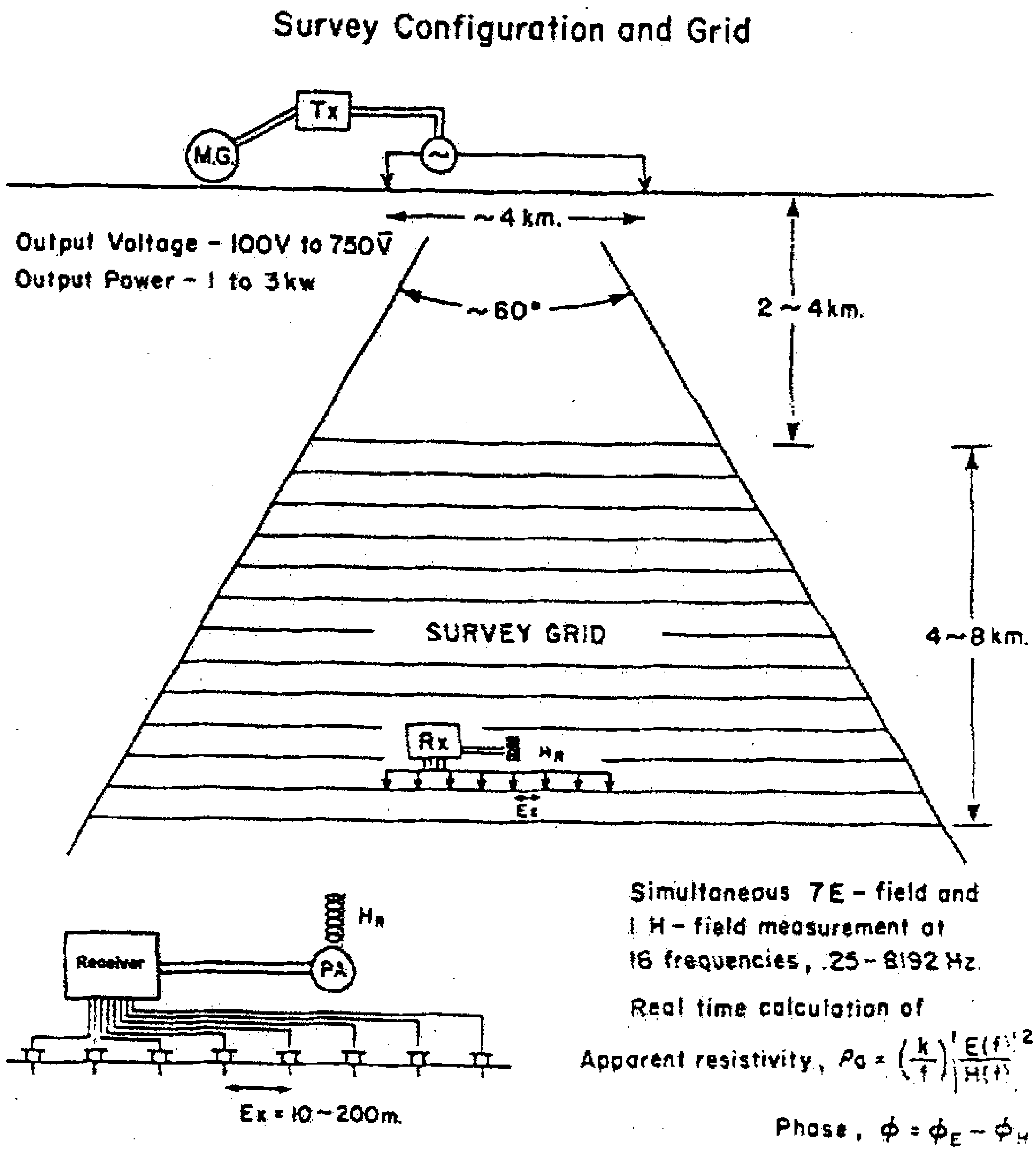


Figure 2-2. An eight-channel receiver used for CSAMT. The survey configuration and grid is illustrated. The standard scalar configuration performs six (or seven based on the employed instrument type) Ex-field and one Hy-field measurements at range of binary-step frequencies (0.25 to 8192 Hz). As illustrated, the E-fields are measured with a dipole using non-polarizable electrodes, in the same way as for IP measurements. The survey traverse line, for the series of equally spaced E-dipoles, is parallel to the transmitter bipole. A horizontal magnetic sensor coil is placed on the ground (three coils in tensor measurements), approximately at the centre of the series of E-dipoles.

meters wire. In the study area, transmitter was put in the most optimum available place in the region to get the best possible signals. Furthermore, in order to keep high signal/noise ratio we forced to move transmitter in different stages.

2 Apparent Resistivity

The apparent resistivity is calculated from the ratio of the electric field and magnetic field magnitude using the well-known Cagniard equation (1) for MT. It should be noted that this Cagniard equation is exactly valid only in the plane wave region of the electromagnetic field; i.e., when the distance between transmitter signal source and receiving location is sufficiently large. The effective survey area, a trapezoidal shape, indicated in Figure 2-2 is thus optimally placed within the maximum possible field strengths and by avoiding areas that are too close to the transmitter bipole.

IV. Applications

CSAMT has been applied extensively to mineral exploration, including Massive sulfides, hydrothermal altered and sulfide targets, and over various base metal, and precious metal deposits. Moreover, it has been used for geothermal studies including mapping reservoir fluids, thermally caused resistivity anomalies, and mapping subsurface structures, evaluating groundwater mixing, delineating hydrothermal alteration, and monitoring reservoir changes.

Mapping structures related to oil entrapment has been also studied by CSAMT in oil exploration. In general, the technique is most useful for reconnaissance mapping prior to final detailing with reflection seismic, but it is also useful in areas where seismic is not effective, such as over basalts. Investigating groundwater quality, mapping brine plumes from leaking tanks, pipelines, and waste-containment ponds and groundwater contamination studies are also surveyed by CSAMT effectively.

Geotechnical applications have been abounded in recent years. Some typical examples include structural analysis in mine planning, detecting voids in underground mines, mapping burn fronts in underground coal gasification projects and coal mine fires, and monitoring enhanced oil recovery operations. It has been used even for mapping spilled petroleum products at refinery sites and for monitoring leachate solution in in-situ copper recovery projects. CSAMT is a relatively new and undeveloped technique and the applications listed will surely be expanded in the near future.

V. Cultural Noise

Despite a generally strong signal source, CSAMT measurements encounter significant amounts of electrical noise. Noise sources can be broken into five categories: operator error, instrumentation noise, cultural noise, atmospheric/telluric noise, and wind noise.

"Cultural" noise arises from man-made metallic structures and radio stations, which carry electric signals. Small town is set up in the Woxi deposit area that cause main source of cultural noise. In general, culture can be grouped into three categories, according to severity of contamination. The first and worst group includes 24-hour underground mining work, grounded metal pipelines, underground phone and power cables, and major cross-country powerlines. The intermediate group includes medium-duty powerlines, train tracks, steel well casings, and culverts. The least disruptive group includes metal fences and overhead telephone lines. While these features vary radically in disruptive effects, those in the first two categories caused the most concern in establishing the survey line locations. A particularly troublesome situation is an interconnected network of pipelines, powerlines, cased wells, and other features. This "mass culture" effect can made accurate measurements very difficult in some stations and longer time spent to acquire data in order to get better average results.

Moreover high-voltage, cross-country powerlines and under ground cables were a major source of cultural noise. In the study area, 50 Hz and its odd harmonics (150, 250, 350, 450,... Hz) are the noise sources; in some cases, specific modes of transmission and modulation may produce unexpectedly high noise levels at higher harmonics, necessitating custom notch filtering. An additional problem is load shifting in the powerlines which can produce time and frequency varying AC signals. These signals make extended stacking-and-averaging very difficult. A relay radio station in the area and powerlines constitute other sources of radio interference. Some signals, such as powerline often occur within the top portion of the CSAMT data band. Vibration from vehicular traffic on nearby roads, as well as effects from the moving metallic mass of vehicles, were also affected data acquiring in some stations. Figure 2-3 shows culture along the lines C1-C4 in the study area.

Cultural noise was best dealt with by avoiding the noise source or by laying out the wires in a minimum coupling configuration (perpendicular to culture). However, this configuration was not possible using Tensor CSAMT. Cultural noise was then countered by notch filtering, band-pass and low-pass filtering, and by extended stacking-and-averaging whenever time was enough. Generally, in tensor measurements, time is a main problem in stacking and averaging. Other sources of cultural noise such as atmospheric noise, and wind noise are not considerable in the study area. The explained cultural noise effect damaged acquired data, specially tensor data, in some stations. Therefore, although lots of processing work was done to reduce its effect, but finally data of two tensor CSAMT lines (lines C3 and C4) was not suitable for 2-D data processing.

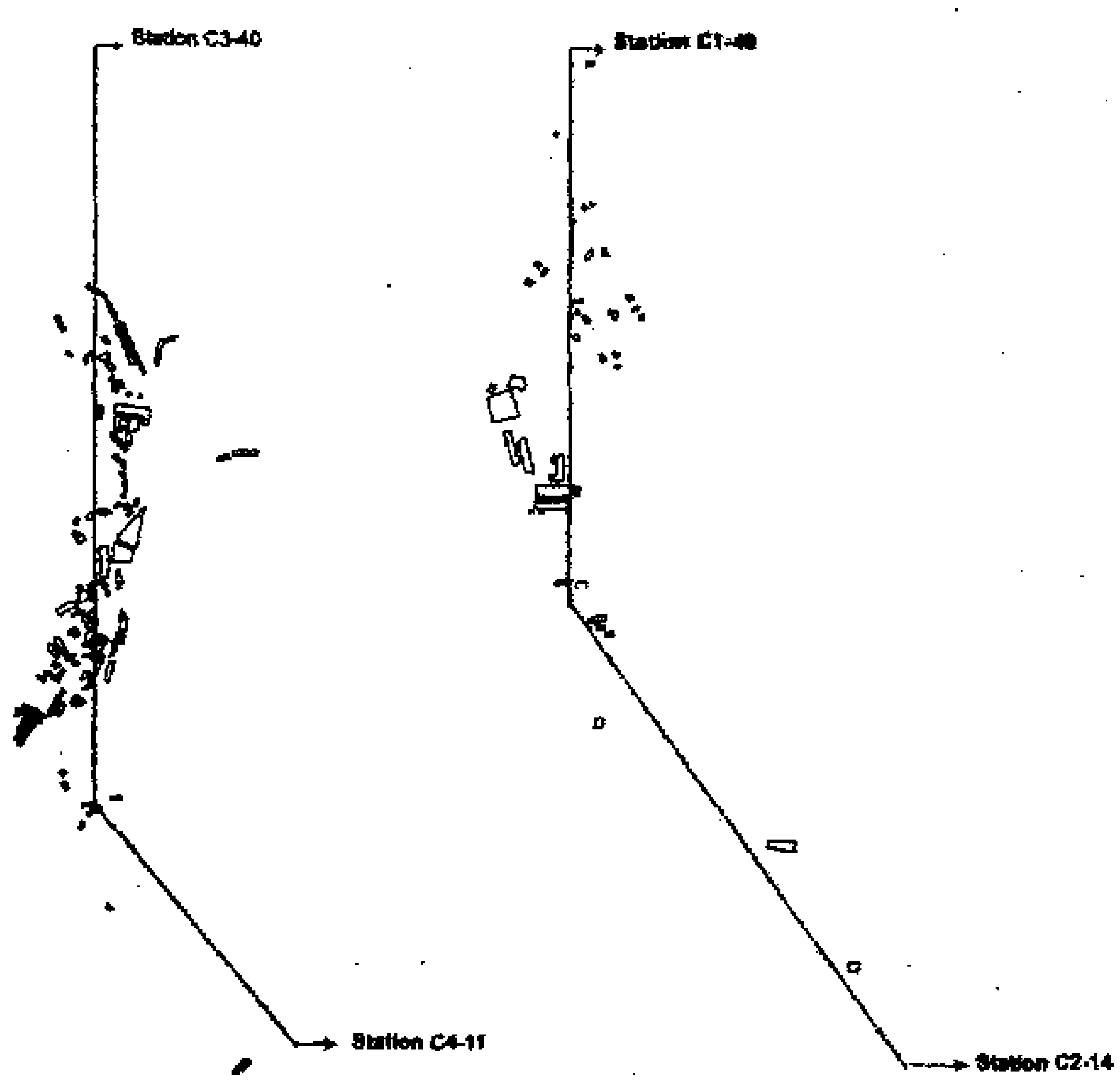


Figure 2-3. Culture along the designed tensor CSAMT lines is shown. Because of high culture damaging effect on data quality, more time was spent to acquire data.

VI. Tensor and Scalar Measurements

CSAMT measurements can include from two to as many as many ten individual-component measurements, depending upon the geologic complexity and economic constraints. Based on the number of components measured and the number of sources used, as shown in figure 2-4, two scalar and tensor measurements were carried out in the region.

Tensor CSAMT is defined as a five-component (Ex, Ey, Hx, Hy and Hz) measurement using two source polarizations. Two sources are required because, unlike natural source magnetotellurics, the CSAMT source is not omni-directional, but fixed in a specific polarization. CSAMT must employ at least two distinct source polarizations to fully define the impedance tensor. A total of ten component measurements, five with each source, are required.

Tensor measurements, *for first time in China,* were used in the area to study the structures. Potential dipole size was chosen large enough so that soundings were far apart relative to the size of geologic features under investigation. As soundings are made closer together, tensor quantities such as tipper become less important because geologic features are mapped directly by virtue of the higher survey resolution. Besides studying the structures, comparison between implemented tensor and scalar surveying in the study area will be undertaken to find out their efficiency.

Tensor CSAMT imposes certain limitations on where the measurements can be made with respect to the sources. This imposition places greater logistic restrictions on tensor CSAMT than tensor MT.

The coincident-source configuration was chosen instead of separated-source configuration because it is more efficient logistically, since the transmitter need not be moved to energize both source antennas. In addition, a differential source overprint is less likely if the sources are coincident. As trend of main structures of the study area are NE to NEE therefore, tensor sources were oriented almost parallel and perpendicular

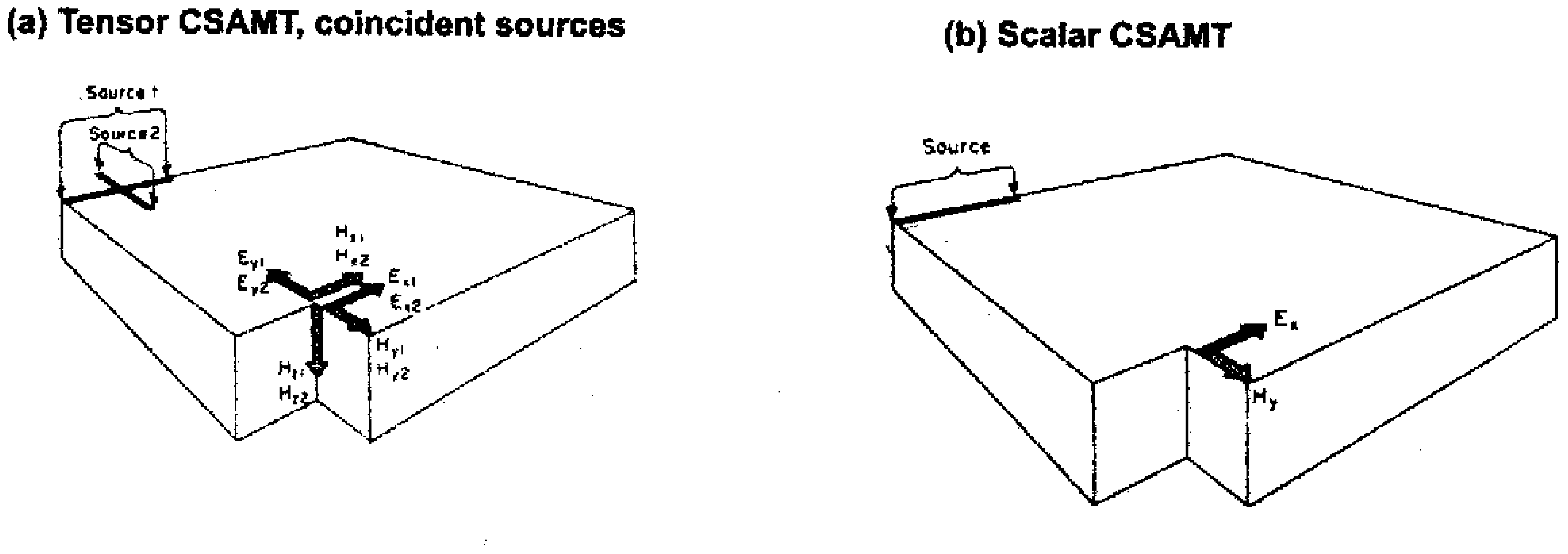


Figure 2-4. Schematic image of implemented Scalar and tensor CSAMT techniques in the study area. (a) Full-tensor surveys use two sources and make five measurements per source in a coincident- source configuration. (b) Scalar CSAMT makes two measurements from a single source.

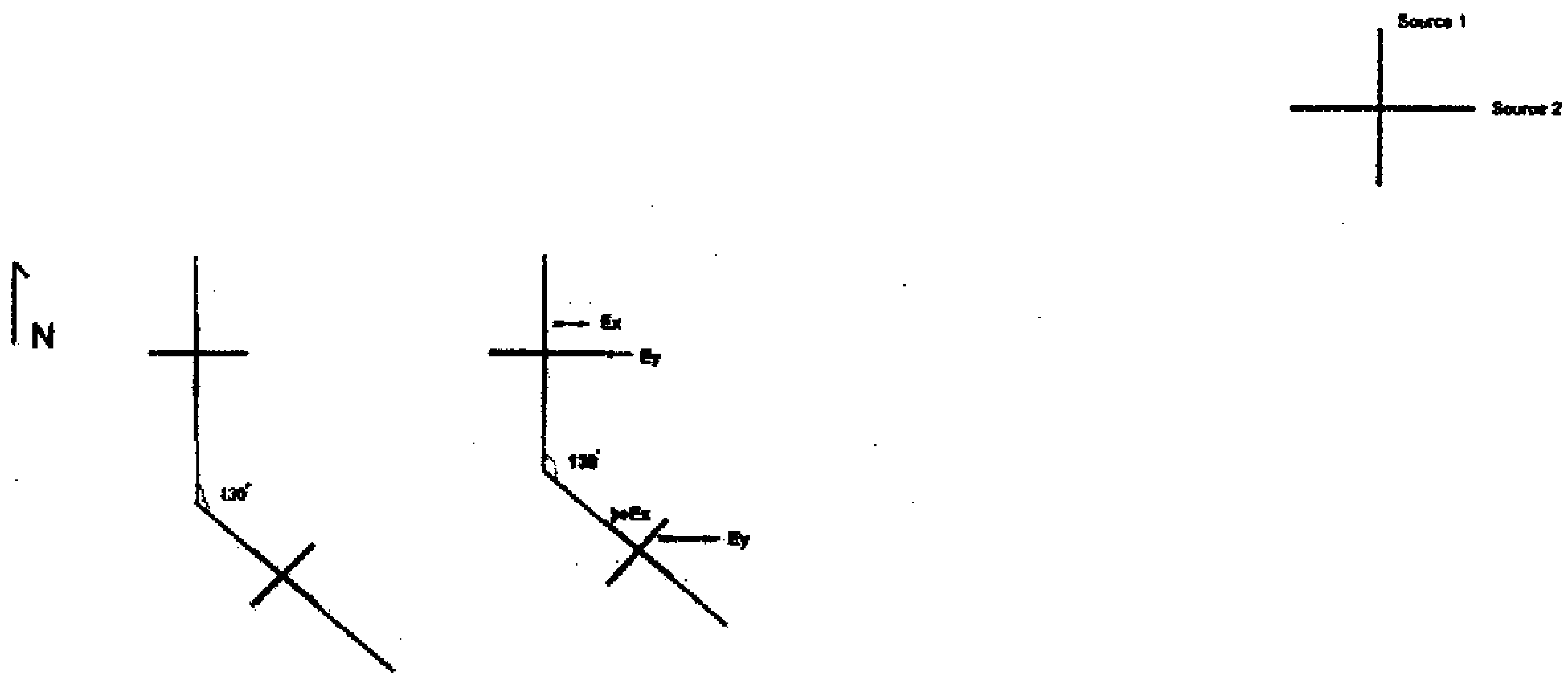


Figure 2-5. Designed tensor CSAMT survey lines C1-C4 and coincident bipoles are shown.. Two independent bipole transmitters were set in the NE of the surveyed lines C1-C4. Transmitter bipoles and survey lines are almost parallel along the N-S direction and perpendicular to general trend of the regions structure (NE). Lines are almost perpendicular to general trend of structures in the study area (scale 1:10000).

to geologic strike, in N-S and E-W directions (Figure 2-5). This orientation simplified the interpretation by directly providing TE and TM mode information.

Tensor quantities are useful for interpreting 2-D and 3-D structure when high-density data are not obtained. Tipper, which is related to the ratio of horizontal H (H_x and H_y) to vertical H (H_x, H_y, and H_z), indicates the presence of and direction to local inhomogeneities.

Scalar CSAMT is defined as a two-component (Ex and Hy or, alternatively Ey and Hx) measurement using a single source polarization. Scalar is adequate in 1-D layered environments or in 2-D environments where the direction of strike is known such as the study area. In more complex environments, scalar may or may not be sufficient, depending upon data density. Single survey lines are especially risky in complex geology when using scalar CSAMT. For example, a linear, steeply dipping fault will be easily detected by scalar data if the dipole orientation happens to be perpendicular to the fault (TM mode); however, if the E-field dipole is in a parallel (TE mode) orientation, fault interpretation and location becomes difficult. As a result, scalar measurements over 2-D and 3-D areas are usually made on a dense grid network. A densely gridded survey partly overcomes the lack of multi component data inherent in vector or tensor. Again, the main exception is where regional anisotropy is strong. In such cases tensor or vector data may be preferred. The main attraction of scalar CSAMT is its relatively low cost and high production speed. These attractions probably explain why the vast majority of CSAMT data obtained so far have been scalar. Hence, since tensor CSAMT is relatively expensive (Uchida et a!., 1987), therefore study of deposit in stages a and b was undertaken by less expensive rapid Scalar CSAMT while in stage c, tensor measurements were made in order to study the structures and mineralized zones in detail and finding out its efficiency comparing scalar method.

VII. Data Density Considerations

The issue of data density is related to resolution both in a vertical and in a horizontal sense. Adequate density is needed to properly define the exploration target, but there is always a trade-off between full target definition and economic limitations.

Vertical density is normally dictated by the frequency range (e.g., 1, 2, 4, 8, . . . Hz) typical of modern digital instrumentation. Survey designed to obtain a sufficient quantity of data over both the features under investigation and over "background". The data was sufficient to fully define anomalous features. Selecting a dipole size smaller than feature to be resolved was very important issue. In stage A and B dipole size was selected as 50 m that was enough for mineralization detection. While in Stage C, dipole size was increased to 100 and in some stations even to 200 m that was enough to study the main structures of the area. Therefore based on the study objective, dipole size and consequently data density were defined. The data density should allow true geologic effects to be distinguished from spurious effects due to culture, topography, differences in TE/TM mode behavior, etc. Regular station spacing except in line C2 were set in order to avoid biasing effects in interpretation.

VIII. Resolution Considerations

The vertical resolution of subsurface features depends upon their lateral size, thickness, depth, and resistivity contrast with respect to background. Thick buried layers, such as basement, are better resolved than intermediate layers. 2-D and 3-D bodies are often harder to resolve than layers at an equivalent depth. Both TE and TM mode (E-field parallel and perpendicular to structure) data are some times needed to properly define 2-D and 3-D structures. The horizontal or lateral resolution is related primarily to the size of the E-field dipole. As a general rule, the lateral resolution for TM mode is roughly equal to the dipole size. Smaller targets may be detectable, but resolution of their position is still dependent upon the E-field dipole size. Hence, effective survey planning requires optimization of the dipole size to the expected minimum target size and to the degree of resolution desired in defining the edges of interesting features. In the study area, surveying were undertaken using two different resolutions. In stages A and B, dipole sizes were chosen 50 meters and resolution is high but in Stage C (tensor) dipole size set to 100 meters (and in few stations 200 meters). The main reason for decreasing resolution in tensor CSAMT was using it for structural studies. While in stages A and B focus was in mineralization zones detection.

IX. Geologic Considerations

Careful consideration of geologic trends prior to designing the survey is important. This consideration is particularly critical in areas where a prevailing geologic strike exists. In the study area, the trend of main structures is northeast. The geology will dictate whether scalar measurements will suffice, or whether more complex measurements are needed. Geology also determines the orientation needed to achieve TE or TM mode measurements, whichever is desired. Highly conductive faults, steeply dipping beds and veins, low carbonaceous strata, thin mineralized veins, homogeneous lithology and high quartz rich veins are examples of geologic features requiring extra care in the study area.

X. Topographic Considerations

The study area has a rough topography. Any E-field measurement will be affected by current density changes resulting from topography, and CSAMT is no exception. Higher current densities are found in valleys, resulting in artificially high resistivities, while lower current densities are found in hills, resulting in artificially low resistivities. While these effects are not easily avoided in the field, it is prudent to consider them when laying out survey line locations. Because identifying all stations on a topographic map is helpful in data interpretation, therefore all the stations elevation and geographical position were exactly mapped by surveying and was used in a simple GIS system. Evaluation of all stations were undertaken to distinguish topographic and non-topographic anomalies. Topographic effects can strongly influence the interpretation process, therefore they were tried to be removed in data processing.

XI. Instrumentation

CSAMT measurements require modern digital instrumentation. The receiver should be capable of both magnitude and phase data acquisition in the 0.1 Hz to 10 kHz frequency range, and should have two or more measurement channels for maximum efficiency. Analog to digital conversion rates of 25 *us* or less per conversion ensure adequate signal digitization at high frequencies. Proper antialias filtering is an essential part of receiver performance. Multi-notch filters for powerline frequency rejection and telluric-tracking filters are often required for acquisition of high quality data. High input impedance, low noise in active components, and low cross-talk between channels are also necessary characteristics.

The receiver should be capable of real-time apparent resistivity calculation and data display, some type of immediate viewing or hard-copy output of data and statistics, and display of field signals on an internal or external oscilloscope. These provisions enable the operator to maintain data quality in adverse noise conditions, to monitor proper equipment function, and to optimize logistical procedures. At least daily plotting of the data (and preferably real-time plotting) is an essential part of maintaining control over data quality. In addition, the equipment should permit convenient data transfer to a separate computer at the end of each field day.

The transmitting unit should be current controlled and provide a stable waveform across a broad frequency range. Output of 10 to 100 A and 1000 V desirable. In order to obtain quality phase data, the transmitter should be synchronized the receiver by high-precision crystal clocks or a radio link. Since transmitters are subjected to high currents and volt ages, ease of maintenance is an important consideration.

A high-gain, field-durable magnetic antenna is required for the H-field measurements. The usual type is a ferrite or iron core, dB/dt design with its resonant frequency above I kHz. A minimum antenna response of 0.1 mV/-y-Hz (0.1 mV/nT-Hz) is normally required for low frequencies, and a maximum response of 200 mV/-y for frequencies above 100 Hz. The

CSAMT transmitter system consisted of a transmitter console, a frequency control device, a motor generator and peripherals. For the transmitter bipole, a long magnet wire is laid on the ground, in an approximate straight line. Both ends are grounded.

The transmitter was a Zonge Engineering and Research Organization model GGT-6, generating 7.5-kW output power. This generates an electromagnetic field at the same frequencies. The transmitter bipole was located several kilometers from the surveyed area. A large trapezoidal survey area, on either side of the transmitter bipole, can be covered from a single transmitter setting. This makes the CSAMT technique very cost-effective for mapping. The receiver was a Zonge Engineering Organization model GDP-16 eight-channel receiver console, which measures both the E- and H-fields in synchronizing connected to an IBM-compatible PC. Magnetic field sensors were obtained using model BF-10 magnetic sensor coil from EMI. The H-field coil was centered on this dipole and moved approximately 6 to 7.5 m off line, away from the receiver operator, to minimize noise generated by motion of the operator.

XII. Laboratory measurements

Some samples from different Formations were picked to undertake physical parameters test in lab. The same tests were done on in- Situ samples in the field. As it can be seen in table 2-1, rocks and ores in the deposit show significant outdoor electrical resistivity differences.

Although samples had been soaked in water at least 24 hours before measuring, but still there are differences between the samples' electrical parameter with field observational value. The observation result in field reflects the real state of rock and the comprehensive regional electrical

Table 2-1. Results of laboratory and field tests on electrical properties of rocks of the study area.

Strata Lithology \ Physical parameter / Quantity		PFE (%): Measured in lab. Total sample	PFE (%): Measured in lab. Range of variance	PFE (%): Measured in lab. Mode	PFE (%): Measured in field	ρ_s (ohm-m): Measured in lab. Total sample	ρ_s (ohm-m): Measured in lab. Range of variance	ρ_s (ohm-m): Measured in lab. mode	ρ_s (ohm-m): Measured in field
Gray, purple slate		17	1.0-2.4	1.2	0.8-1.0	17	397-1857	1047	800-1600
Pale, altered slate (without mineralization)		13	0.4-4.9	2.9	1.6-2.3	13	239-2988	1135	500-1000
Mineralized zones	altered slate	28	2.8-9.3	4.5	6.0-8.0	28	430-1180	670	100-600
	Scheelite-rich	15	2.3-4.6	3.8	6.0-8.0	15	508-1270	720	100-700
	Stibnite-rich	18	4.7-14.0	9.0	6.0-8.0	18	80-636	213	100-300
Silicified, pyritized slate		8	3.3-7.0	5.4	3.0-5.0	8	700-2198	1543	1600-2000
structural breccia		9	1.8-5.4	2.6	2.0-3.0	9	1080-2908	1728	1500-2000
Red strata		-----	---------	---------	1.5-2.0	-----	-------------	--------	100-300

properties of fillers in their pores and joints. Moreover, sample and original rock are in different physical and chemical conditions, while electrical parameters are easily influenced by factors like humidity, temperature, pressure, fillers in the holes and mineralization degree. Besides, sample determining with small dipole and determining direction have big differences with what actually observed in the field. Therefore, through electrical parameters we can only approximately know the relative differences of ore resistivity.

From table 2-1, the resistivity of the veins and host rock slate show obvious differences. While mineralized zones have low resistivity and high PFE, slates have high resistivity and low polarization and both are easily distinguishable by CSAMT. The problem is dealing with low-resistivity, low-PFE Cretaceous red strata in the region. They cannot be distinguished from ore veins by CSAMT. Fortunately they are just in the shallow depth of the area and are detectable from anomalous deep mineralized zone. Moreover, IP method can be helpful in distinguishing them from high-PFE mineralized zones.

XIII. Source field behavior

Receiving a plane wave at a survey site is Cagniard's condition for the source field. It can be fulfilled when the source is more than three skin depths away (Goldstein M.A. et al., 1975), using the largest resistivity between receiver and transmitter and lowest frequency of sounding (Sandberg and Hohmann 1982).

The working parameters of the CSAMT method include the minimum and the maximum distance of the transmitter, working frequency range and transmitter position. These are determined by the objectives and the physical properties of the study area. The objective in the Woxi deposit is to determine the extending depth of deep mineralized zones about 1000 meters underground. The outcropped strata of the survey area are Proterozoic slate rock and Cretaceous red strata. The average resistivity of the slate rock and red strata is about 1000 and 300 ohm-m respectively. Minimum working frequency is

$$f_{\min}=p.\ (356/\delta)^2=300*(356/2000)^2 \approx 9.5\text{ Hz} \qquad (5)$$

Where

$\delta=503\sqrt{p/f}$ (δ: skin depth)

Therefore, in the survey, the minimum and maximum working frequencies were 8 and 4096 Hz, respectively. In the Cretaceous red strata area, the approximate penetration depth of the maximum frequency was about 100m. Thus, the investigating depth of the implemented CSAMT method ranged from 100m to 2000m. The investigation depth limits the minimum distance of the transmitter. Investigation depth of a plane wave depends on skin depth

$$D= \delta/\sqrt{2} \qquad (6)$$

To get a plane wave in receiver sites, the transmitter should be placed in a farther distance, normally more than 3 to 5 skin depths at the lowest frequency of interest so that far-field conditions exist at the site. The maximum distance of the transmission was limited by the signal /noise ratio and sensibility of the instrument. However, for some reasons like applying CSAMT in highly resistive terrain, the required transmitter-receiver separation may be so large that obtaining data too close to the transmitter is necessary in order to receive the signal. Moreover, in some areas like the study area, high N/S ratio aborted attempts to locate the transmitter dipole at far distances to satisfy the far-field condition and forced to acquire data in the transition zone. Therefore, according to the parameters above, in the slate rock area, data of frequencies lower than 32 Hz was put in the transition zone and the apparent resistivity in the all-zone calculated.

XIV. Survey grid, and data acquisition

Figures 2-5, 2-6, and 2-7 illustrates the survey configuration and grid for stages a, b, and c respectively. The standard configuration performs six (or seven) Ex-field and One Hy-field measurements at each of 10 frequencies, (8 to 4096 Hz) at stage A, and B and 11 frequencies at stage C. As illustrated, the E-fields are measured with a dipole using non-polarizable electrodes, in the same way as for IP measurements. The survey traverse line, for the series of equally spaced E-dipoles, is parallel to the transmitter bipole. The measurement dipole length is determined by the desired scale of the survey; this may also be influenced by the E-field signal strength, which is in turn determined by the transmitter-receiver distance, the transmitter bipole current and the earth resistivity. The receiver dipole length may be in the range from 10 meters to 200 meters.

A horizontal magnetic sensor coil is placed on the ground, approximately at the centre of the series of E-dipoles. It was placed several meters away from the E-dipole line and the receiver console, to avoid interference, as well as to reduce inductive coupling due to operator movement. Only one H-field measurement is required for each group of six E-field measurements but at stage C, two H-field were measured.

The transmitter (powered by a suitable motor generator) sends current into a long wire, grounded bipole. The length of the bipole may be varied from several hundred meters to several km, depending upon signal strength requirement. This will also be discussed later, as will be the transmitter-receiver distance and the survey area to be covered with one transmitter bipole location.

A. Stage A

In the survey, the maximum distance of the transmitter was 8380 m (line A4) where he acquired data had good S/N ratio. The minimum distance of transmitter in the study area was 6380m (line A8) where the acquired data had high S/N ratio. Positions of transmitter and survey lines are shown in table 2-2 and figure 2-6.

Table 2-2. Transmitter bipoles and surveyed lines location and other specifications.

Line no.	Total stations'	Distance from Transmitters	Largest angle	Bipoles' length and coordination
A1	10	7840		
A2	13	7130		Tx1
A3	27	6620	27°	AB=1160 m (
A4	33	8380		
A5	40	7880		Tx2
A6	40	7380		AB=1860 m
A7	39	6880		
A8	35	6380	24°	

From this figure, the maximum skew angle of the stations in the field for Tx1 and Tx2 was less than 30°. They both are smaller than the skew angle of the theoretical Zero-field value line (45°) and are smaller than the working maximum skew angle in practice (30°). Therefore, the layout of the work is reasonable and practical, and the acquired data is authentic.

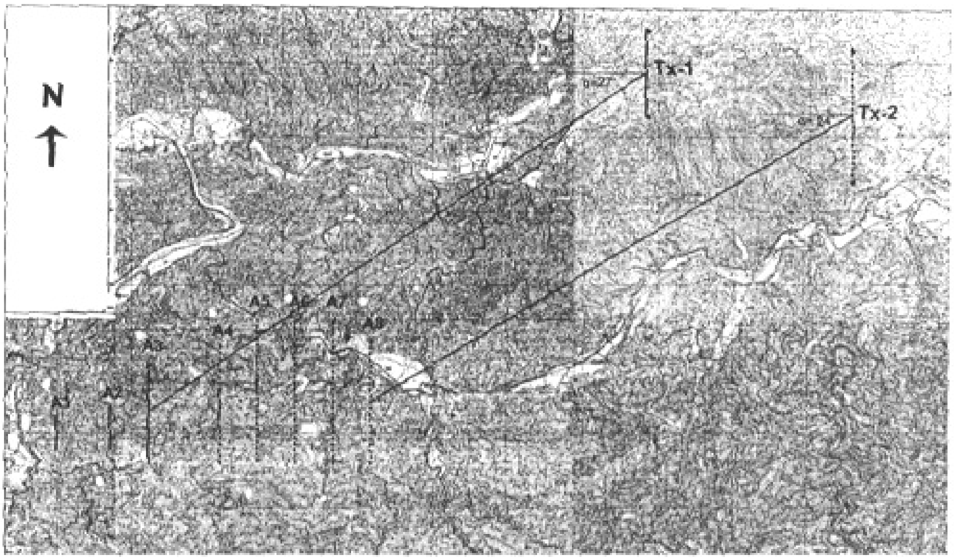


Figure 2-6. CSAMT survey layout of Woxi Deposit. Two independent bipole transmitters were set in the NE of the surveyed lines A1-A8. Transmitter bipoles and survey lines are parallel along the N-S direction and perpendicular to general trend of the regions structure (NE). CSAMT data was acquired in two independent stages, first stage; lines A1-A3 using the TX1 and second stage, lines A4-A8 using the TX2 bipole (scale 1:10000).

B. Stage B

Second stage (stage B), CSAMT sounding was carried out by adding 6 lines to the west of the Stage A lines in order to: i) to know the extending situation of singularity and blind mineralized zones, ii) to get a clear image of the mineralization in western area and iii) to find out the spatial relationship between eastern and western zones. In this section by analyzing all CSAMT cross sections of Cagniard resistivity, by analyzing apparent resistivity section plots and apparent resistivity contour plots in different depths and 3-D apparent resistivity plot, we can get the characteristics of ground electric field attribution and the relationship of geological regularities, and also we can determine the mineralized zones' location.

Table 2-3. Transmitter bipole and surveyed lines location and other specifications in stage B.

Line no.	Number of stations	Distance from Transmitters	Largest angle	Bipoles' length and coordination
B1	20	5500		
B2	20	6100		
B3	40	6600	27°	Tx1 AB=1500 m
B4	38	6800		
B5	38	7100		
B6	20	9500		

In the survey, the maximum distance of the transmitter was 9500 m (line B6) where he acquired data had good S/N ratio. The minimum distance of transmitter in the study area was 5500m (line B1) where the acquired data had high S/N ratio. Positions of transmitter and survey lines are shown in table 2-3 and figure 2-7.

From this figure, the maximum skew angle of the stations in the

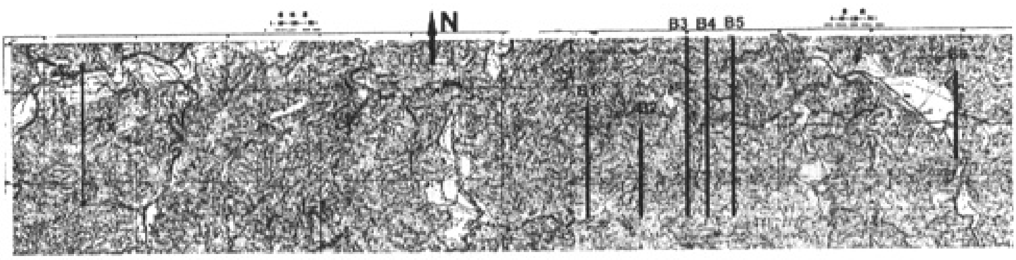


Figure 2-7. CSAMT survey layout of Woxi Deposit in stage B. Bipole transmitter was set in the west of the surveyed lines B1-B6. Transmitter bipole and survey lines are parallel along the N-S direction and perpendicular to general trend of the regions structure (NE), (scale 1:10000).

field for Tx1 was less than 20°. They both are smaller than the skew angle of the theoretical Zero-field value line (45°) and are smaller than the working maximum skew angle in practice (30°). Therefore, the layout of the work is reasonable and practical, and the acquired data is authentic.

C. Stage C

Tensor CSAMT sounding in the study area was carried out on four lines. (Figure 2-5, table 2-4). Each line is between 1000 to 2200 m (C1 1600 m, C2 2150, C3 2250, and C4 1000 m). Lines 1 and 3 are in NS direction, line spacing is 1450 meter, and dipole spacing is 100 m. Lines 3 and 4 are in NW-SE direction, line spacing about 1900 meter, and dipole spacing 100 and in some stations in line 2, 200 meters. Maximum used current was 7.5 A with the 8 to 4096 hertz frequency range. Magnetic field (Hy) amplitude, electric field (Ex) amplitude and phase (φ_M) were measured applying Scalar CSAMT. X-axis is defined normal to each profile, almost parallel to the assumed strike of major structures of the study area. Y-axis is CW along the profiles then.

Table 2-4. Transmitter bipoles and surveyed lines location and other specifications in stage C.

Line no.	Number of stations	Bipoles' length and coordination
C1	17	Tx1 AB=2000 m Tx2 AB=1900 m
C2	14	
C3	23	
C4	11	

XV. CSAMT working parameters

Working parameters of the CSAMT method including the minimum and the maximum distance of the transmitter, working frequency range and position of the transmitter, are determined by the study target, and physical properties of the area (table 2-5). The objective was to determine the extending depth of deep mineralized zones of about 1000 meters underground.

XVI. Data quality and accuracy

The landform of surveyed area is complicated and vegetation on the ground and buildings are concentrated. The main factors that affect accuracy of geophysical works are 24 hours underground mining work and cultural noise. Repeating data acquisition in some random stations was done to check the data quality. In order to reduce the operational error at every receiver station, we chose the average value to limit the random interruption. Using this method the required accuracy was achieved. To get the most possible and satisfied accuracy repeating was done beyond the 10% of the original work.

According to the geological task and the web degree of different survey areas, different accuracy requirements are as follows: (table 2-7)

The formula of the mean-square relative error is

$$M = \sqrt{\sum (\delta i / \bar{x}_i)^2 / 2n} \qquad (7)$$

Where

$\delta i = Xi - X'i$, $\quad \ddot{X}_i = (Xi + X'i)/2$

Table 2-5. Working parameters and surveyed lines characteristics in Stage A, B, and C

	Tx1	Tx2	Max. distance of receives and transmitter	Min. distances of receive and transmitter	Dipoles length (m)	Max. current	Frequency range	Measured parameters
Stage A	1160	1860	8380	6380	50	10 A	8Hz~ 4096Hz	Hy & Ex amplitude and phase (φM)
Stage B	1500	----	9500	5500	50	10 A	8Hz~ 4096Hz	Hy & Ex amplitude and phase (φM)
Stage C	2300	1900	10000	7900	100& 200	10 A	4Hz~ 4096Hz	Hy, Hz & Ex amplitude and phase (φM)

Where M is the mean-square relative error and x_i and x'_i are the original measurement value and tested measurement value at the no i station; n is the total tested stations. From table 2-7, we can see that test work is beyond the 10% of the original work, and the accuracies are all satisfying.

In the area of Woxi deposit, the main factors that affect the accuracies of geophysical working methods are ground condition, human interruption, topography, station positioning error and sensibility of the instrument. Ground condition will affect the current and measurement electrodes ground impedance, so to change the power supply and measurement of the potential difference. In order to avoid or reduce the affection, many acquiring have been taken, such as changing the multi-channel data acquiring to the individual station sounding. In order to reduce the operation error at every receiver point, we chose the average value to limit the random interruption. Using this method the required accuracy was achieved.

Tables 2-6 and 2-7 are respectively geophysical Exploration workload arrangement plot of western Woxi and prospecting areas' assignment plot of central and eastern Woxi. Table 2-6 is the statistics of material explore scientific research workload; indoor workload has been employed into term detective point.

A. Precise evaluation of geophysical work

Table 2-7 is the control table between precise projects of material-explore work, which each research area plunged into, and the precise work, which has been achieved in practice. Considering the strong cultural and mine noise, some work done to guarantee the high quality of data including long data acquisition and enhancing efficiency of error

Table 2-6. Geophysical exploration workload in the study area.

Method/ Stations/ area	*Multi-frequency IP*	*Scalar CSAMT (Stage A)*	*Scalar CSAMT (Stage B)*	*Tensor CSAMT*	*MIP Sounding*	*Near-ore IP Profiling*	*Lab. studies*	*Sum*
Woxi	*608*	*268*	*237*	*65*	*5*	*104*	*475*	*1762*
Tanghuping	*340*	--------	--------	---------	--------	-------	*34*	*374*
Total	*949*	*268*	*237*	*65*	*5*	*104*	*509*	*2136*

monitoring and correlation coefficient and at the site till acquiring the satisfied data.

XVII. Data processing, and interpretation

Modern data processing capabilities allow a great deal of flexibility in manipulating data. But they also open the door to confusion or incorrect interpretation if great care is not taken to understand the processing procedure and its inherent limitations. Processing can modify the appearance of data, hence care should be taken to understand and validate the wide range of processing choices available to the interpreter.

Survey data were measured, the parameters were calculated, and transferred into permanent mass storage media at the end of each day, in the field base camp. The data were immediately processed by computer then. The field presentation of the results was in the form of a contoured apparent resistivity pseudosection plot or profile plot. Other parameters such as phase, or E and H field magnitude, were also presented in the same form.

The immediate data processing and presentation was helpful in modifying the survey plan and also to verify acquired data quality in the field. In case of any problem, data acquiring was repeated on the following day. A 1-D inversion program was also developed for use in the field base camp.

Processing was done in two stages: preprocessing and interpretive processing. Preprocessing involves an examination of the data set for errors and noise. The work utilizes magnitude and phase for all E- and H-field components measured, the calculated apparent resistivities and phase differences, and data scatter (preferably both standard deviation, and coefficient of variation).

Once the data were in acceptable condition, interpretive processing began. Interpretive processing involves optimizing plotting conventions for the particular data set, as well as enhancing certain effects by various processes such as normalizing, static correction, filtering, and derivative calculation.

Static corrections were very helpful in removing the erratic appearance of static-affected frequency data. There are different static-correction processing techniques to use. In order to estimate the concrete position of static effect, several typical abnormal positions were chosen and singularity was calculated using "Wavelet method". The results are listed in table 2-9. Processed results finally were compared to the original resistivity and phase information in order to maintain a sense of reality in processing.

Table 2-7. Accuracy and reliability of the implemented techniques.

Applied method	*Designed accuracy*	*Exact accuracy*	*Total work*	*Test work (checking)*	*Precision method*
Dual-frequency IP	*10%*	*4.3%*	*372*	*43*	*Mean-square relative error*
Dual – frequency resistivity	*10%*	*3.1%*	*372*	*43*	''''''''''
CSAMT (stage A)	*10%*	*3.8%*	*268*	*41*	''''''''''
CSAMT (stage B)	*10%*	*6.2%*	*206*	*25*	''''''''''
Near-ore IP	*10%*	*3.6%*	*41*	*8*	''''''''''
Pseudo-random multi-frequency IP	*5%*	*3.7%*	*508*	*100*	''''''''''
Near-ore IP Profiling	*55*	*4.4%*	*94*	*10*	''''''''''
Multi-frequency IP sounding	*5%*	*3.5%*	*5*	*5*	''''''''''

A. Resistivity model and geological implications

Woxi deposit has a large sloped extension. Depth of sloped vein in tunnel no. 28 is about 700 meters (from the ground), while single-layer vein thickness is only one to three meters. Within the prospecting zone, the ratio of the burying depth of single-layer vein is less than 1/100, which is far beyond the high-resolution ratio of CSAMT. This is the negative factor of finding ore layers with CSAMT in the zone. With careful analyzing, we found that Woxi deposit has geological and geophysical characteristics as follows:

1 Veins are in Proterozoic Madiyi Formation of Banxi Group slates so the lithology of host rock is single and resistivity in lack of carbonaceous slates is relatively optimal, hence is greatly decreasing the geological nonambiguity of geophysical prospecting.

2 Every vein has strong alteration zones specially pyritization in both sides. There are similarities in electrical properties between altered wall rocks and veins. They have significant apparent resistivity differences with unaltered wall rock, and are the target of CSAMT prospecting in this zone.

3 Veins and altered wall rocks have group-output characteristics. All the vein layers in the group are parallel with unequal distances. Based on field observations and borehole data, average thickness of veins with their altered wall rocks is about five meters. Their dip angle is 45°, resistivity 100 ohm-m, the average buffer is among layers 50 meters, and slate resistivity is 1000 ohm-m. Modeling as a two-mineral layer group, it can be considered as a 60-m thick low-resistivity (300 ohm-m) stratum. Obviously, with decreasing the veins spacing, resistivity of the model stratum decreases as well. In order to simplify the model, we constitute it with horizontal layers. CSAMT apparent resistivity-sounding curve (Figure 2-8) shows a typical type H (Telford, W.M., et al., 1995). It proves applied CSAMT method's ability in finding such a parallel arrayed underground multi-layer vein group. However, it cannot distinguish the position of individual veins clearly.

According to above discussion, in Woxi deposit, the geological and geophysical prospecting target is composed of low- resistivity, high polarization multi-layer veins and altered wall rocks.

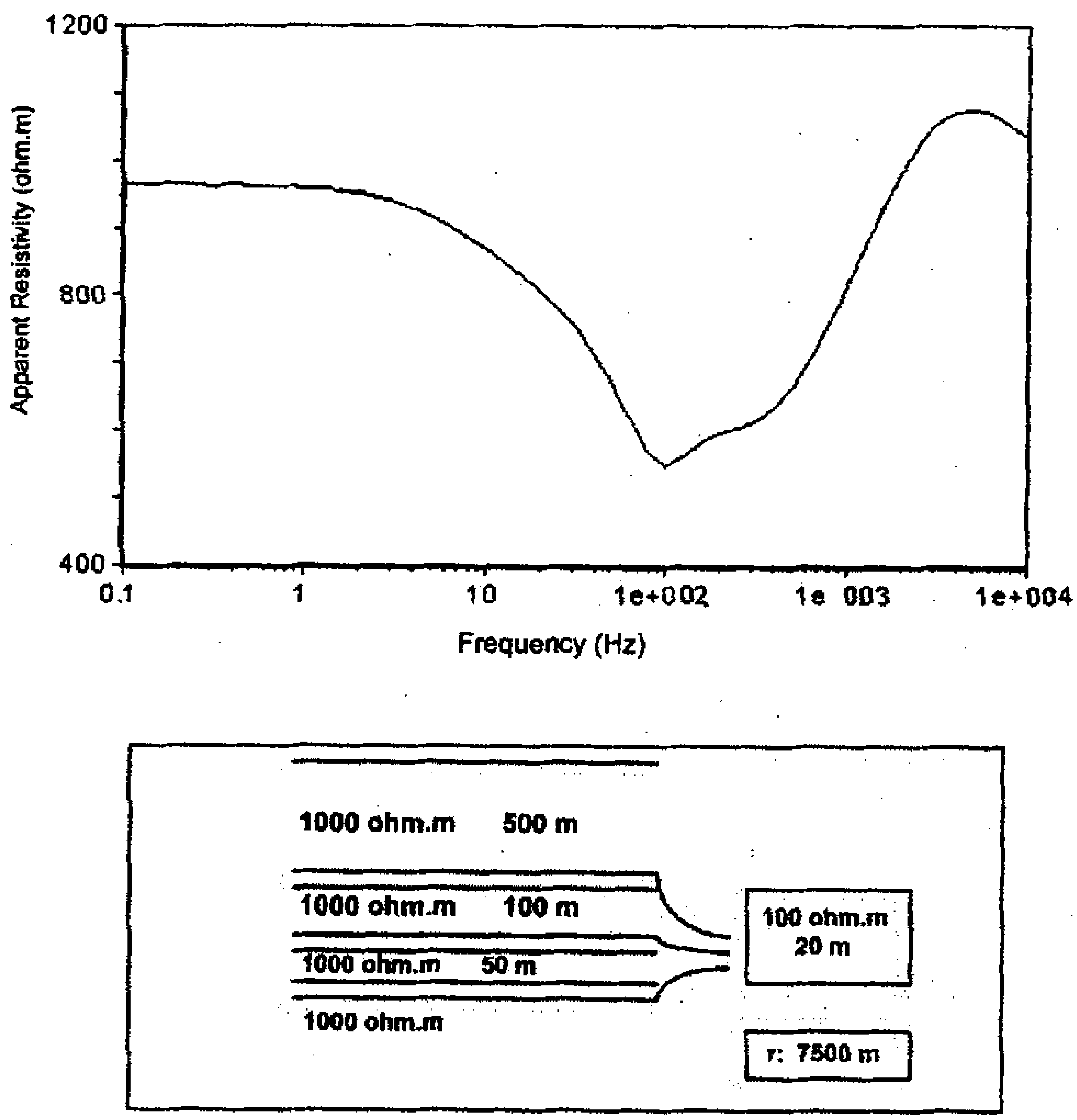


Figure 2-8. Two horizontal mineral layer group modeling. The apparent resistivity-sounding curve shows typical type H. R is distance between transmitter and receivers.

B. Near field and static shift correction

In order to reduce near-field effect, the all-zone apparent resistivity technique was implemented (Tang J., et al., 1994). The raw apparent resistivity pseudosections are shown in Figure 2-9a. There were some vertical anomalous strips on the all zone apparent resistivity pseudosections. Those are mainly caused by static shift, which changes the electrical character and cause fatal errors in depth inversion.

Two methods, wavelet and relative matrices, were used to estimate the static shift at different frequencies (Song S., et al., 1995). The correlation coefficient matrixes of lines are listed in table 2-8. Taking correlation coefficient of line A5 as an example, it is clear that correlation coefficient is small between low and high frequencies, which reflects that electrical properties are not correlative in different depths. It means there is an obvious electrical property difference between deep and shallow strata ground. Nevertheless, the coefficient in large frequencies is increasing, which shows the analogy of electrical properties. The coefficients in low frequencies increase rapidly, reflecting the homogeneity of the deep electrical properties. Therefore, based on the correlation coefficient matrices it is deduced that in the Woxi deposit electric inhomogeneity exists. This electrical inhomogeneity sometimes is strong enough to reduce static shift seriously. Moreover implementing large potential dipole (50m) was helpful to reduce the static shift effect. Even though, where the correlation coefficient between minimum and maximum frequencies is generally large, static effect is still obvious in some profiles. In such phenomena, such as profiles (such as A1 and A2), the electrical property in depth is not revealed clearly.

Table 2-8. Correlation coefficients of matrixes of lines A1 to A8.

A1										
	F14	F13	F12	F11	F10	F09	F08	F07	F06	F05
F14	1									
F13	0.35	1								
F12	0.59	0.7	1							
F11	0.82	0.55	0.9	1						
F10	0.92	0.49	0.83	0.97	1					
F09	0.67	0.43	0.84	0.94	0.84	1				
F08	0.52	0.34	0.73	0.81	0.67	0.95	1			
F07	0.59	0.33	0.77	0.87	0.75	0.97	0.98	1		
F06	0.57	0.56	0.9	0.88	0.76	0.94	0.94	0.95	1	
F05	0.69	0.45	0.84	0.93	0.84	0.98	0.96	0.98	0.97	1
A2										
	F14	F13	F12	F11	F10	F09	F08	F07	F06	F05
F14	1	0	0	0	0	0	0	0	0	
F14	1									
F13	0.5	1								
F12	0.5	0.84	1							
F11	0.59	0.68	0.94	1						
F10	0.59	0.62	0.89	0.92	1					
F09	0.59	0.47	0.82	0.89	0.94	1				
F08	0.56	0.44	0.77	0.82	0.93	0.98	1			
F07	0.5	0.54	0.87	0.93	0.98	0.97	0.95	1		
F06	0.54	0.54	0.87	0.94	0.97	0.97	0.95	1	1	
F05	0.46	0.59	0.9	0.97	0.89	0.87	0.8	0.93	0.94	1
A3										
	F14	F13	F12	F11	F10	F09	F08	F07	F06	F05
F14	1									
F13	0.9	1								
F12	0.93	0.9	1							
F11	0.79	0.55	0.71	1						
F10	0.64	0.48	0.7	0.84	1					
F09	0.85	0.77	0.93	0.78	0.87	1				
F08	0.4	0.46	0.48	0.17	0.18	0.35	1			
F07	0.93	0.86	0.92	0.7	0.67	0.89	0.26	1		
F06	0.94	0.84	0.91	0.76	0.67	0.87	0.27	0.98	1	
F05	0.85	0.76	0.91	0.81	0.88	0.95	0.45	0.86	0.87	1
A4										
	F14	F13	F12	F11	F10	F09	F08	F07	F06	F05
F14	1									
F13	0.88	1								
F12	0.77	0.65	1							
F11	0.94	0.88	0.89	1						
F10	0.21	0.21	0.58	0.31	1					
F09	0.14	0.11	0.56	0.26	0.97	1				
F08	0.16	0.15	0.58	0.3	0.97	0.99	1			
F07	0.23	0.33	0.58	0.39	0.88	0.89	0.92	1		
F06	0.21	0.3	0.55	0.36	0.88	0.9	0.93	0.99	1	
F05	0.29	0.52	0.44	0.45	0.57	0.54	0.59	0.83	0.8	1

Table 2-8. to be continued.

A5										
	F14	F13	F12	F11	F10	F09	F08	F07	F06	F05
F14	1									
F13	0.22	1								
F12	0.43	0.26	1							
F11	0.21	0.14	0.68	1						
F10	0.15	0.17	0.71	0.9	1					
F09	0.21	0.11	0.63	0.97	0.81	1				
F08	0.2	0.11	0.63	0.94	0.79	0.97	1			
F07	0.19	0.13	0.64	0.97	0.82	1	0.97	1		
F06	0.2	0.14	0.66	0.97	0.83	0.99	0.97	1	1	
F05	0.21	0.15	0.69	0.95	0.83	0.98	0.96	0.99	0.99	1
A6										
	F14	F13	F12	F11	F10	F09	F08	F07	F06	F05
F14	1									
F13	0.86	1								
F12	0.55	0.52	1							
F11	0.42	0.32	0.89	1						
F10	0.62	0.42	0.82	0.87	1					
F09	0.32	0.31	0.85	0.94	0.69	1				
F08	0.31	0.3	0.83	0.94	0.69	1	1			
F07	0.32	0.31	0.85	0.94	0.71	0.99	1	1		
F06	0.34	0.33	0.87	0.87	0.69	0.95	0.95	0.97	1	
F05	0.35	0.33	0.88	0.87	0.69	0.94	0.94	0.96	0.99	1
A7										
	F14	F13	F12	F11	F10	F09	F08	F07	F06	F05
F14	1									
F13	0.6	1								
F12	0.63	0.93	1							
F11	0.55	0.85	0.93	1						
F10	0.73	0.87	0.93	0.93	1					
F09	0.64	0.88	0.94	0.96	0.96	1				
F08	0.62	0.87	0.93	0.97	0.96	0.99	1			
F07	0.58	0.84	0.91	0.97	0.95	0.99	0.99	1		
F06	0.53	0.84	0.91	0.96	0.93	0.98	0.98	0.99	1	
F05	0.52	0.84	0.9	0.95	0.93	0.98	0.98	0.96	1	
A8										
	F14	F13	F12	F11	F10	F09	F08	F07	F06	F05
F14	1									
F13	0.93	1								
F12	0.46	0.72	1							
F11	0.35	0.62	0.98	1						
F10	0.38	0.56	0.65	0.66	1					
F09	0.14	0.33	0.59	0.65	0.88	1				
F08	0.08	0.26	0.56	0.63	0.83	0.99	1			
F07	0.08	0.26	0.58	0.65	0.83	0.99	1	1		
F06	0.08	0.26	0.55	0.62	0.81	0.98	0.99	0.99	1	
F05	0.08	0.27	0.59	0.67	0.83	0.97	0.98	0.99	0.99	1

Table 2-9. Singularity results calculated by wavelet method in all eight surveyed lines and their related explanations. Static shifts are detectable from other geological features based on singularity calculation results.

Line no.	Stations	Singularity	explanation
A1	8~10	0.1354	Vein
A2	15~16	- 0.1846	Static shift effect
A3	8~12	0.2344	Vein group
	14~17	0.1896	Vein group
	19~20	- 0.2137	Static effect, local power lines
A4	18~24	0.3041	Vein group
	30~32	- 0.1095	Static effect, fault shear zone
	8~10	0.0341	static effect , wide shear zone
A5	10~12	- 0.0543	Static effect, shear zone
	18~26	0.4038	Vein group
	32~34	- 0.1364	Static effect, shear zone
A6	6~8	- 0.1031	shear zone
	18~24	0.2832	Vein group
	30~32	- 0.2043	Static effect
A7	21~24	0.2376	Vein group
	32~34	- 0.2345	Static effect
A8	22~28	0.4273	Vein group
	30~32	- 0.2168	Static effect

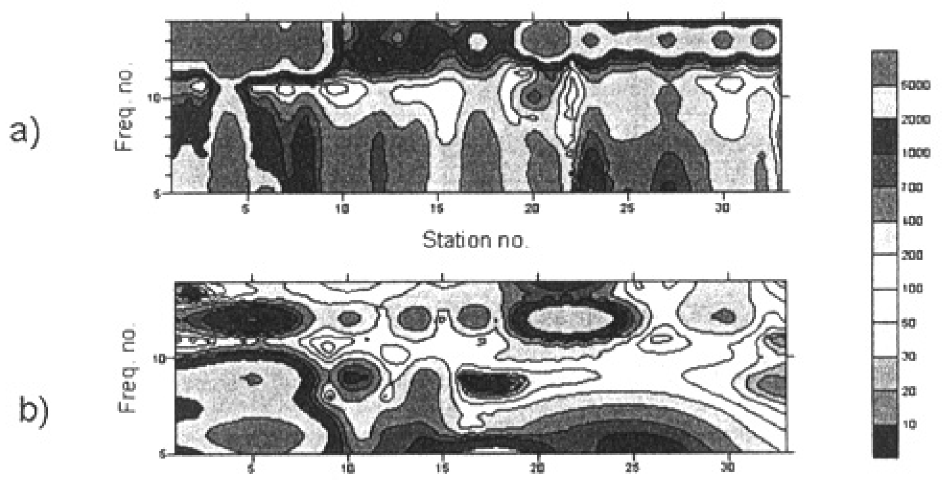


Figure 2-9. (a). Pseudosection of original Cagniard apparent resistivity of line A4 in Woxi Deposit. (b) Pseudosection of all zone near-field corrected Cagniard apparent resistivity of line A4 in Woxi Gold Deposit

In order to estimate the concrete position of static effect, several typical abnormal positions were chosen, and the singularity was calculated using wavelet method (Song S., et al., 1995). The results are listed in Table 2-9. In theory, singularity of deep large abnormal structure or geological objects is positive, while shallow inhomogeneous ones are negative. From Table 2-9, it is shown that there are large low resistivity objects in deep Woxi deposit, with positive singularity. For the anomalies caused by static effect, singularity is negative. Figure 2-9b shows the resistivity pseudosection of line 4 after near-field and static shift corrections using all-zone wavelet method where the scale is 2^3

C. 1-D Interpretation

1-D interpretation is generally sufficient in simple stratified geology, or where the areal extents of lateral resistivity changes are large with respect to the survey grid density. When such conditions are met, use of existing computer modeling routines can give relatively fast inversions. In addition, 1-D interpretation is simplified enough that standard computer-generated type curves can also be used for rough interpretation. This is not the case for 2-D and 3-D environments.

A useful characteristic of a 1-D environment is that, in the far-field zone, apparent resistivity never changes by more than the corresponding change in frequency. Hence, a log-resistivity versus log-frequency plot al ways shows slopes $\leq|\pm$ **45°**|. Larger far-field slopes indicate that the environment is 2-D or 3-D. However, note that even in a 1-D earth, larger apparent resistivity changes and negative phases can occur in the transition zone. This subject is expanded upon in the Case Histories section.

D. 2-D Interpretation

Data interpretation over 2-D structure is much more complex than in the 1-D case because "edge effects" must now be considered. Edge effects are encountered when a lateral resistivity contrast is present between two bodies in the earth. This edge effect results in a discontinuous normal electric field across the boundary between the two bodies; there are two important ramifications of this. First, resistivity soundings with the E-field dipole oriented parallel to the 2-D strike (TE mode) differ from those with the dipole perpendicular to strike (TM mode). Second, when sounding through a conductive body, TM mode resistivities will not recover to their true values beneath the body, as they do in l-D soundings.

The difference between TE and TM mode behavior is illustrated by modeling a conductive slab as done in Wannamaker et al. (1984). Although TE mode produces a smoother edge response, TM mode produces a more exaggerated and thus more diagnostic response. Hence, if the main objective of a survey is to delineate lateral contacts, then the TM mode is preferred.

1 RRI 2-D inversion

The RRI (rapid relaxation inversion), widely recognized as a fast and robust algorithm by the geophysical community, is a forward method that was developed at the university of Washington by Drs. Torquil Smith and John Booker. This method was used to build 2-D models of the study area using CSAMT data. The built models will be discussed later.

E. 3-D Interpretation

Interpretation of 3-D structure is complex and until recently has been unsupported by modeling. Therefore, 2-D modeling was used to approximate 3-D features, as shown in Figures 5-8. However, 3-D modeling would be required to solve problems with finite strike lengths, when operating in the TE mode. In general, the 3-D problem requires that all measurements be regarded as TM mode, with their attendant effects due to discontinuous normal E-field. The resolution of 3-D features is subject to the same criteria mentioned in the discussion of 2-D interpretation. In general, resolution is no worse than the 2-D case but modeling complexity is significantly greater.

F. Comprehensive interpretation

1 Stage A

(i) Sounding pseudosections' interpretation (stage A)

The original apparent resistivity pseudosections of lines A1 to A8 are shown in figure 2-10. In the figure if apparent resistivity pseudosection, with the reducing of the near-field effect (smaller than 7 frequency point), the apparent resistivity is adding up to twice, which is due to the warpion of the near field affection. Thus, the transition zone data should be corrected for near-field effect. All-zone apparent resistivity method was used to do the correction and produced pseudosections are shown in figure 2-11. To analyze and compare, the electric field has been used to calculate the apparent resistivity pseudosection of the all-zone (figure 2-12).Thus, in the apparent resistivity

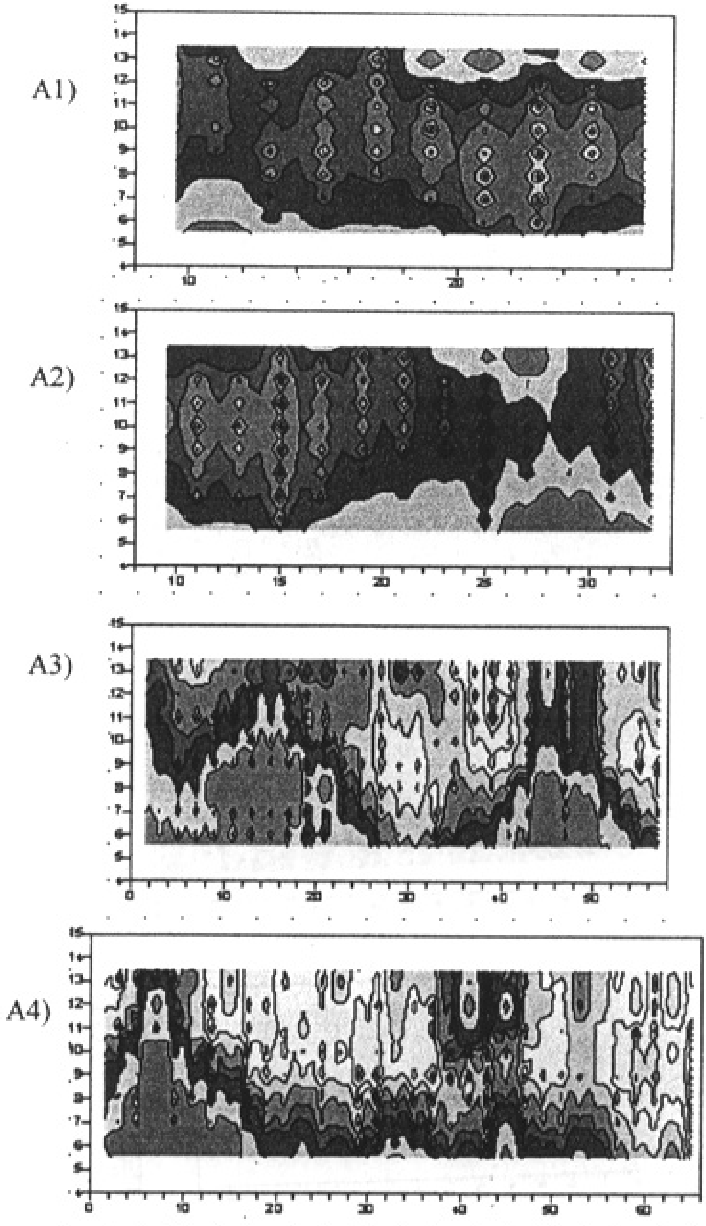


Figure 2-10. The original apparent resistivity pseudosections of lines A1 to A8. X axis is station number and Y axis is frequency number.

A5)
A6)
A7)
A8)

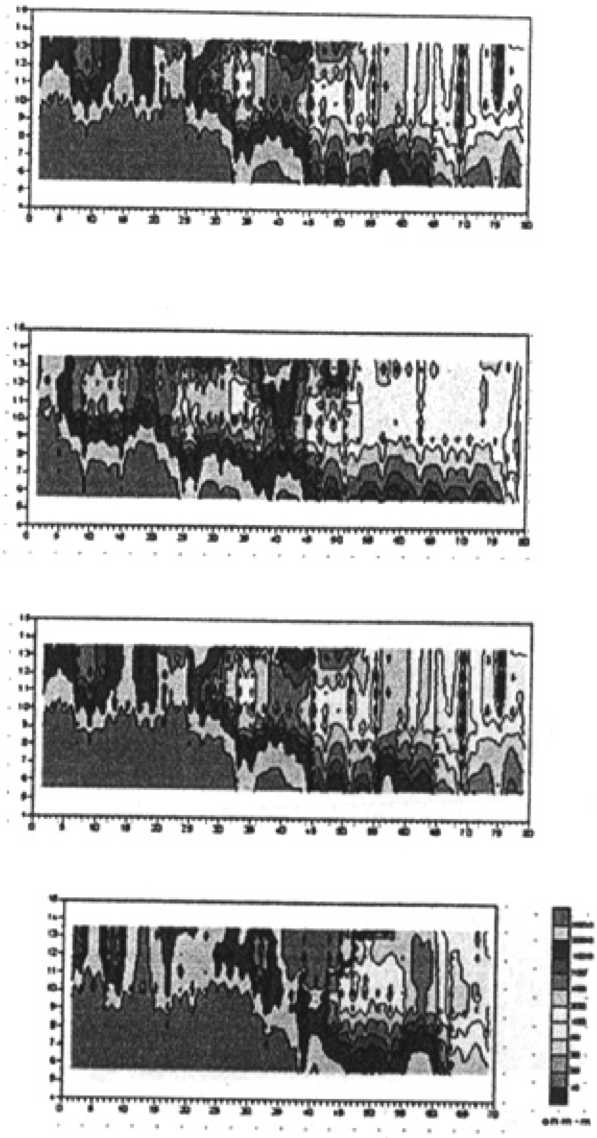


Figure 2-10. to be cont.

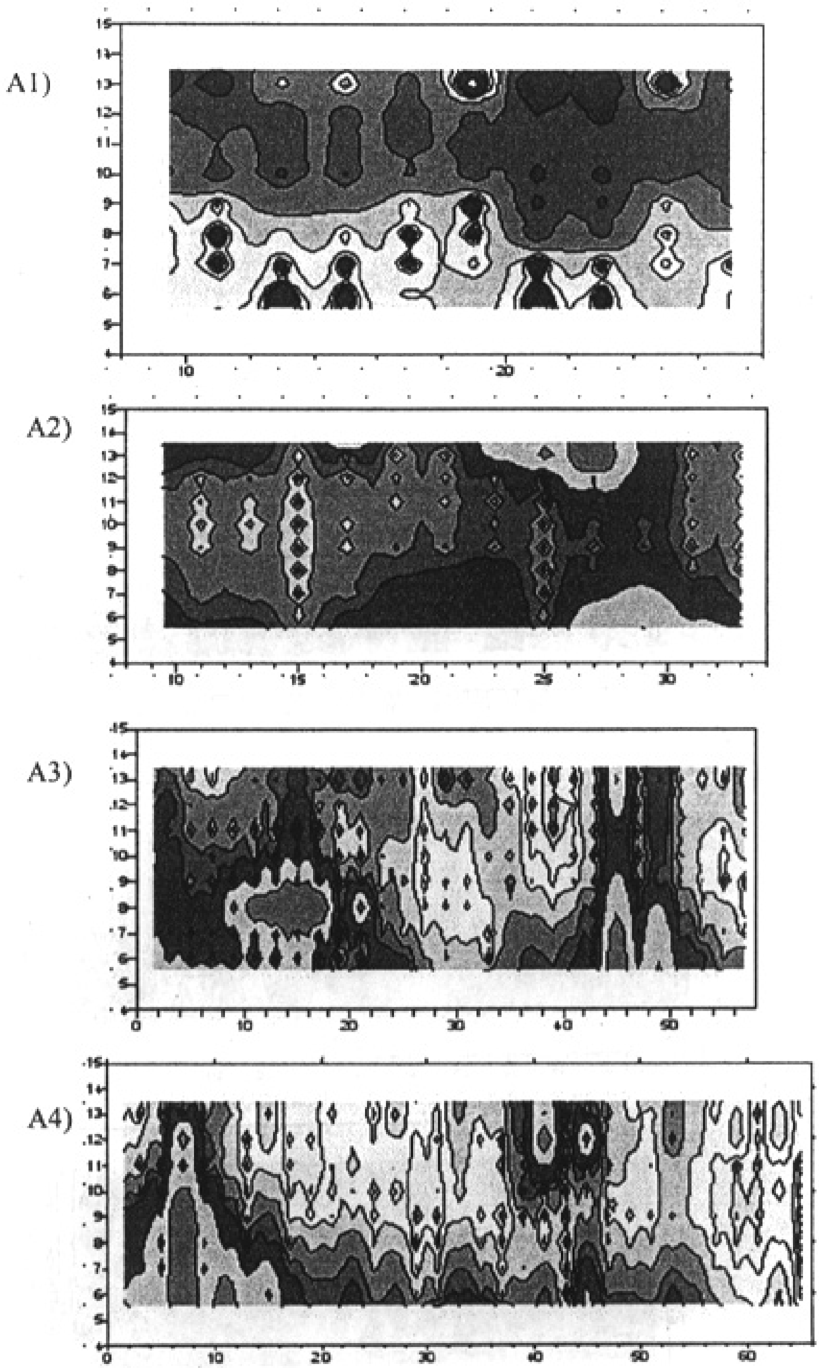


Figure 2-11. Pseudo-section of all zone Caginard apparent resistivity of lines A1-A8.

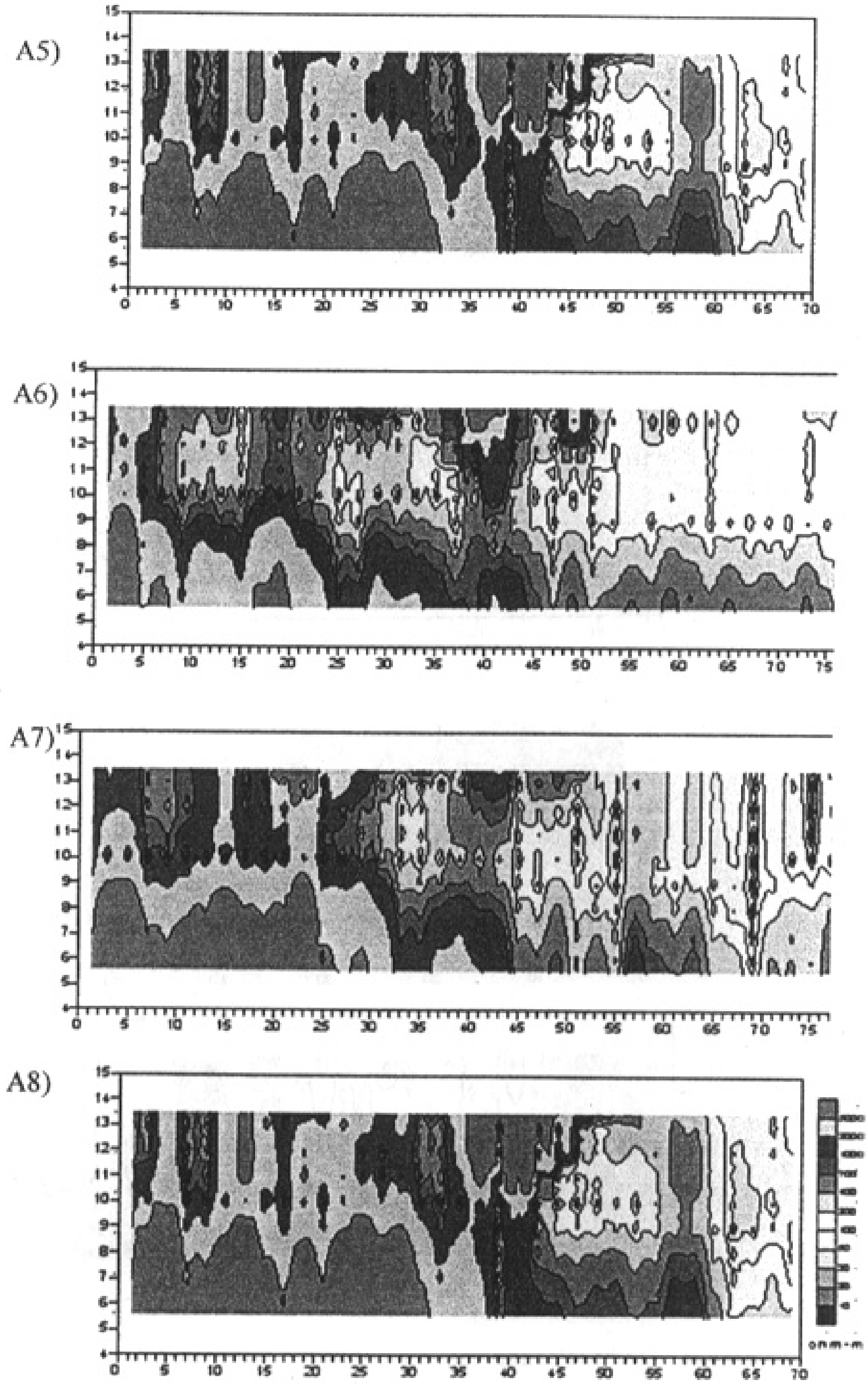


Figure 2-11. to be cont.

A1)

A2)

A3)

A4)

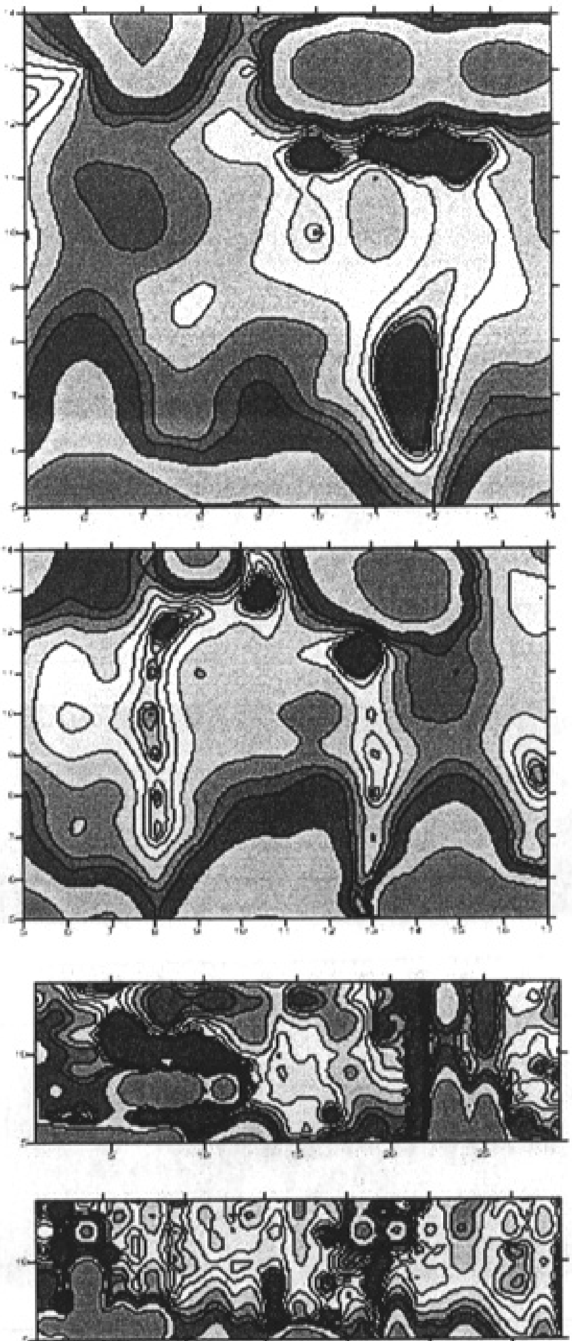


Figure 2-12. Pseudosections of all zone electrical field apparent resistivity of lines A1 to A8.

A5)

A6)

A7)

A8)

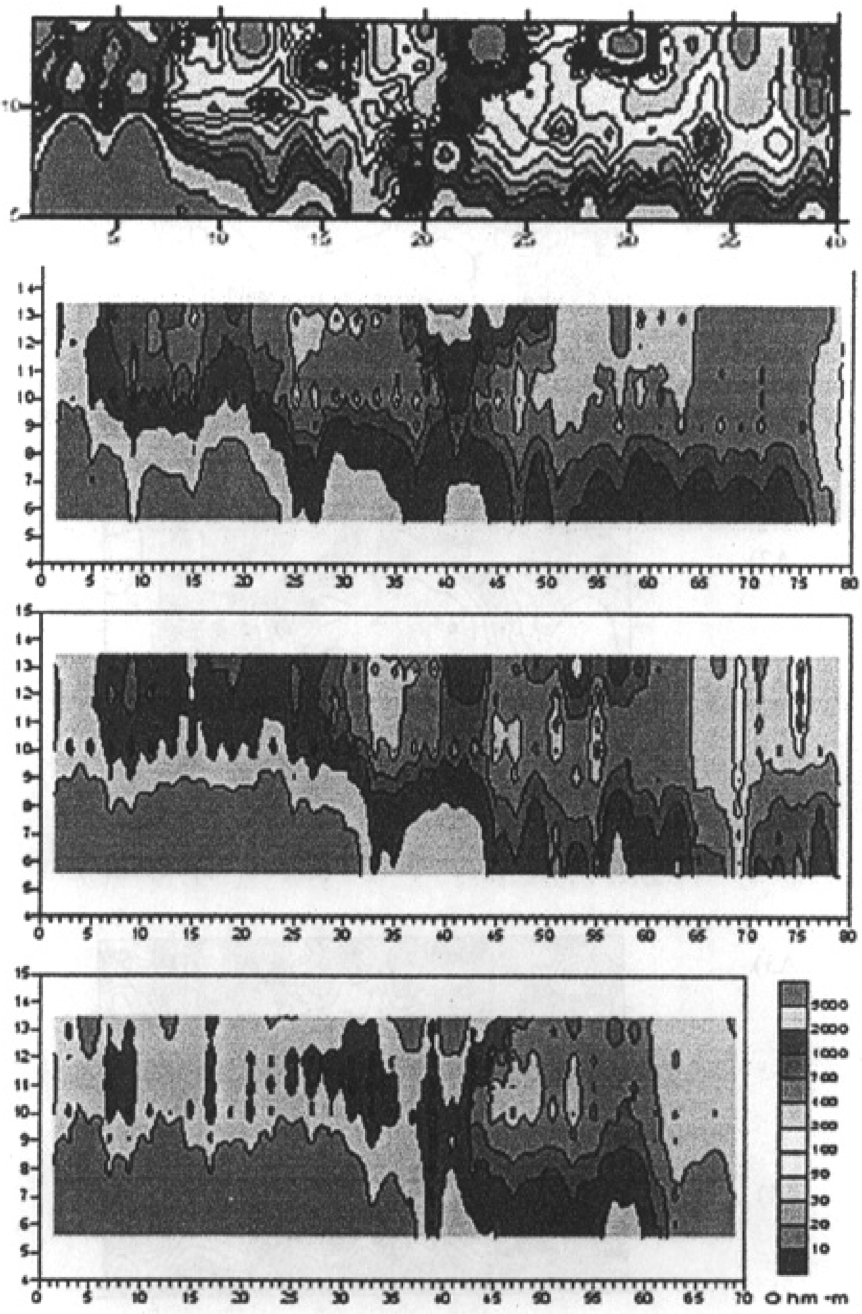


Figure 2-12. to be cont.

pseudosections of the all-zone, the warpion of the low frequency stations has been removed to tell the character of electrical property of the deep layer. By analyzing these figures, we can get:

1. Mineralized zones with the intense wall-rock alteration characterized by low apparent resistivity (about 100• Ω •m) are obvious in the geophysical profiles. On the whole, ore vein group is trending north, dip angle is small at line A4 to A6 it strikes NE but at line A6 to A7 it turns to strike about NEE and the depth of bury turns to be low the range towards line A8 trends to constrain.

2. The layers of the survey area have been clearly mapped by the CSAMT method. The Proterozoic and Banxi group slate rocks have the apparent resistivity ranging from 700~3000 ohm-m which is affected by electrical property of the surface and groundwater. Cretaceous Red strata has apparent resistivity ranging from 200 ohm-m to 500 ohm-m. The Quaternary strata cover big area of low resistivity (lower than 100 ohm-m).

3. Different joints and faults have been clearly showed on the pseudosections. Unclosed narrow low resistivity strips in the pseudosection indicate faults including water. The narrow high resistivity strips opened upward are joints and faults without water. Contacted strips with different rock properties are thick apparent resistivity gradient belts. When there are joints or unconformity interfaces, the apparent resistivity constant value line shows distortion. There are similar characteristics for phase difference. Significant faults are listed in table 2-10

Table 2-10. Distinguished structures in stage A.

Line no.	Station no.	attitude	Surface sign (mark)	Character, quality, proper
A1	6	Incline to north	nothing	Normal fault
A2	14	Incline to north	nothing	Normal fault
A3	3	Incline northward	Woxi Fault	Normal Fault
	12	Incline to north	nothing	Normal fault, fault displacement about 300m
	25	Incline to north	nothing	Normal fault, fault displacement about 100m
A4	5	Incline northward	Woxi Fault	Normal Fault
	10	Incline to north, obliquity more than 40°	nothing	Normal fault, fault displacement about 200m
	23	Incline to north, obliquity more than 40°	nothing	Normal fault, fault displacement about 100m
	15	Incline to north	nothing	Normal fault
	29	Incline to north	nothing	Normal fault, fault displacement about 500m
A5	6	Incline northward	Woxi fault	Normal Fault
	16	Incline northward	nothing	Normal fault
	22	Incline to north	nothing	Normal fault, fault displacement about 100m
	30	Incline to north	nothing	Normal fault, fault displacement about 300m
	37	Steep dip to south	presumed Ganziping faults' extension to east	Normal fault
	11	Incline northward	Woxi Fault	Normal Fault

A6	16	Incline to north	nothing	Normal fault
	22	Incline to north	nothing	Normal fault, fault displacement about 200m
	32	Incline to north	nothing	Normal fault, fault displacement about 200m
	37	Steep dip to south	presumed Ganziping faults' extension to east	Normal fault
A7	12	Incline northward	Woxi Fault	Normal fault
	20	Incline to north	Nothing	Normal fault, fault displacement about 200m
	30	Incline to north	Nothing	Normal fault, fault displacement about 200m
	36	Steep dip to south	presumed Ganziping faults' extension to east	Normal fault
A8	11	Incline to north	Nothing, presumed Woxi faults' extension to east	Normal fault
	16	Incline to north	nothing	Normal fault, fault displacement about 100m
	20	Incline to north	nothing	Normal fault, fault displacement about 200m
	32	Incline to north	nothing	Normal fault

Note that there are some important faults in this table. It shows that upper boundary of the Woxi deep mine is Woxi fault and northern boundary is Ganziping fault. Under Woxi Fault there are many trending north and big trending angle normal faults between layers. Cutting the layers and deep mineralized zone they create complex structure. Mineralization on line A8 probably is in the lower block of the normal fault at point 22 and contacts with Woxi fault where it is weathered.

4. Presence of very strong anomalous zone in the north of Ganziping Fault in A5-A8 profiles indicates an important mineralized zone.

5. On the strip composed of line A4 stations 3-6, line A5 stations 1-7, line A6 stations 0-9, line A7 stations 0-16, and line A8 stations 0-18, there is an extremely high resistivity geological object is impossible in this area, it probably caused by a thick quartz vein nearby (figure 1-4').

(ii) Characteristics of sounding curves

Apparent resistivity sounding curves of all lines are shown in figure 2-13. We can conclude that:

1. On every survey point, the sounding curves lower than frequency eight have the similar character: the curves appareling to each other and elevating slowly. It shows that in deep the electrical property are similar on the bottom base. They are still old layers of Proterozoic Formation. Affected by the transition belt or close area, the apparent resistivity sounding curves are elevating deep electrical property. But in the all-zone the apparent resistivity has removed the anomaly change, which can reflect the character of the deep electrical property.

2.All the sounding curves on high frequency point (above 12 frequency), and the apparent resistivity excluding the inhomogeneity of the resistivity on the surface, are in the same resistivity range, about 300-700 ohm-m. Therefore, in depth about 300 meters, the changing of the electrical property is homogeneous.

3. The most obvious difference of all the sounding curves are in the middle frequency range (frequency 8 to 11). Concretely, the curves of this part can be divided into model H and HKH show the existence of the system of ore veins in the middle depth. Straight-lines indicate

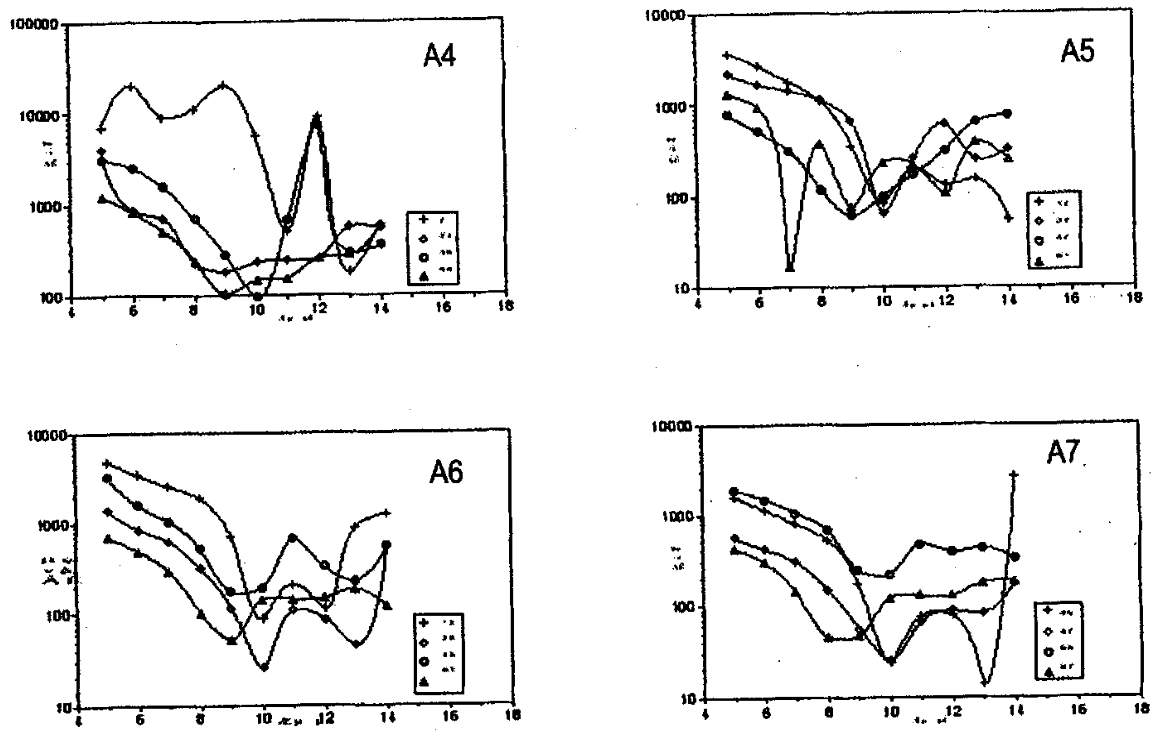


Figure 2-13. CSAMT sounding curves of all zone Cagniard apparent resistivity of lines A1 to A8. Y axis is resistivity (ohm-m) and x axis is station number.

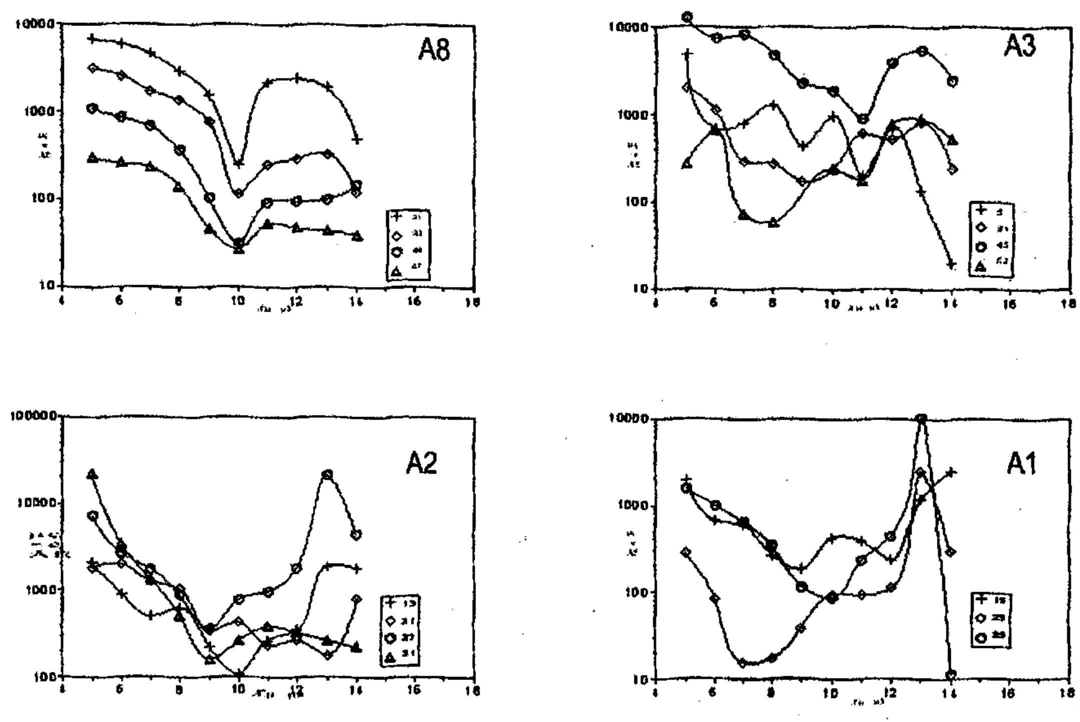


Figure 2-13. to bee cont.

homogeneous electrical property of the strata. Additionally, there are other models of the sounding curves, such as model K and the complex models, which point to local difference of electrical properties.

4. Because of low station density and large investigation depth, position of individual ore veins cannot be specified on H and HKH type curves. They can only show "group ore vein along with their altered wall rocks" and their depth range.

5. On the sounding curves, the static effect is not serious except some local survey stations.

6. The character of line A3 and line A4 to A8 can be compared. But there is a small low resistivity anomaly with the narrow range, comparing to line A4, it belongs to the edge of the mineral area.

7. The electric properties of line A1 and A2 are similar, relatively homogeneous and small anomaly range. Due to the limitation of the length of the survey line, there is no data on north of station 15, so it is difficult to estimate the changing of the underground electrical property. However, the electric properties of stations 4 to 15 and station 4 to 15 of line A4 have something in common, and they have the same anomaly depth. All discussed above shows a good mineralized prospect in the survey area.

(iii) Apparent depth interpretation

The final goal of the geophysical survey is to get an image of the geological features of the surveyed area using electrical properties. To do so, it is necessary to convert the frequency pseudosections into pseudo-depth pseudosections. The so-called Bostick formula

$$H = 356 \sqrt{p/f} \qquad (8)$$

where h is depth, and P and f are apparent resistivity and frequency, respectively is used for conversion. These pseudosections are representation of vertical-sounding plots. They are not "cross-sections" and should not be viewed as such, since the data represent a bulk average of effects from the surface to some depth h. The data must be inverted to yield a true cross-section. Advantage of the apparent depth conversion is rapid calculation. It, in some circumstances, can echo the depth pseudosection. As this formula is deduced under some specific condition (two-layer type D), the results should be corrected for the surveyed area. Nevertheless, there is no unique method for the correction and finally the inverted apparent depth is only show a relative depth range.

Depth inversion can be in one, two and even three dimensions. But this kind of inversions need more time, and the original model should be determined by experience. The basic inversion result can be compromised to be the original inverse model. However, because the ore vein are multi-system in Woxi area, and the CSAMT method can not distinguish every layer, therefore to avoid the misleading of the multi-solution of ore veins positions, Bostick method was used to get the depth results.

Figure 2-14 is the all-zone apparent resistivity-apparent depth inversion pseudosections of line A1 to A8 after static effect correction. They are plotted with station number on the horizontal axis and increasing depth on the vertical axis. These figures are helpful to draft the complicated geological profiles. Analyzing these figures, give same conclusions as next section.

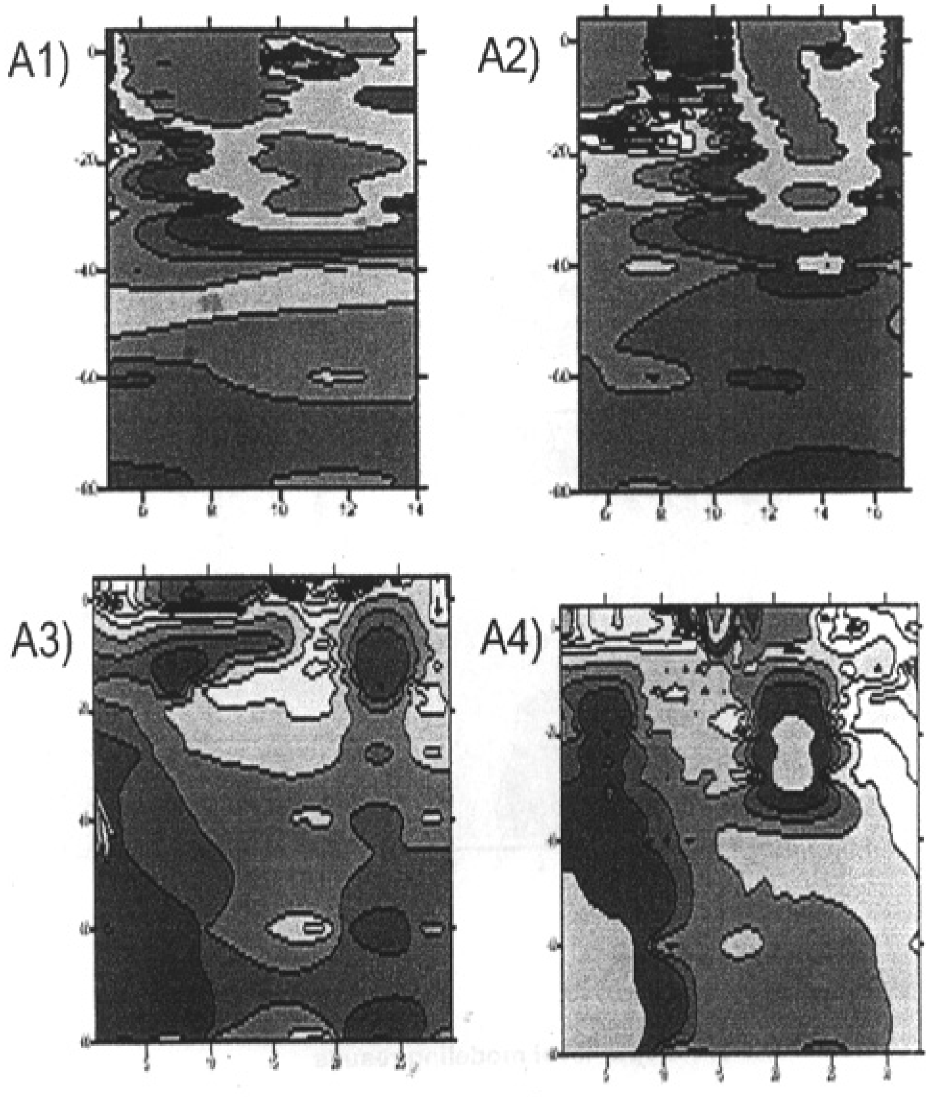


Figure 2-14. Apparent depth Pseudosection of of all zone Cagniard apparent resistivity of lines A1 to A8.

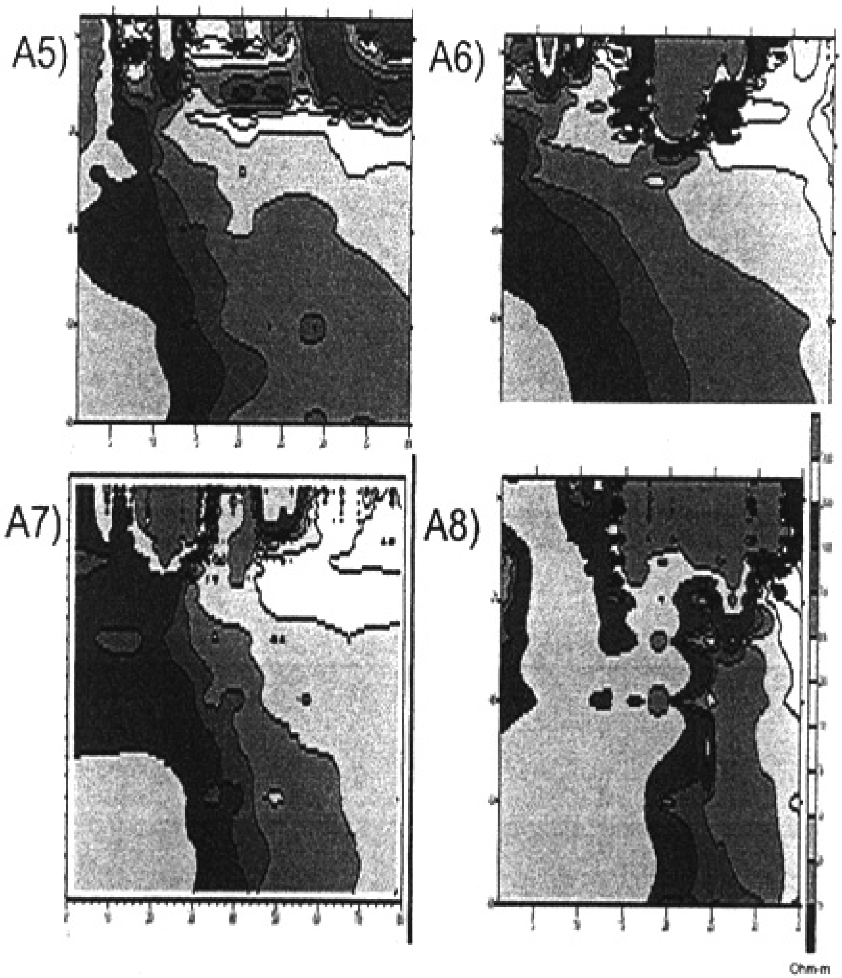


Figure 2-14.. to be cont.

(a) C. Depth-level modeling results

In order to know the electrical property changing characteristics in different depths of lines A4 to A8, all-zone apparent resistivity-apparent depth inversion pseudosections (figure 2-14) were used to produce apparent depth plan views of 200m, 500m, and 1000m (figure 2-15). It is similar to the geological plan view map in different depths. From the figures we can get that:

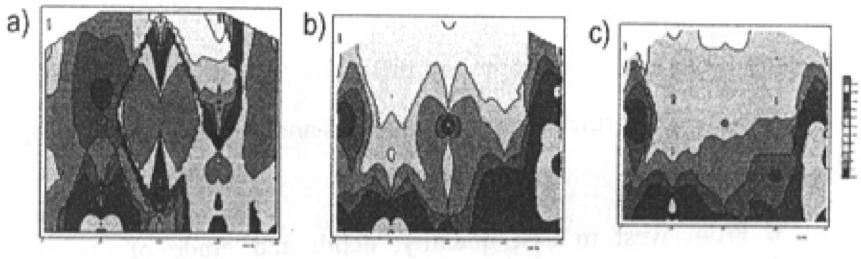


Figure 2-15. 2-d view of all zone Cagniard apparent resistivity of lines A4-A8 in different depths, a (h=-200m), b (h=-500m), and c (h=-700m).

1. In shallow depth (h=200m) the electrical property of the study area can be divided into two parts; the old layers on the south and Red strata on the north. The large Woxi fault runs NEE from line A4 to A8, line A4 station 5, line A5 point 6, line A6 point 10, line A7 point 13 to line A8 point 15. It has the wavy undulate characteristics in this direction. Ganziping Fault trends NWW from line A5 station 35; line A6 stations 33 to line A7 station 32. Besides existence of a fault in the contact of Proterozoic and Cretaceous Red strata, some other faults exist between the Proterozoic strata.

2. When the depth is between 500~700 meters, the resistivity of the whole survey area are almost homogenous. The resistivity is low, and low resistivity anomaly belts are always cut or warped by the constant value line gradient belts. It shows that in depth, the mineral veins were sheared by a series of faults, and formed different enriched mineralized zones.

Mineralized zones in the study area can be divided to three zones: 1.Northern zone that is located in the north of the Ganziping Fault. 2. Middle zone that extends from line A4 stations 15-25, line A5 stations 17-28, line A6 stations 20-29, line A7 stations 22-27 and line A8 stations 23-

26. 3. Southern zone strikes from line A4 stations 10-15, line A5 stations 10-17, line A6 stations 11-16 to line A7 stations 14-17. Depths of the mineralized zones in three zones are different. The middle one is a little deep, and the both northern and southern parts are sheared by the Xiaozuxi Fault.

3. From west to east quantity, depth and grade of ore veins gradually decrease. It is probably caused by faulting, uplifting and weathering. When the depth gets up to 1000 meters, the electrical property of Woxi deposit stabilized and it indicates the Proterozoic strata with resistivity of 1000 to 3000 ohm-m.

2 Stage B

(i) Pseudosections interpretation

(a) a. Original Cagniard apparent resistivity log frequency pseudosections

Figure 2-16 is shows the original apparent resistivity pseudosections of lines B1 to B6. Pseudosection plots show data plotted with station number on the horizontal axis and decreasing frequency on the vertical axis. Pseudosections are representations of vertical-sounding plots. Several scales are available for the vertical scale in pseudosections. As plots with data positioned at evenly spaced log-frequency intervals are the easiest to read, therefore all the plots are plotted in log-frequency format. Following is the interpretation of the plots, reminding that except some stations, the near field effect is weak but we cannot ignore its role and we will discuss produced new plots after near-field correction.

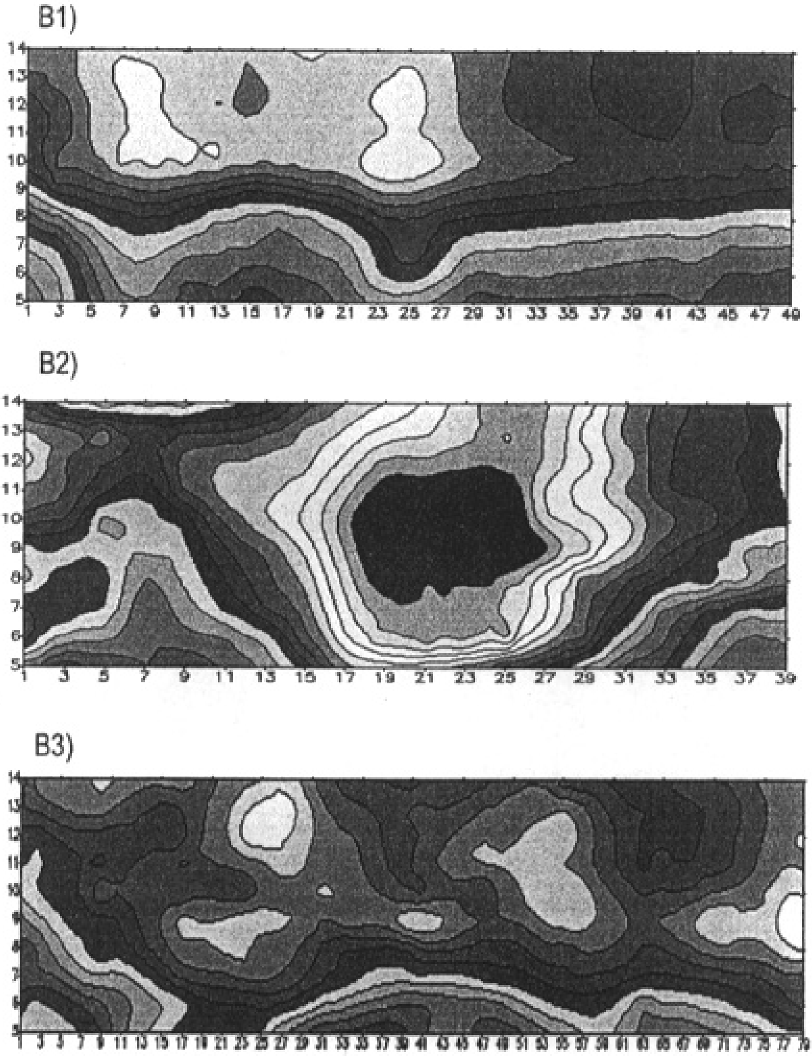


Fig 2-16. CSAMT pseudosection maps of line B1 to B6. X axis is station number and Y axis is frequency number. Resistivity is shown in log scale.

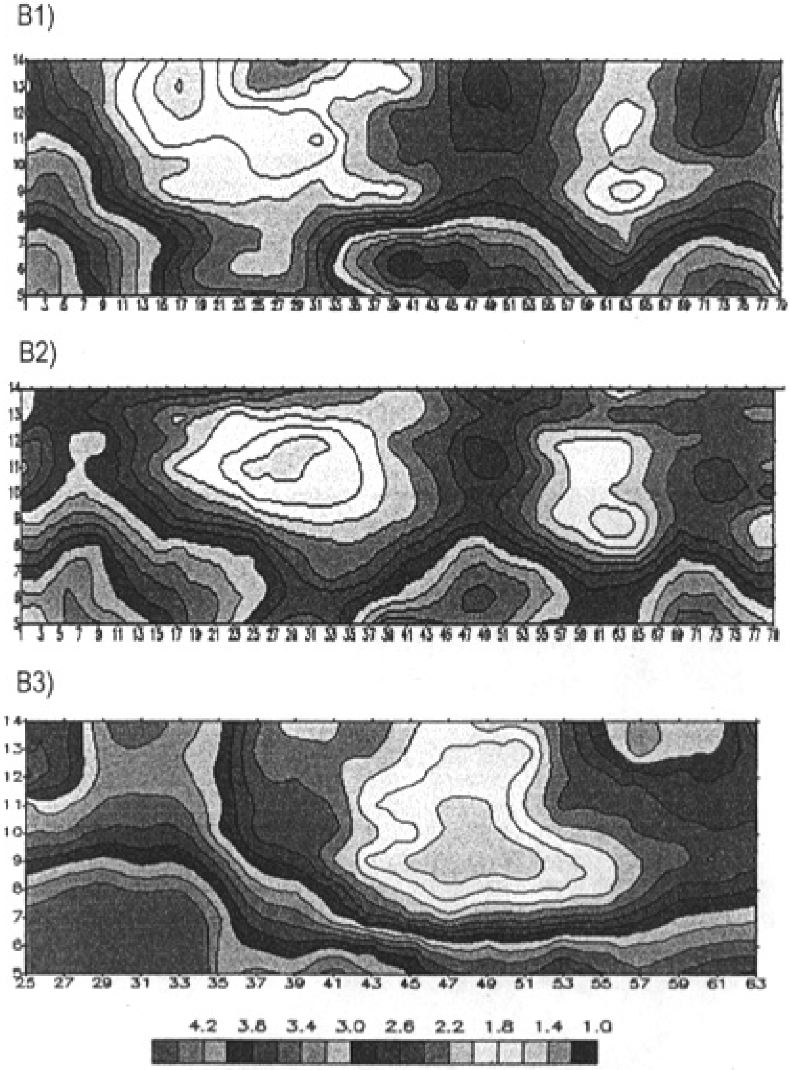


Figure 2-16. to be cont.

1) In general, low apparent resistivity, about 200 ohm-m to 300 ohm-m, indicate strongly altered and brecciate rich mineralized zone. While high resistivity anomalies, to thousands of ohm-m, stands for groundwater containing mineralized zones. Low resistivity anomalous zones runs in the direction of NEE, partly having obviously curving phenomena, the reason of this curve is firstly to be thought as the result of the fractured and faulted strata. Anomaly slopes to NEE with small dip. From lines B1 to B6, low resistivity anomaly commonly changes from shallow to deep, the anomaly width has the gradually contractive trend of facing east.

2) Cagniard apparent resistivity larger than 1000 ohm-m area, fundamentally reflect Proterozoic Banxi Group strata. Electrical properties of Banxi Group strata are comparatively stable and clear. These strata generally run towards north with mild slope.

3) Areas with Cagniard apparent resistivity between 300~600 ohm-m fundamentally reflects the Cretaceous red strata. This strata distribution is concentrated in the western part of the deposit. During section analysis sometimes we face dispersing character of resistivity distribution that possibly cause by underground mining works. Therefore, using geological data was helpful in interpretation. In addition, the widely distributed zone with low resistivity of less than 100 ohm-m in each section reflects the Quaternary strata.

4) Appearance of transverse mutation of Cagniard resistivity appears with static shift effect, is indication of fractured zone existence in the pseudosections. Faults fundamentally cause static effect that can be

seen as low resistivity vertical strips. Most important faulting of the surveyed lines are listed in table 2-11.

Table 2-11. Major faults of the region

Line number	Station	attitude	Ground sign	Geological character	Remarks
B1	5	Sloping North	Nothing	faulting	
B2	2	Sloping North	Woxi Fault	Normal Fault	
B3	6	mildly Sloping North	Woxi Fault	Normal Fault	
	22	Sloping North	Nothing	faulting	
	53	mildly Sloping North	Nothing	faulting	Ganziping Fault
B4	4	Sloping North	Woxi Fault	Normal Fault	
	55	Sloping North	Nothing	faulting	Ganziping Fault
B5	59	Sloping North	Nothing	faulting	Ganziping Fault
	9	mildly Sloping North	Woxi Fault	Normal Fault	
B6	42	Sloping North	Nothing	faulting	
	54	Sloping North	Nothing	faulting	Ganziping Fault

(ii) Depth inversion

(a) a. Plot analysis:

1) Figure 2-17 shows apparent resistivity at frequency point 12 (1024 Hz); it shows depth of about 150 meters. From the plan view resistivity, it can be seen that high resistivity zone, between 1500 and 2000

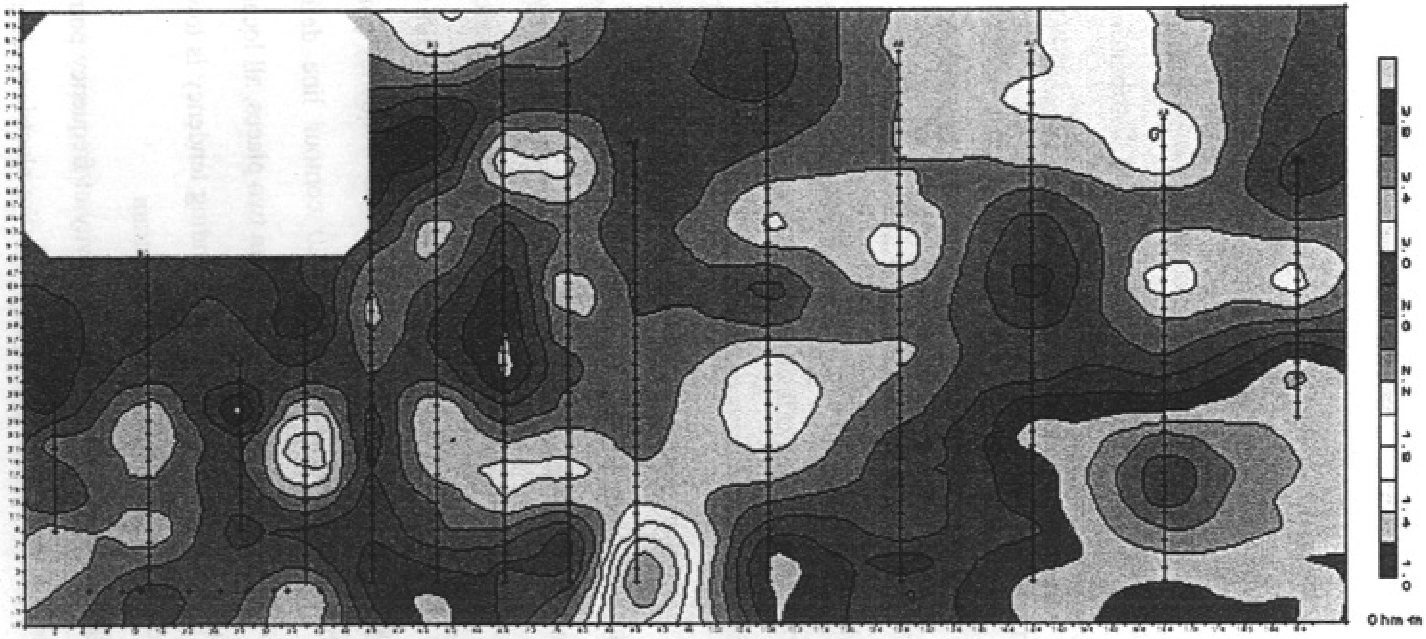


Figure 2-17. Contour map of ρa of CSAMT of frequency no. 12

ohm-m, mainly located at northwest and southwest parts and probably are indication of Proterozoic Banxi group. Low resistivity zones, from tens to 150 ohm-m, in the west, middle and east parts of the plan view, is somewhat due to Quaternary strata but for the most part are probably sourced by Cretaceous red strata. Whole resistivity distribution on the plan view shows a low resistivity line between two high resistivity middle zones with resistivity under 150 ohm-m that runs from southwest towards northeast. Furthermore, the resistivity grades strip between the high and low resistivity zone is predicted as faults.

2) Attached figure 2-18 is plan Cagniard apparent resistivity plot at frequency point 9 (128 Hz), reflecting approximate depth of 500 meters. In the plot, the contour line in the lowest south is the high electrical zone equal to or more than 3.00 which is sign of Banxi group strata that runs from west to east. Unfolded area is gradually bigger, and the distribution indirectly reflects the Woxi layer that its run is NEE. Resistivity contour lines less than 2.20 reflects mineralized zone, the vein fundamentally runs toward NEE. While it expands to NE it obviously shrinks towards south.

3) Figure 2-19 shows Cagniard resistivity at frequency point no. 7 (32 Hz), about 1367 m underground. 2.20 contour line defines the mineralized zone that has been shrinked within two places, all locate in the northern part of the plan view plot. Its expanding tendency is toward the northeast, and the rest part is all Banxi group strata.

4) Figure 2-20 shows Cagniard resistivity at frequency point no. 5 (8 Hz) about 3000 meters underground. From the electricity distribution character, it can be seen that there is no veins below this depth. It just relates to distribution of Banxi Group.

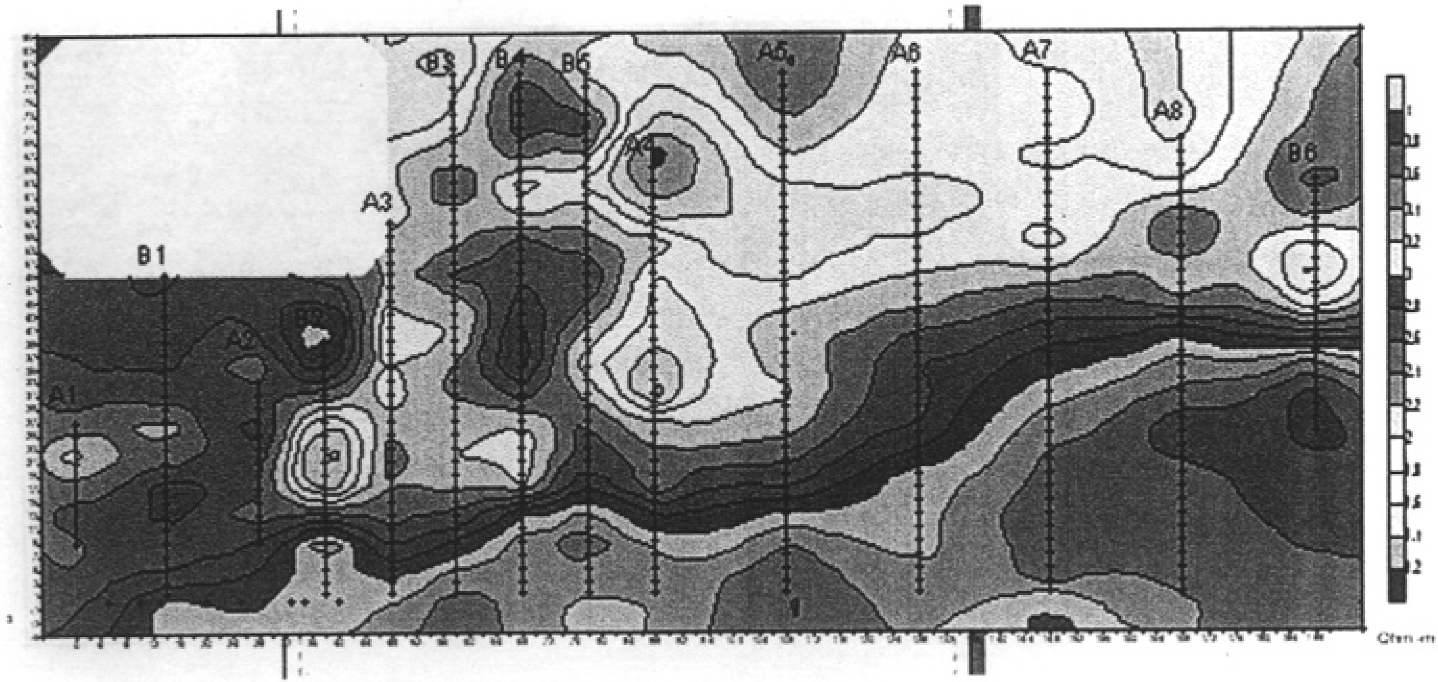


Figure 2-18. Contour map of ρa of CSAMT of frequency no. 9.

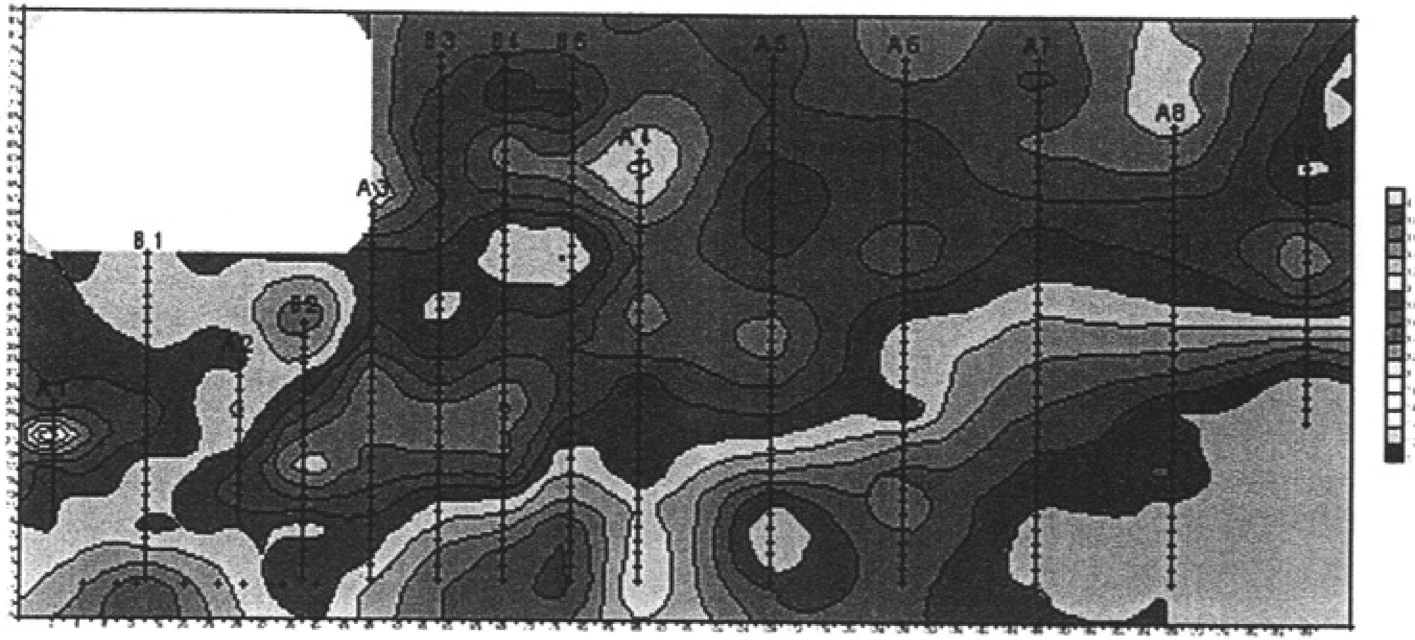


Figure 2-19. Contour map of ρa of CSAMT of frequency no. 7.

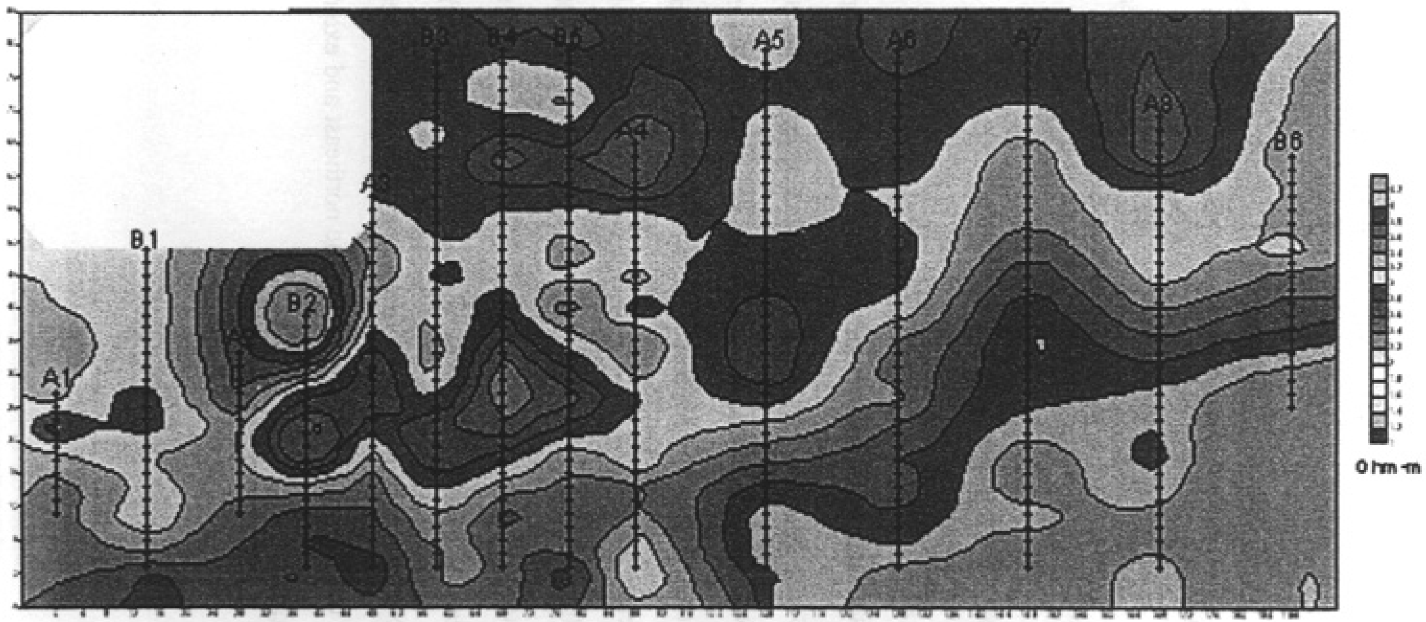


Figure 2-20. Contour map of ρa of CSAMT of frequency no. 5.

From the above analysis, we can make a conclusion as following:

(1) The veins mainly locate between 400 and 1000 meters and there are no obvious underground veins striking towards north or east more than 1000 meters.

(2) The northern part of Ganziping Fault is considered as very important mineralized zone.

(3) There is still good mineralized zone in the eastern part of Woxi deposit runs toward nearly NE.

(b) b. Depth-level modeling results

Figure 2-21 shows results of modeling data which shows resistivities at fixed levels. It shows the relationship between the resistivity and mineralized zone. According to the measured results of some rocks and ores electric parameters in the lab and field (Table 2-1), contour lines 2.4 (equal to 250 ohm-m) are boundary indications of the mineralized zones. Its distribution is shown in the depth-level modeling results, which covers from frequency point 9 (about 500 meters) to 7 (about 1300 meters). Biased by the Woxi fault, the mineralized zone displaced towards the north. Different from the other frequencies, no mineralized zone is seen in frequency point 5. The area covers from west to east and gradually turns to the northeast and extends to north of Ganziping Fault.

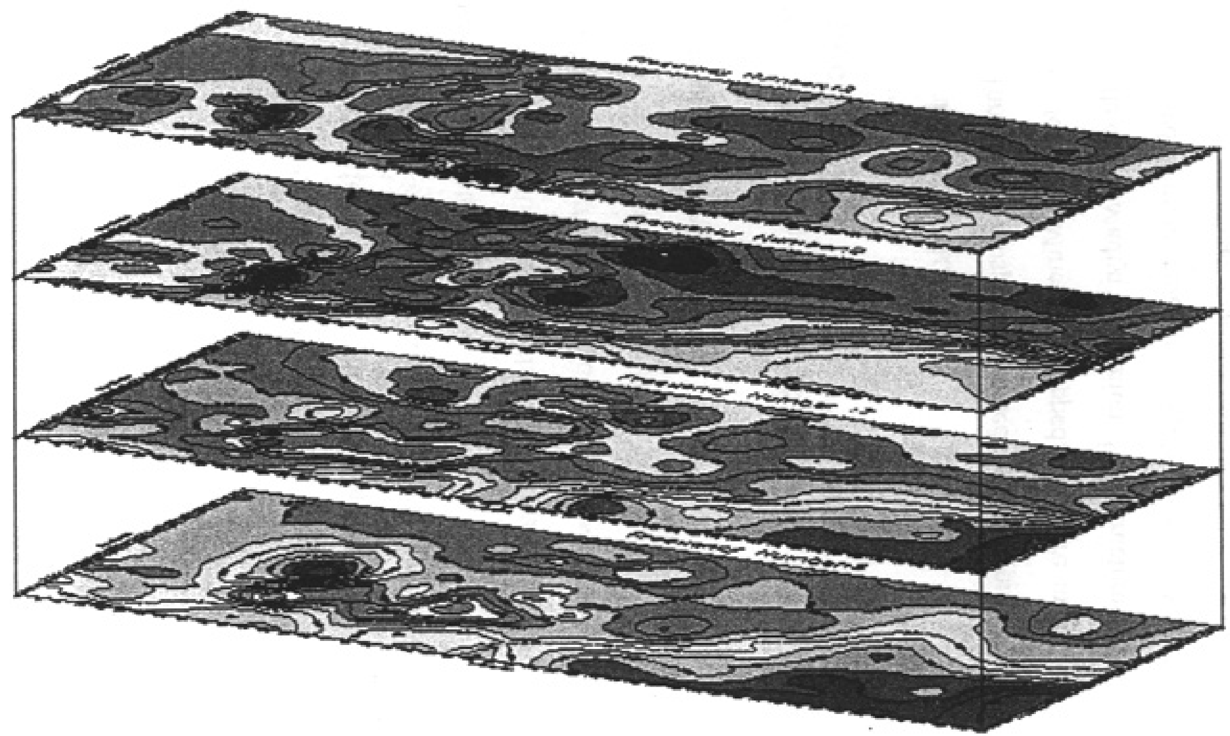


Figure 2-21. CSAMT depth-level modeling ρa results map for frequency numbers 5, 7, 9 and 12.

Chapter 3
Induced Polarization

I. Introduction

The IP method is very sensitive to a small amount of polarizable particles in the medium. However in the case of existing of clay minerals it is a disadvantageous as clay minerals in the shallow alteration zones can be troublesome and misleading.

In the IP method, the depth of investigation is a function of the distance between the transmitter and the receiver dipoles and consequently the lateral resolution has to be decreased if one attempts to investigate the deeper part of geological section.

II. Dual-frequency IP over the Tanghuping Prospect

A. Introduction

The Tanghuping prospect is located in Tanghuping Village, about 6 km southeast to Woxi Mine. Outcropped strata are mainly purplish-red Proterozoic slates of Madiyi Formation. [1,2,] Structures in the area are mostly composed of many faults in different sizes that have had activities for several times. [2, 6] These fault sets are distributed along the east-west and northeast directions. Mineralization of gold mainly is developed in the fractures, and accompanied by alterations such as silicification and pyritization. [2,4,7] As concentrations of 0.5-1.0 percent sulfides can give rise to appreciable IP effects, therefore IP is an effective and reliable method for investigation disseminated sulfide zones. Thus, it was employed to target the pyritized gold-bearing quartz veins in the study area.

B. Electrical properties of the prospect and geophysical model

Table 3-1 shows the testing results of electrical nature of typical ore samples in the Tanghuping prospect. From Table 3-1, it can be seen that apparent resistivity of Proterozoic slates is about 1000 Ω.m, Percent Frequency Effect (PFE) is about 1.0, and the feature of low polarization is shown. Relatively, apparent resistivities of silicified and pyritized slates

Table 3-1 Measured results of electrical nature of typical rock samples of the Tanghuping Prospect in the laboratory and field

Rock type	Number of samples	PFE (%)			Apparent resistivity (Ω.m)		
		Measured in Lab.		Measured in the field	Measured in Lab.		Measured in the field
		Range of variance	mean		range of variance	mean	
grey, purplish-red slates	17	1.0~2.4	1.2	0.8~1.0	397~1857	1047	800~1600
silicified and pyritized slates	8	3.3~7.0	5.4	3.0~5.0	700~2198	1534	1600~2000
structural breccias	9	1.8~5.4	2.6	2.0~3.0	1080~2908	1728	1500~2000

and structural breccias are 1500-2000 Ω.m, and PFE is 3.0-5.0, feature of high resistivity and high polarization is displayed, which are different from the host rocks physically to a certain extent. Gold-bearing quartz veins are thin steep layers with uniform host rock lithology [2,8]. Steep tilting sheets, well-distributed in half space, can be used to simulate the feature of such kind of veins. Figure 3-1 shows a group of computer simulation results. While calculating using integral equation method, let's suppose that thickness of a buried vein is 10m, its hanging wall's depth is 50m, its dip angle 60°, resistivity is 1500 Ω.m, polarizability is 20%, and resistivity of the host rocks is 1000 Ω.m. The results indicate that abnormal apparent

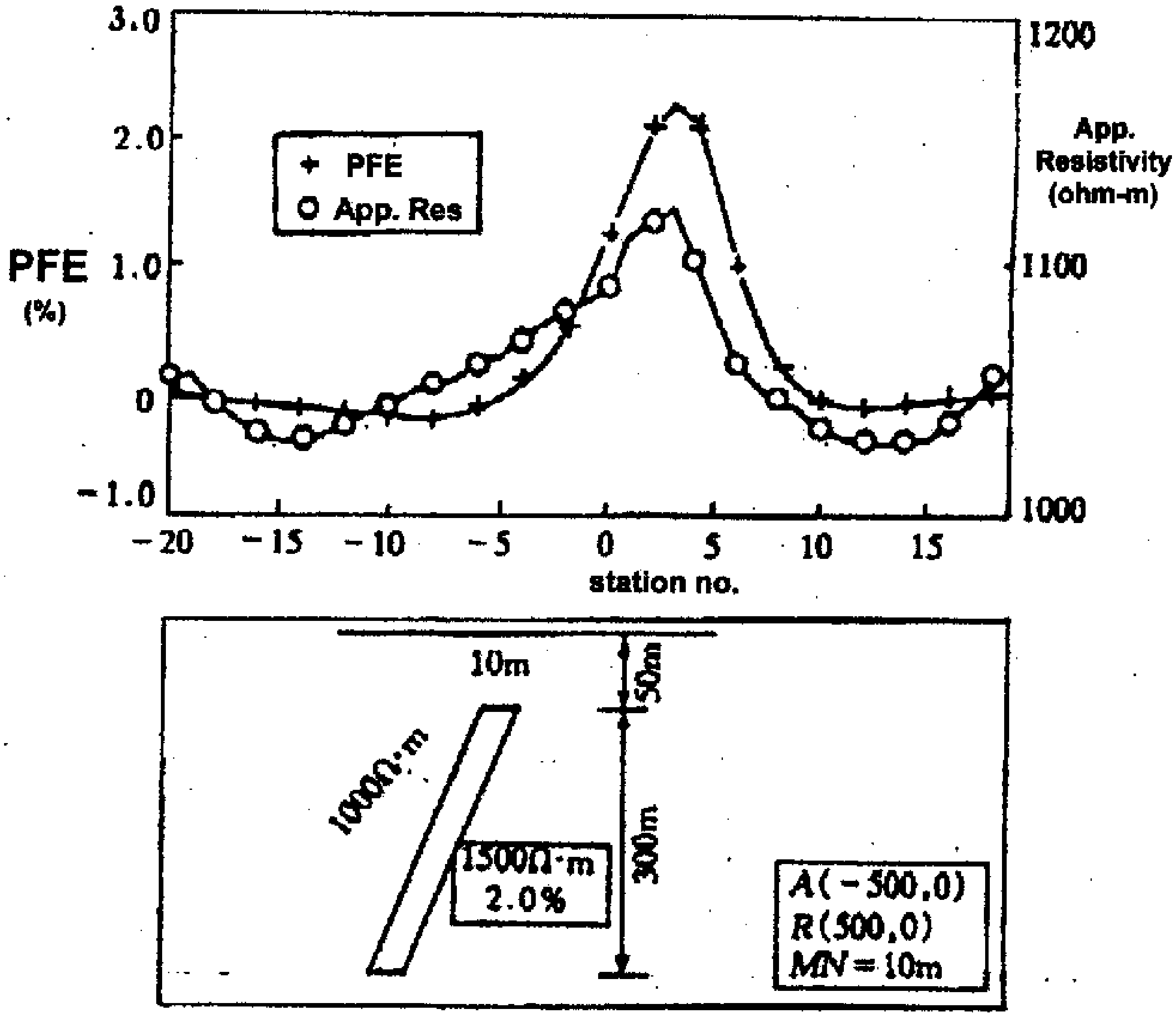


Figure 3-1. Numerical calculation results of the model vein in Tanghuping prospect.

resistivity produced in this type of ore veins is not obvious, about 10%, while the PFE is clear. Therefore, we can use high-precision dual-frequency IP to survey and evaluate silicified and pyritized gold-bearing potential difference due to dual-frequency current is measured synchronously [9]. Hence, three proper electrical arrays, gradient, pole-dipole, and reverse pole-dipole, were made use to study the deposit

C. Surveying, interpretation and geological explanation

1) Gradient array

Six lines designed to dual-frequency IP sounding in the west of the surveyed area. Figure 3-2 is a plane view of PFE. Gradient array was used in work, length of current (AB) and potential (MN) electrodes are 300 and

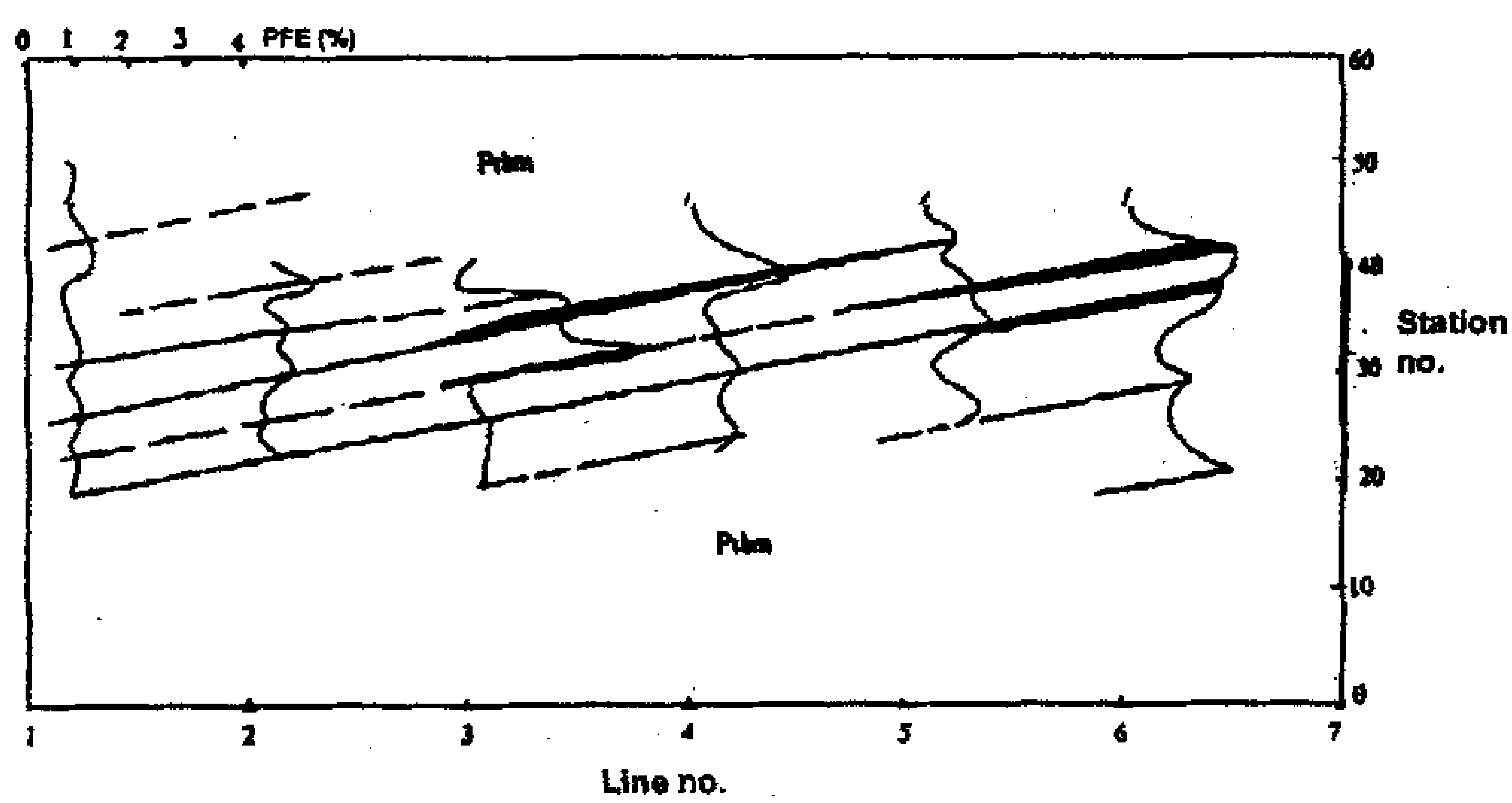


Figure 3-2. Plan view of PFE of lines 1 to 6 and explanatory geologic diagram of Tanghuping Prospect (scale 1:2000). Ptbm= Madiyi Formation

10 meters respectively. Theoretically, anomaly shown by dual-frequency IP is explained as follows: resistivity of gold-bearing quartz veins is weak

while their polarization is high. Characteristics of water-containing shear zones, mountain ridges and shear zones are low resistivity, low polarization. Well pyritized, water saturated, shear zones and veins in mountain ridges show low resistivity and high polarization anomaly. Finally, combining the geologic and topographical map of the area, and by analyzing figure 2, we can infer:

(1) There are nine veins in the prospect. On the whole, sloping to the north, their strike is NEE, and their dip angle varies 60° to 75°. Generally, there is a vein diverging and converging phenomena in the area. After diverging, veins gradually disappear while in converging cases, veins gradually close to each other and form rich mineralized zones at their intersections.

(2) From station 30 to 40 in Line 3, it is a converging zone. Here, the ore vein appears on the ground, where gold grade is about 3-6 ppm. It diverges and extends to Lines 2 and 4, then gradually disappears.

(3) From Station 36 to 44 in Line 6, it is another converging zone of veins 4 and 5. It extends to Line 5 and disappears to west. It extends at least 100 m from line 6 to the east.

(4) Based on above explanations and figure 2 it can be deduced that emphasis should be laid on the east of Tanghuping, not on the zone along lines 3 and 4 as designed.

2) Pole-dipole and reverse pole-dipole array

Following stage one operation, combination of pole-dipole and inverse pole-dipole arrays with two different spacing were used to study the two most prosperous lines 3 and 4. Figures 3-3 and 3-4 show a combination result of both arrays' PFE curves in different spacing.

On the combined section curve of PFE in Line 3, a counter-crossing station of high polarization is displayed near stations 32 and 36. Analyzing the area between the two points, we see the vein slopes towards the north. The result when dipoles spacing is 155m (Fig. 3-4) shows that the veins converge in depth and extends over 200m. The combined section curve of Line 4 indicates that the vein extends little to the depth, not more than 150m, and has no economical values.

Figure 3-5 is the sounding curve of station 36 in Line 3. In a plate-like structure, the sounding curve is not only related to underground electrical nature, but also to the implemented configuration. Results of experiments show that: on an upright plate of high resistivity, when the dipole line is perpendicular to the plates strike, shape of the ρ_s curve is "G", when it is paralleled, there is no change on the ρ_s curve. In regards to the IP sounding on the plate of high resistivity and high polarization, when the dipole is parallel to the strike, the shape of PFE curve is "K". In this case, increasing the plate's slope, anomaly becomes stronger. When the dipole axis is perpendicular to line of strike, the shape of PFE curve is similar to a sphere, resembling a "G". Measuring PFE as the main geophysical parameter in the survey, dipole axis was configured parallel to local structures' line of strike.

Analyzing Figure 3-5, it can be seen that when the AB dipole axis is parallel to the strike, the curve of apparent resistivity is almost a straight line with a little undulation. While among 10 and 60 meters AB/2 spacing, PFE curve is KHK-type in general. Corresponding to two vein groups

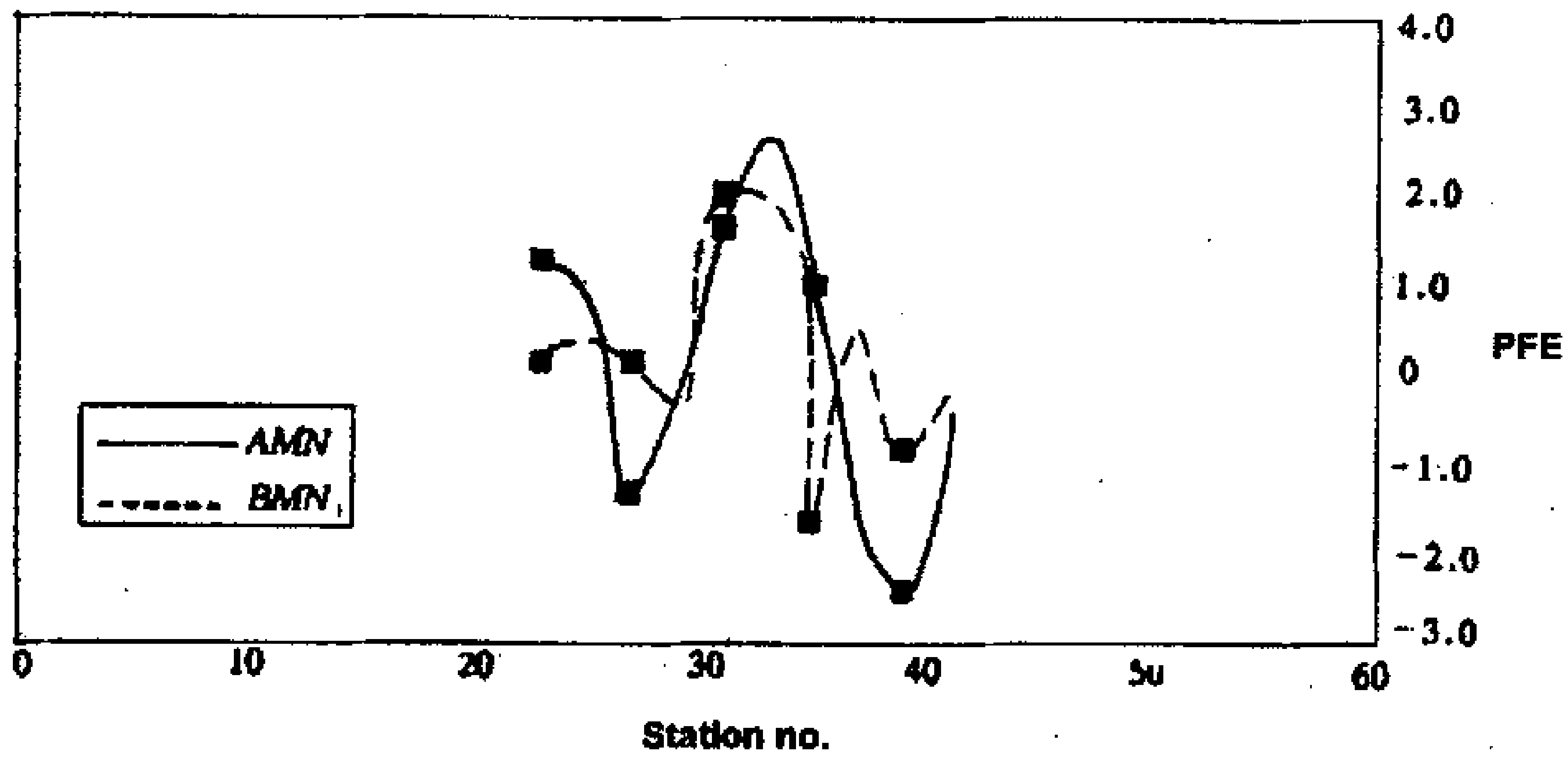


Fig. 3-3 Pole-dipole (AMN) and reverse pole-dipole (BMN) PFE curves with dipole spacing (AO=105 m) in line 3 in Tanghuping Prospect (scale 1:2000)

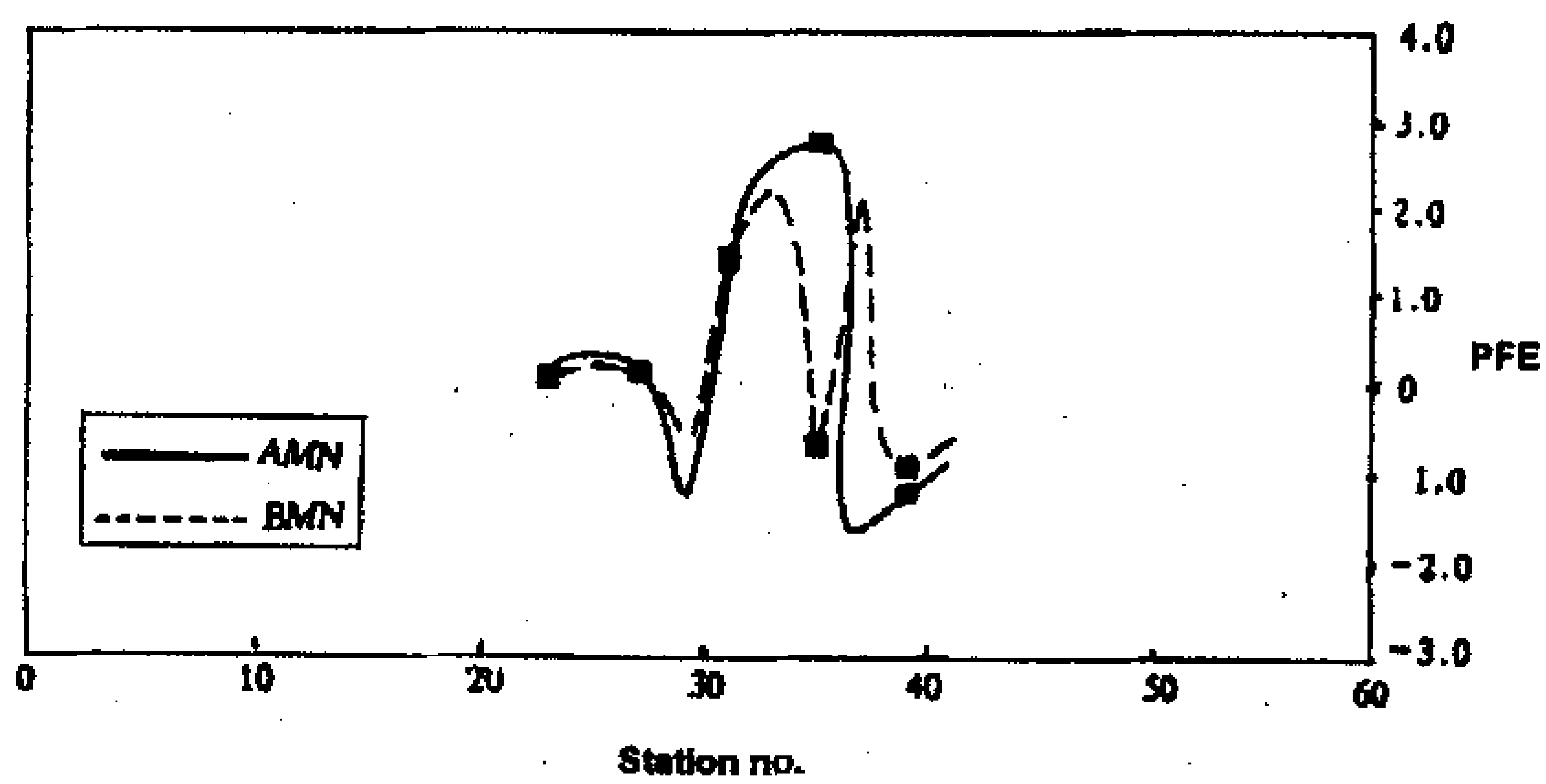


Fig. 3-4: Pole-dipole (AMN) and reverse pole-dipole (BMN) PFE curves with dipole spacing (AO=155 m) in line 3 in Tanghuping Prospect (scale 1:2000)

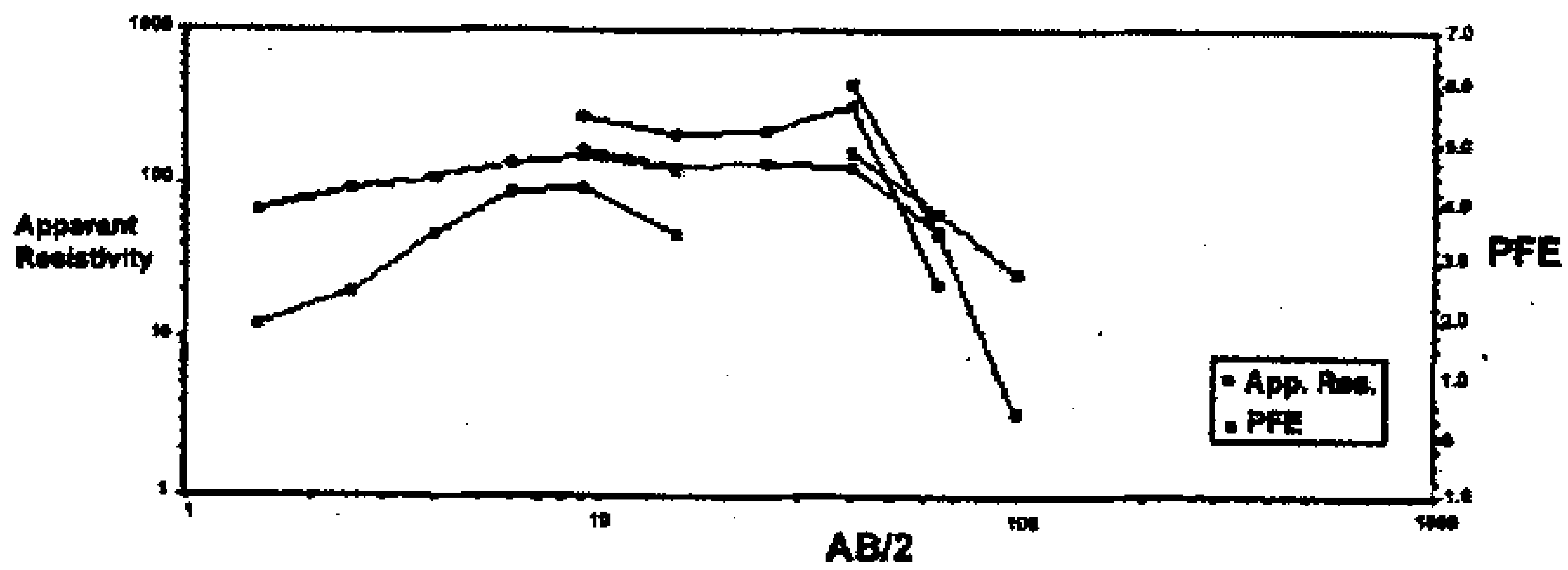


Fig. 3-5: Sounding curves of station 36 of line 3 in the Tanghuping Prospect

which converge at depth, the above is consistent with the result of the combined section. The two veins with 6 and 26 meters thickness are buried at 6 and 35 meters depth respectively at their start point. Eventually sounding results of all the sounding stations were used to draw the explanatory geological sections of the lines. Figure 3-6 is the explanatory geologic section of Line 3.

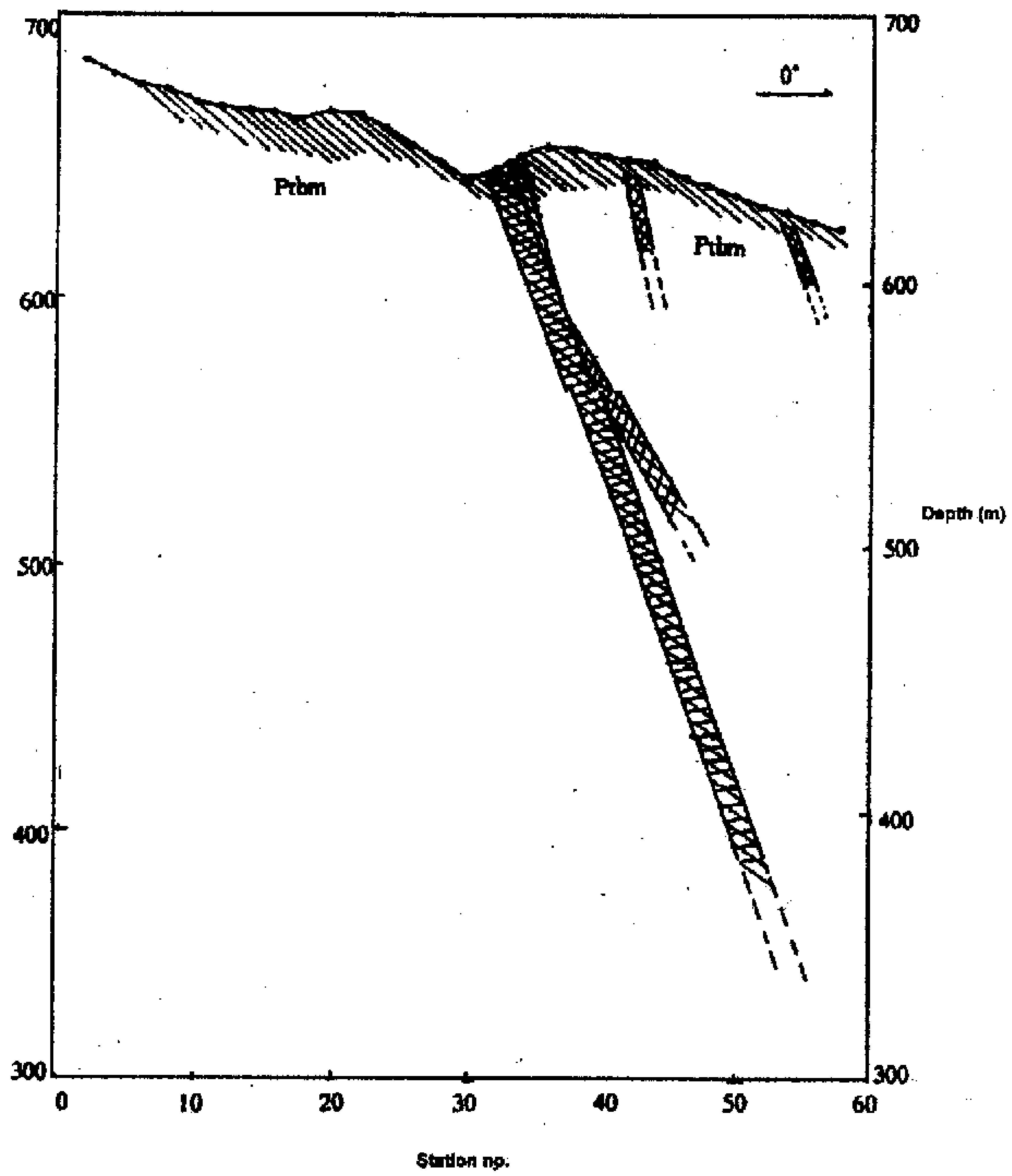


Fig. 3-6: Explanatory north-south geologic cross-section of Line 3 in the Tanghuping Prospect

III. Near-ore Induced polarization profiling

A new IP configuration, near-ore IP array, is introduced and its efficiency is evaluated implementing it over a strata-bound gold deposit. Particular geological conditions, such as locating of gold veins below 700 meters in the study area, makes results of popular and well-known IP configurations non-reliable and ineffective. Therefore field and laboratory studies along with computer simulation and modeling resulted in designing and implementing the near-ore IP array. Comparison and contrasting of field experiments results along three survey lines subjected to intensive drilling and geological studies shows efficiency and reliability of the implemented method.

A. Introduction

Scalar CSAMT and IP surveys were undertaken over Woxi Au-sb-W deposit. Introductory studies for performing IP and CSAMT geoelectrical exploration work at a strata-bound gold deposit in North West of Hunan revealed that because of large depth of ore bodies, ordinary IP methods are not helpful. As many nonferrous deposits in China have the same condition, therefore it was decided to design and introduce a new array to be applied in such situations practically. After doing field and lab,. Works, computer modeling was done and finally "near-ore IP array" was designed. This paper discusses the method, data acquiring and results and compare them with CSAMT results. Field studies, and geological and borehole data was used to check the results. Comparison shows that near-ore IP was successful in its mission considering its precision, reliability and efficiency.

B. Technique description

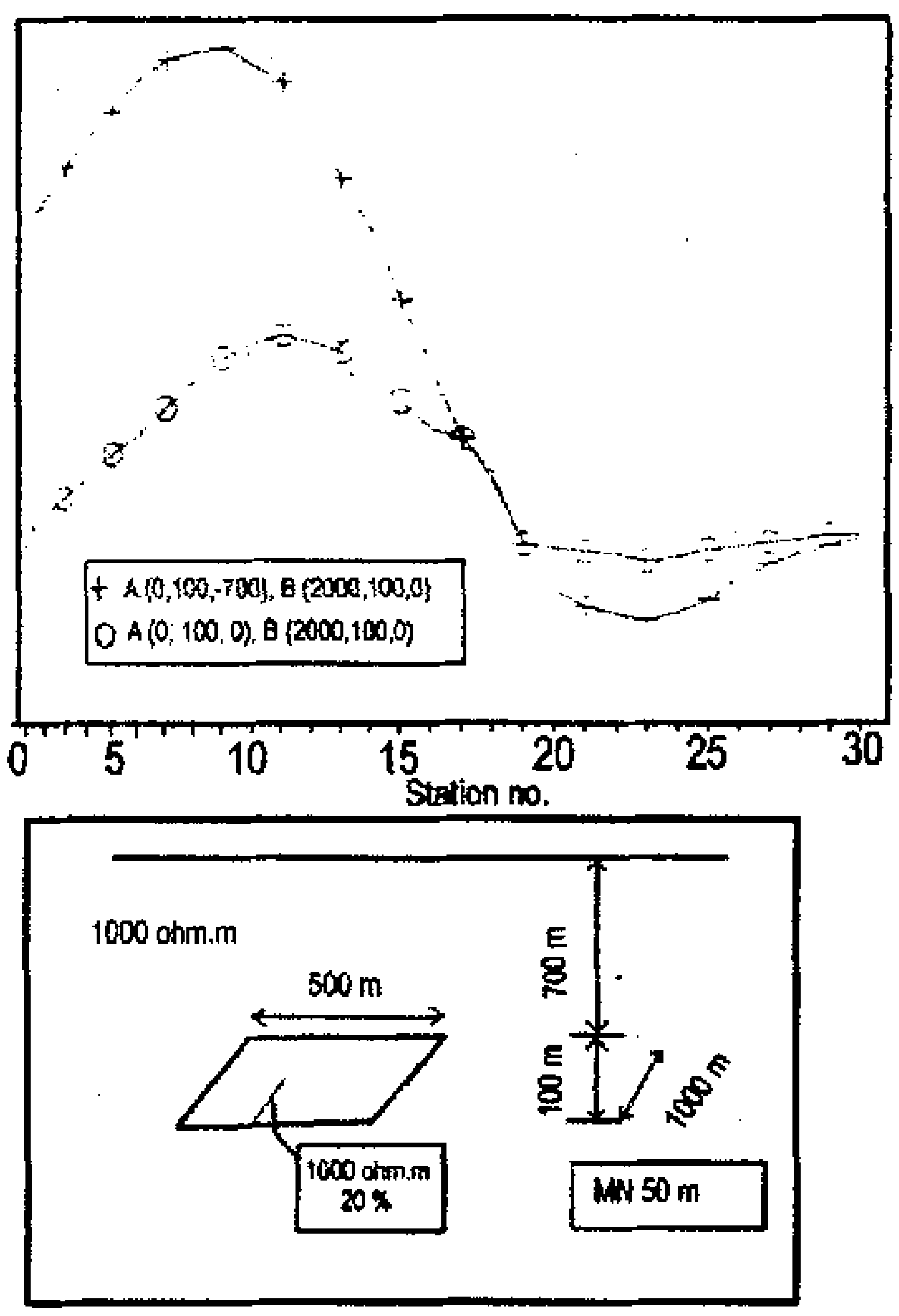
In this technique like pole-dipole, both potential and one current electrode are placed on ground along the surveying line. The other current electrode is placed in the footwall in vicinity of the boundary between ore vein (including the alteration) and its host rock in pits or tunnels. Transmitting the electrical signal to the mineralized body directly increases the effective stimulated induced current, signal to noise ratio, and decreases the effect of the induced electric field. This technique is quite helpful in locating the mineral veins in depth and tracing their distribution and extension in the area.

C. Laboratory studies and modeling

Resistivity and IP phase measurements on the core altered and fresh host rocks samples were undertaken to get relative differences between the background host rock and the veins. The resistivity of the host rocks varies between 400 to 1200 ohm-m and provides enough contrast with surrounding host rocks.

Some computer simulation works including developing a computer program in FORTRAN were carried out for modeling. Figure 3-7 shows the calculation results over a 3-D plate with volume integral method. The plate is of 500*100*1000 m whose resistivity and polarizability are 1000 ohm-m and 20%, respectively. The host rock is homogeneous half space with 1000 ohm-m resistivity and no polarization. The assumed station intervals are 50m. It is clearly shown in the figure that the traditional pole-dipole's IP anomaly is too weak to judge the existence of the plate. But with the near-ore IP method which one current electrode is buried near the plate, the IP anomaly is too weak to judge the existence of the plate. But with the near-ore IP method which one current electrode is buried near the plate, the IP anomaly becomes strong and obvious, the plate can be distinguished and its attitude also can be judged.

Figure 3-8 is an experiment result of near-ore IP method over an inclined copper plate in a 1.5*1.5*1.0 meter water tank. The copper plate is of 15*21*0.2 centimeter and its top buried depth is 10cm. It is clear that result of near-ore IP method is comparable to traditional gradient configuration IP except the break point in the left that is produced when the measure electrodes pass through the current electrode.



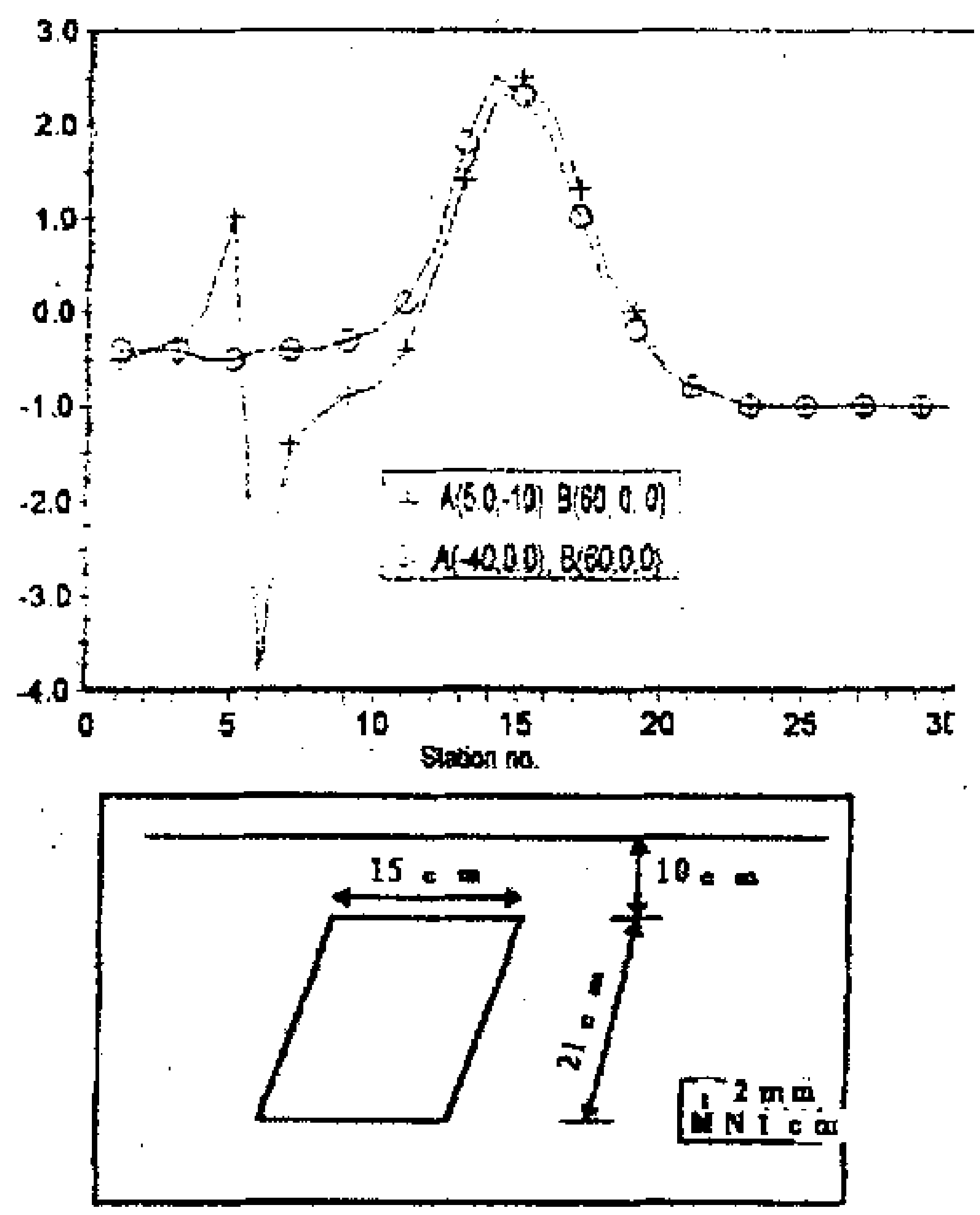


Figure 3-7. Modeling over a 3-D plate Using volume integral Figure 3-8. Near-ore IP experiment over an inclined copper plate.

D. Field Survey and data acquisition

While using CSAMT many geological structures and bodies can produce low resistivity anomalies but in absence of carbonaceous layers in the study area, IP is great help in distinguishing the anomalies caused by mineralization. Therefore, near-ore IP survey carried out in six lines, A4, and B1 to B5. Transmitter was a GGT-6 and data was acquired by a GDP-16 instrument. For example in line A4 while one current electrode was buried on the ground at (4600, 6550), the other one was set about 700 meters underground (4800, 6760) in tunnel no.27.

E. Qualitative interpretation

The results of near-ore IP profiling with a 20 m dipole length is presented in the form of apparent resistivities and phase pseudo-sections.

Among six lines, just three lines, A4, B2, and B4 were chosen to be explained here. Figure 3 shows cross sectional curves of the near-ore IP and 2-D CSAMT over the lines. Regarding 2.2(%) as PFE threshold,, strong anomaly is found at 700-1000 m on line A4 (figure 3a). Its peak is 16% indicating existence of a rich sulfide zone. The curves shape shows sloping veins to the north and its half-peak width shows their depth around 200 m. 2-D CSAMT shows low resistivity almost all along the line but based on IP anomaly, the mineralization is bounded between 700 and 1000 m. Borehole information approves the idea. Borehole 14/CK7 in the west of line A4 at 875 m offset shows rich ore-veins V1~V4 at depth 150-360m that lies in IP and CSAMT anomaly range. Based on boreholes 4/CK4 and 20/CK9, weak IP anomalies at 600 and 1200 m are related to a blind vein and vein V4 with 1-2 m thickness respectively. Low resistivity zones in the

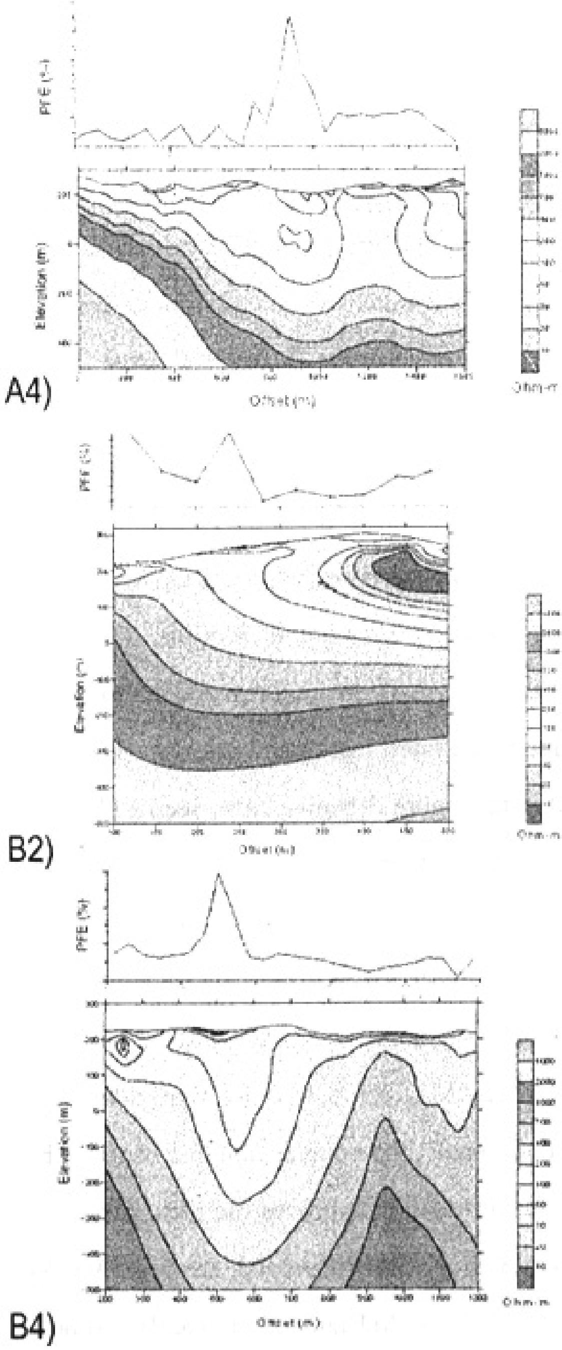


Figure 3-9. Cross sectional curves of the near-ore IP and 2-D CSAMT over the lines.

rest of the line are caused by rice-fields and Woxi and Ganziping Faults in the south and north.

IP anomalies on line B2 (figure 3b) are also well corresponding to CSAMT anomalies. PFE peak at 100to 150 is related to dolomitize, pyritized, and silicified breccia zone in the south of Woxi Fault. This anomaly at 150-250 is related to presence of ore veins in shallow. and after 400 meter's caused by presence of other ore-vein group in depth. Strong IP and CSAMT anomalies on line B4 (Figure 3c) is also caused by the same reason. Anomalous zones shift from south to north and decreasing their depth from B2 to B4 are obvious.

Figure 3-10 shows the IP PFE changes curves of lines B1 to B5. These are the observing results, which are in the abnormal range of the Cagniard resistivity. As the background PFE in the study area is about 2% and more therefore 4% was considered as PFE anomaly threshold. Based on this definition, as it can be seen in the graphs, anomalous zones were detected in the surveyed lines.

So the near deposit induced polarization method can tell the character of the low abnormal, resistivity of CSAMT and determine the enriched positions, which is the reference of the interpretation of the abnormal resistivity of CSAMT in this area.

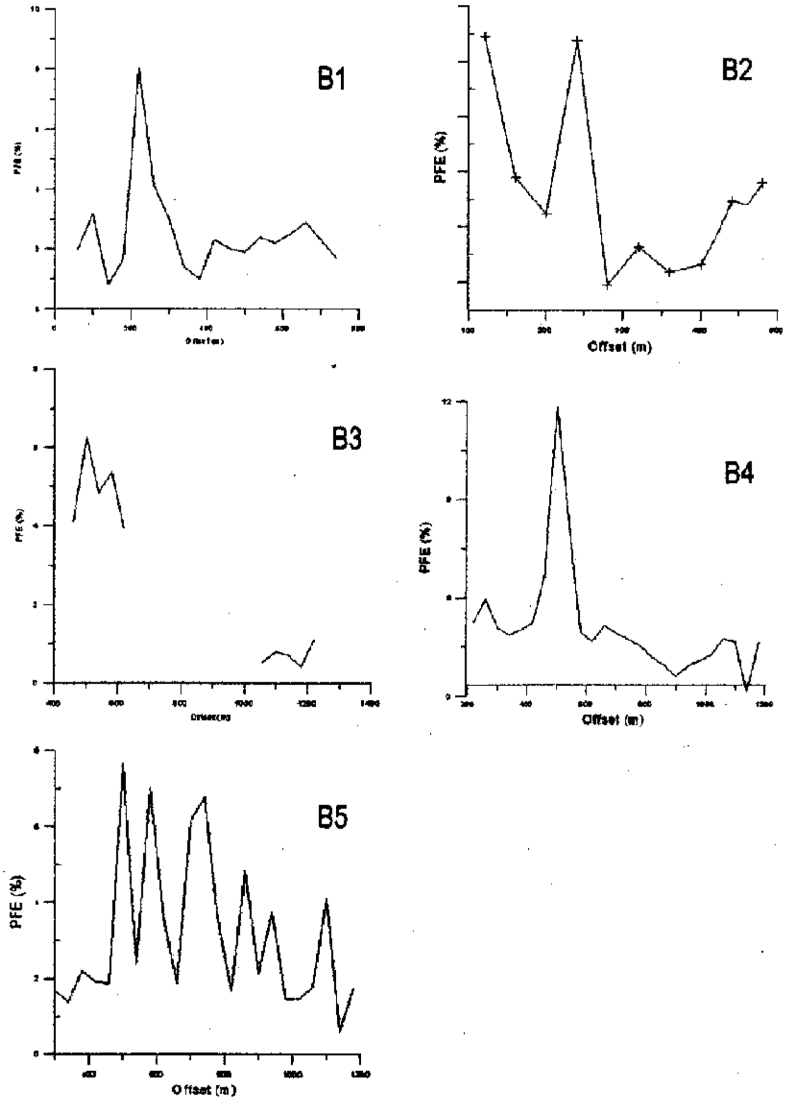


Figure 3-10. IP PFE changes curves of lines B1 to B5 are shown in the plots.

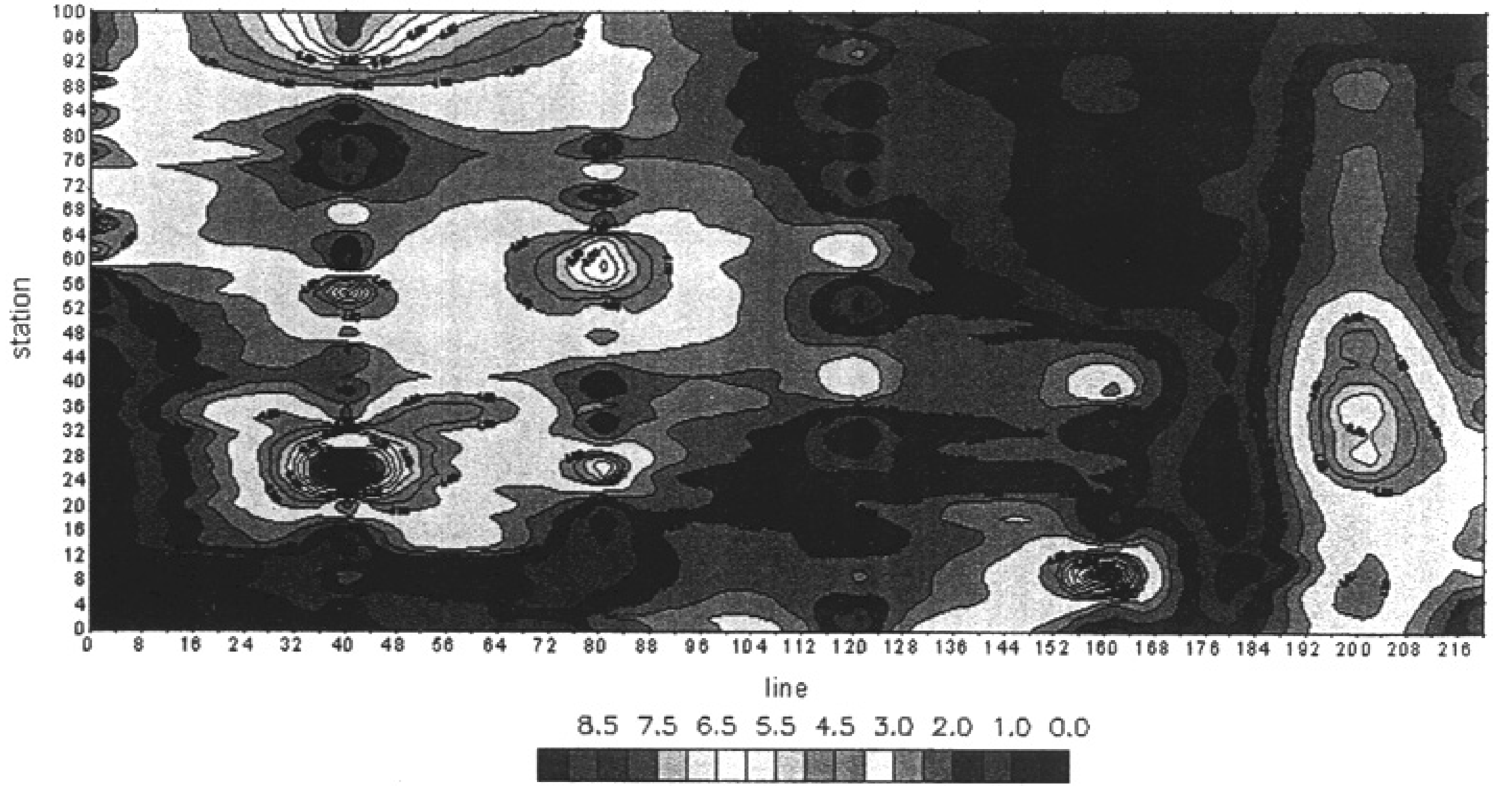


Figure 3-11. Fs contour map of near-ore IP PFE.

IV. Multi-frequency IP sounding

A. Introduction

Multi frequency IP was employed to measure multi-frequency apparent and complex resistivity at the same time. According to the geology and the geophysical condition in research area, in few stations, gradient array frequency-domain Pseudo-random IP sounding was employed to help better analyzing the anomalies recognized by CSAMT. In fact it was used to approve the anomalies found by CSAMT and to find out their dimension and exact location. The main geophysical parameter of identifying mineral anomaly is two parameters PFE and relative phase ($\Delta\Phi$). Table 3-2 is listed working parameters used for the sounding.

B. Accomplished Workload

Geophysical Work mainly includes two stages: first stage is fieldwork and data acquisition and the second stage is data processing, computer simulation and modeling, interpretation and finally deciphering. Tables 2-6 and 2-7 show geophysical exploration workload arrangements of western Woxi and central and eastern Woxi respectively. Data of 508 stations was acquired using Multi-frequency IP instrument. Measurements of 100 stations (about 20% of total stations) were repeated in order to reduce the error and guarantee the high quality of acquired data.

C. Explanation of the *Ps*, PFE and $\Delta\Phi$'s sounding curves

1) Line A5, station 36

Figure 3-13 shows multi-frequency sounding curve of *Ps*, PFE and

Table 3-2 is listed working parameters used for the sounding.

half the current dipole spacing (1/2 AB)	minimum: 40 m maximum: 1500 m	half the current dipole spacing (1/2 AB)
half the potential dipole spacing (1/2 MN)	minimum: 3 m maximum: 20 m	half the potential dipole spacing (1/2 MN)
Detection depth	900-1000m	Detection depth
output	PFE, $\delta\varphi$, and resistivity for each individual frequncy	output
Maximum transmitted electric current (depends on configuration)	1.5 A	Maximum transmitted electric current (depends on configuration)
acquired cycles	4 in normal condition and 8 in noisy areas to insure the data quality	acquired cycles
Measured frequencies	Low=0.125 hz, medium=0.5hz, high=2 hz	Measured frequencies

phase of line A5, station 36. Resistivity plot shows great change in lower region, and trends to equality when it goes deeper and deeper, it conferred that shallow strata could be Cretaceous- strata, and deeper layers could be Banxi strata reflection.

2) Line B5 station 82

Figure 3-14 shows multifrequency sounding curve of Ps, PFE and phase of line B5, station 82. Because there also have the anomaly of PFE and $\Delta\Phi$ when semi-dipole distance is above 500 m, so it can confer that there is possibility for mineralized layers hiding in the deep depth of station 82. It shows that the target may be buried at 500 m underground. Considering that the exploring spot is located to the north of Ganziping

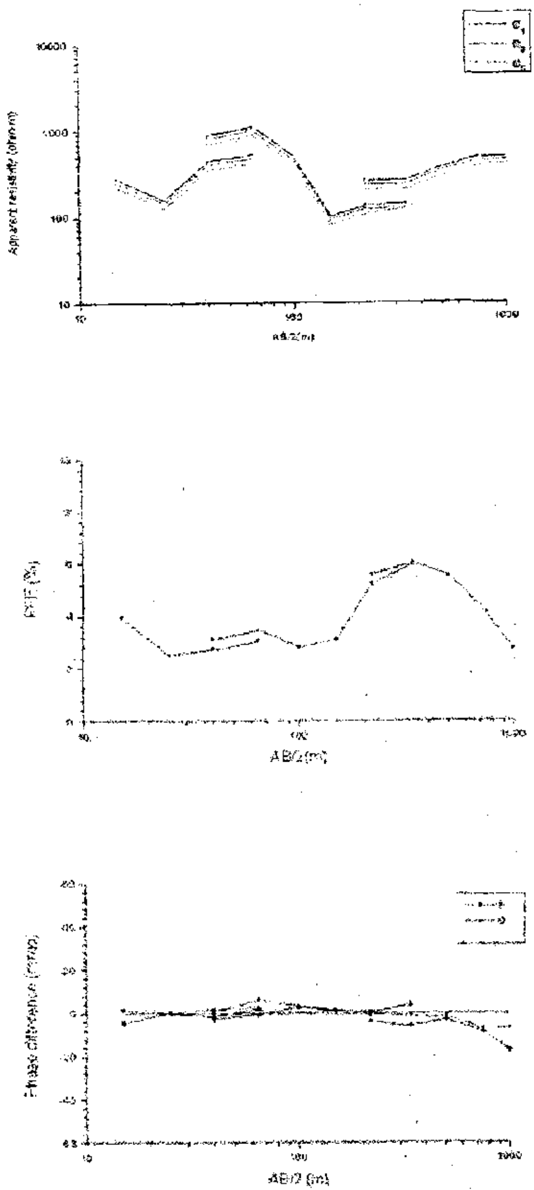


Figure 3-12. pseudo-random 3-frequency IP sounding resistivity, PFE and phase difference results at station 36 of line A5.

fault, no.1 Cagniard CSAMT abnormal resistivity- low resistivity zone, so no. 1 anomaly zone needs further studies.

3) Line B6 station 48

Figure 3-15 shows multifrequency sounding curve of *Ps*, PFE and phase of line B6, station 48. Resistivity curve is a KHAK type, earth-electro position will break when ½ AB dipole distance between 100 ~150 m (low p), deducing that there is trace of passed by faulting, below the fault there's mainly the reflection of Banxi Group strata.

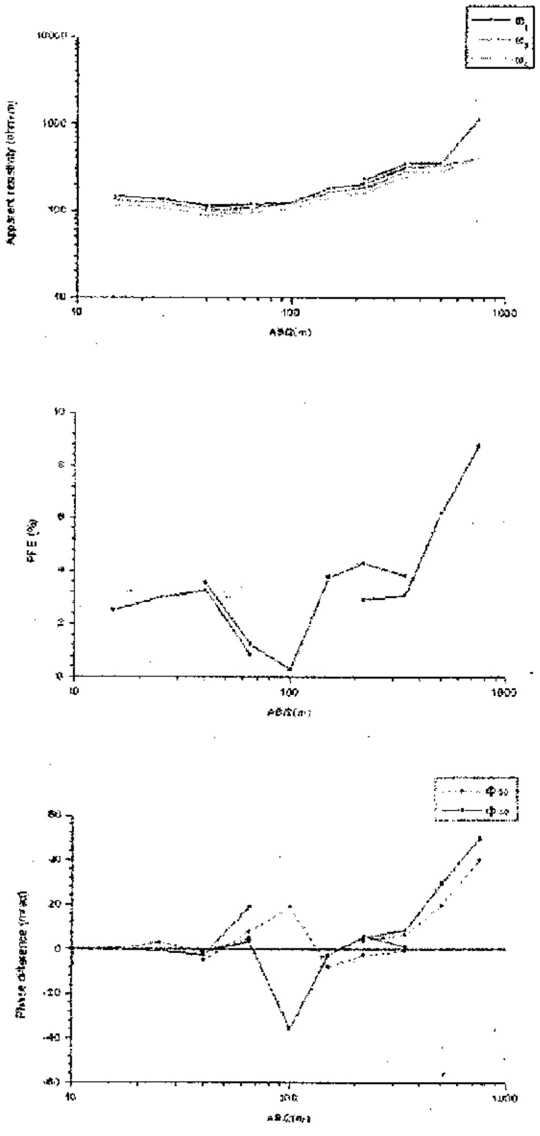


Figure 3-13. pseudo-random 3-frequency IP sounding resistivity, PFE and phase difference results at station 82 of line B5

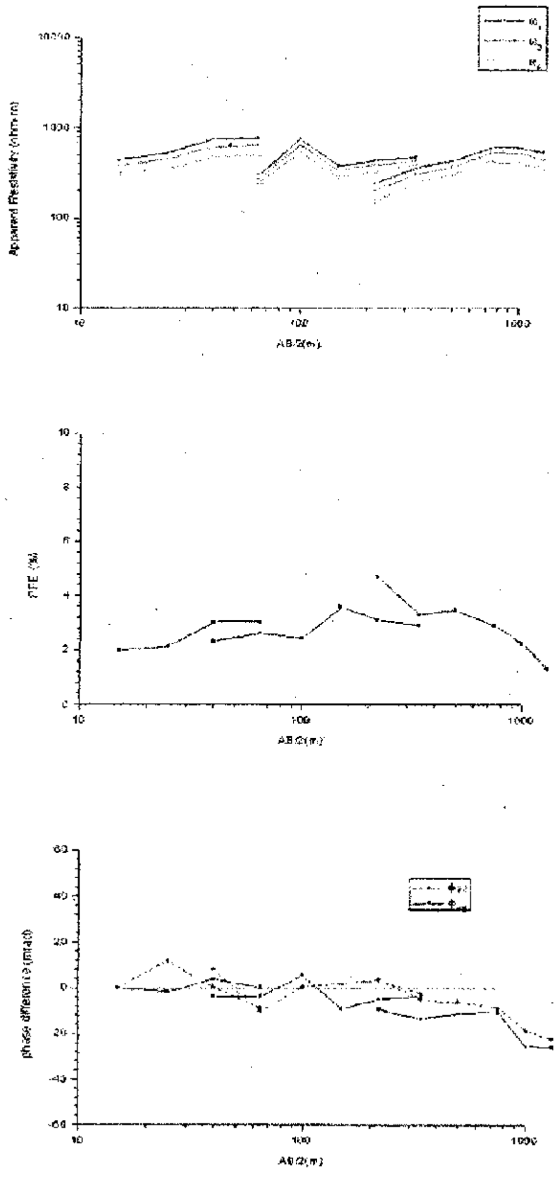


Figure 3-14. pseudo-random 3-frequency IP sounding resistivity, PFE and phase difference results at station 48 of line B6.

Chapter 4

Magnetotelluric (MT)

I. Introduction

MT method is a way of determining the electrical conductivity distribution of the subsurface from measurements of natural electric and magnetic fields on the surface. The MT method depends on the penetration of electromagnetic energy into the earth. Depth control comes as a natural consequence of the greater penetration of the lower frequencies. MT measures electric and magnetic fields at the surface of the earth, across a wide range of frequencies, to provide information about the resistivity vs. depth structure. depth of investigation is approximately proportional to square root of period.

II. Field Survey

MT data were collected at 11 stations using the V5 System (see figure 4-1) for frequencies ranging between 320 and 0.0025 Hz sounding was carried out in 11 stations on a NW-SE line to study the deep structures of the regions.

III. Data interpretation

We adopt planar dimensional continuous inversion method to do the data inversion interpretation. Its principle is: for planar dimensional continuous electric media model, we divide M horizontal layers along vertical Z axis, and in each layer the electric property is tiny variable along vertical direction, and is continuous variable along horizontal direction. If

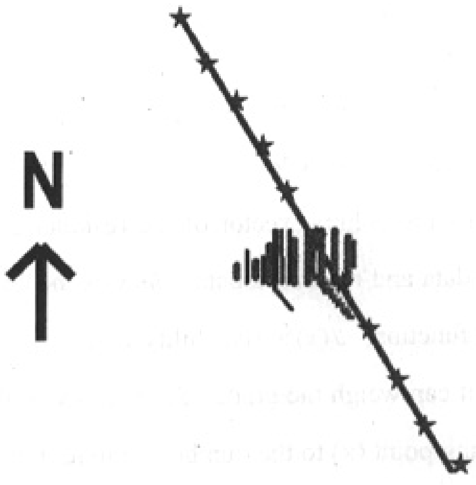


Figure 4-1. MT data collected in eleven stations on a 20 km length NW-SE line, almost perpendicular to the main structures of the region. Station spacing is 2000 meters. Short lines are CSAMT surveyed lines in the study area.

the profile line is along X axis, the model can be depicted as follow:

$$\rho(x) = \{\rho_x(x), \rho_2(x), \dots \rho_M(x)\} \qquad (1)$$

From the forward model we define arithmetic operators which transform the model function space into MT data space

$$d(x) = F[x, \rho(x)] \qquad (2)$$

Where $d(x)$ is the measured MT data. The inversion is finding out $\rho(x)$ from the known $d(x)$. This can be achieved by taking minimal value in the continuous fonctionelle

$$\int \sum_{i=1}^{M} \{d_i(x) - F_i[x, \rho(x)]\}^2 dx \qquad (3)$$

By making use of the fonctionelle increment relational expression, and developing F_i at $\rho(x)=\rho^0(x)$, and reserving the first order variance, and deleting the high order infinite bit, then formula (3) becomes linear fonctionelle which is relative to the model function variance $\delta\rho(x)$. Through the calculative formula which determines the fonctionelle variance, we can obtain:

$$\int[J^T(x)J(x)\delta\rho(x)-J^T\Delta d_i(x)]dx=0 \qquad (4)$$

Where $\Delta d_i(x)$ is the column vector of the residual error between the model forward data and measured data; $\delta\rho(x)$ is the column vector of the correctional function; $J(x)$ is sensibility matrix, whose element is $\frac{\partial F_i[x,\rho(x)]}{\partial\rho_L(x)}$, and it can weigh the grade of sensibility of the number i data of any measuring point (x) to the number L model function, also it stands for the sensitive order of a certain laminar layer of the model. Usually MT is measured pointwise, assume that the surface measuring points are $X_p, p=1,2,...m$, then the solutions which satisfy formula (4) should make the integral function equal to zero, namely:

$$J^T(X_p)J(X_p)\delta\rho(X_p)-J^T(X_p)\Delta d_i(X_p)=0 \qquad (5)$$

The column vector of the model correctional function becomes the column vector of the model correctional function under the measuring point X_p. We take data normalization in formula (5), transform the natural logarithm of model function, and induct proper regularization method, and then we can obtain steady solution gradually in formula (5) in each iteration.

Figure 5-2 shows the 2-D results of the implemented MT method in the study area. As it can be seen obviously, a very important considerable dislocation in the strata of area at depth had happen. This is probably related to suspect buried extension fault in the area on the south of Woxi Fault. More information about the fault and strata can be retrieved by doing some collecting data on some more parallel lines and by decreasing the station spacing to increase data density.

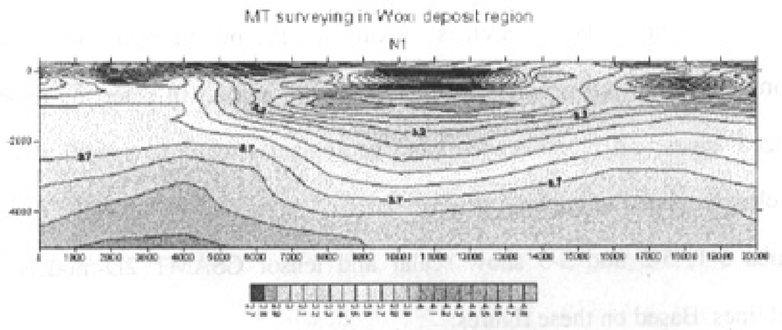


Figure 4-2. Two-dimensional model of the MT sounding in the study area. X axis is offset and Y is depth, both in meter

Chapter 5

Discussion, Results and Conclusion

I. Discussion

(A) Comprehensive geophysical interpretation

The main point of the research is looking for the mineralized zones and their extension. The employed geophysical methods were helpful in detecting anomalous mineralized zones and moreover mapping geological structures, specifically faults, that are closely related to mineralization.

Figures 5-1, 5-2 and 5-3 show scalar and tensor CSAMT 2D-models of the surveyed lines. Based on these figures:

Six main faults (F1 to F6) in the area were detected and mapped that only three of them were mapped in the fieldwork (Figure 5-4). There is a meaningful relation between faults and mineralization in the area.

From A1 to A3 we have two different ore-vein groups located in different depths. Depth changes are in relation with postmineralization movements in the study area such as moving of hanging and footwall blocks of the faults and fault strike slip movements and even fault rotation..

Founded on the 2-D figures, there are three main distinct mineralization zones: Zone between F2 and, north of Ganziping Fault (F6) block, and south of Woxi Fault (F2) block. The first block is controlled structurally between Woxi and Ganziping faults. Based on geophysical and borehole data there are vein groups that part of them were already being exploited. Second mineralized zone in the north of Ganziping is located on upper block of Ganziping fault. The anomaly is strong in A4, A5, A6 and

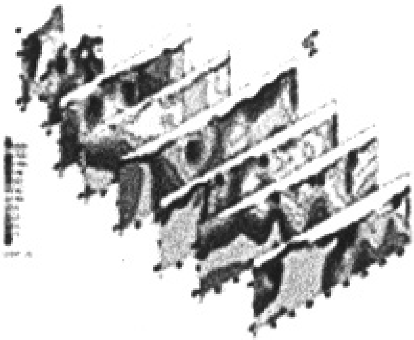


Figure 5-1. Three-dimensional presentation of 2-D CSAMT inversion models over lines A1-A8. Anomalous zones caused by mineralization and faulting are obvious. A very high resistive zone on the south is hint to a potential concealed extension listric fault.

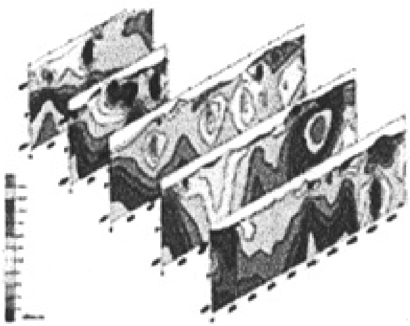


Figure 5-2. Three-dimensional presentation of 2-D CSAMT inversion models over lines B1-B5. Similar to figure 5-1, mineralized zones and faults are clear. Trace of very high zone on the south of the profile is shown.

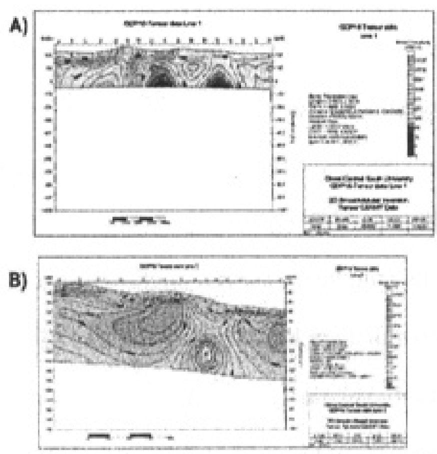


Figure 5-3. Two dimensional inversion models of tensor CSAMT on lines C1 (A) and C2(B). Two very distinct mineralized zones on line C1 and one deep mineralized zone on line C2 are demonstrated.

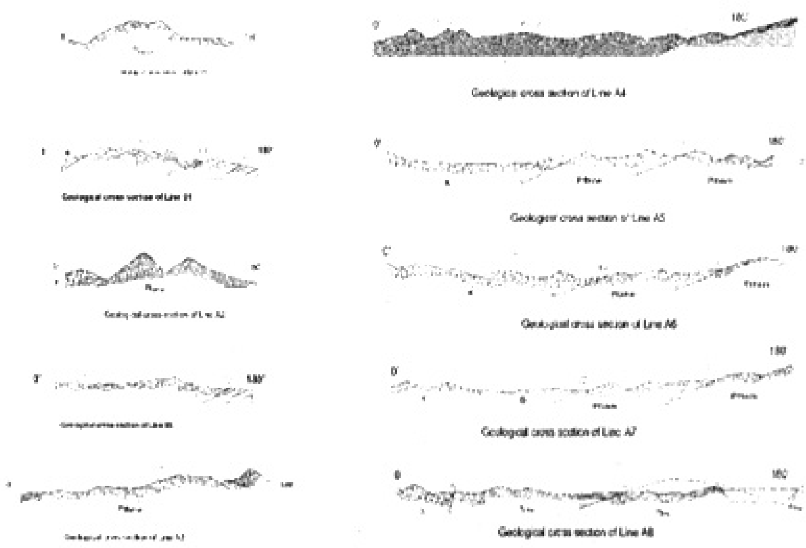


Figure 5-4. Geological cross sections of lines A1-A8. Faults mapped in the field and faults traced by employed geophysical methods are shown. Based on the offered model, all the faults are extension faults including Woxi (F2) and Ganziping (F6) Faults.

partly B3 lines and mineralized zones relatively are deeper than Woxi-Ganziping block. The reason is subsiding of the hanging wall or uplifting of the footwall. The other important anomaly on line B2 in the south is related to rice puddies in the area.

From line A5 a new branch of Woxi fault (F3) (Figure 1-5) appears. While Woxi Fault turns to northeast, this fault keeps striking to the east. From line A5 to A8, three mineralized zones are shown that are located in F2-F3, F3- F5 and north of Ganziping Fault (F6) blocks. A noticeable anomaly zone for first time appears at elevation 150-250 m on the south of Woxi fault on line A8. It locates in 100 to 300 meters offset on the northwest of Xintianwan dextral Fault and shows its structural importance in mineralization.

On line C1 mineralization are clearly demonstrated by 2-D tensor CSAMT model (Figure 5-3 a). In addition, F2, F3 and F5 and two distinct mineralized zones between them are very clear. Figure 5-3b demonstrates the 2-D tensor CSAMT model on line C2. All along the line can be divided to three high-low-high resistivity zones. The first high resistivity zone indeed starts from line C1, station 16 and then reaches its peak at station 1, line C2 where it is over a wide silicified breccia zone belt. This high resistivity zone extends to station 3 on line C2 with more than 600 meters width. This high resistivity zone is probably a sign of a buried fault in the south of Woxi fault that will be explained in detail. A very low resistivity anomaly is observable between stations 4 to 7 at approximate depth range of 450 to 750 m. This anomaly, the same as line A8's southern anomalous zone, is located on the northwest of the Xintianwan dextral fault and again shows structural importance of Xintianwan Fault. Except this anomaly, no any other low resistivity zone is seen on the line C2. However a little farther from station 14 (end of C2), Tanghuping prospect is located that shows good mineralization as discussed in Part 3.

CSAMT results indicate that Ganziping fault is a post-mineralization high-angled listric extension fault in the north of the deposit. Based on the offered model, F2-F6 block was relatively raised by Woxi Fault on the south and subsided and displaced ore veins on the north to depth. As a result there is possibility of finding mineralized zones in north of Ganziping fault that is strongly supported by geophysical anomaly. Although because it is covered by Wuqiangxi Formation and Cretaceous strata, it is buried deeply. Towards north, indeed, ore veins get deeper and layers shrink when extending along its strike direction. Their strike direction changes towards northeast.

Therefore opponent to the previous ideas that all them were considered Woxi deposit as a deposit structurally confined between Woxi and Ganziping faults, the new reserves were explored based on the offered model.

Figure 4-2 shows 2-D MT plot as discussed in Part 4. An obvious high-angle listric extension fault, named Wuhe (武赫) is seen that dips around 2000 meters depth in upper lithosphere and extends the displaced area where it gets flatten. It steep at shallow but gradually becomes sub horizontal and to the north meets other faults at its base. There is no outcrop of the fault and just the silicified breccia vein that in the west changes to "dolomitize, silicified" and "dolomitize, pyritized, silicified" zones can be seen on the surface in the surveyed area.

(B) Comprehensive geological interpretation.

Woxi deposit is classified as strata-bound metamorphic- hot- liquid deposit. Transported metals are mainly sourced from metamorphic calcium rich purplish- red slates of middle- upper parts of Madiyi Formation, Banxi Group. It is source of ore minerals, and metamorphic solution is the main carrier of ore minerals. Calcium, iron, manganese and sulfur which are used for mineralization of scheelite, ferberite and gold-antimony are mainly from host rocks.

Structural Studies show that V_6, V_5, V_2, V_1, V_3, V_4 and V_7, are controlled by a shear zone, which appears as reverse "S". West of the shear zone is slantingly cut by Woxi Fault and its east is cut by Xintianwan and Tanghuping reversed faults (see figure 1-6). Rocks in the zone are altered slates. According to structural studies, the mentioned zone is a subsiding shear zone, and on top of it a buried separate fault, which is a high angle normal fault, called Wuhe (F1) is developed. Then Woxi Fault is a listric normal fault with a large angle in upper block of concealed Wuhe (F1) fault and in depth it converges with Wuhe fault in Xuefeng period. Structural effect of Woxi Straight Fault was formed in this stage. After that, the shear zone has turned into late stage folds.

Yu'ershan –Hongyanxi blocks in the central and western parts of the Woxi deposit were the main study targets in stage B. However, profile B6 was designed to acquire data in the eastern part that shows good qualification as a prosperous target however needs detailed exploration works.

Woxi Fault cuts the mineral veins of Yuershan-Hongyanxi region, in depth and displaces them. Many researchers categorize Woxi fault as a reverse fault but based on the many evidences of the research it is classified as a listric extension fault

Geophysical data processing, interpretation and deciphering along using geological and boreholes data have been undertaken to all the acquired data in designed lines. Figure 5-5 shows all lines interpreted geological cross sections.

Based on near-ore IP method results, Woxi deposit and Yu'ershan ore blocks' electroposition are related, the mineralized zone is obviously shrinking from east to west, so it was established the part of Yu'ershan which extending toward east contains a small scale of ore veins. Possibility studies for western Woxi to contain mineralized zone needs further research.

II. Results

1. Dissimilar electrical characteristics of different strata were used for their distinguishing and mapping. Mineralized zones with the intense wall-rock alteration characterized by low apparent resistivity (about 100• Ω •m) are obvious in the geophysical profiles. On the whole, ore vein group is trending north, dip angle is small at line A4 to A6 it strikes NE but at line A6 to A7 it turns to strike about NEE.
2. Mineralized zones location, dimensions, depth, and extension as well as deep vein-groups structures were mapped by CSAMT

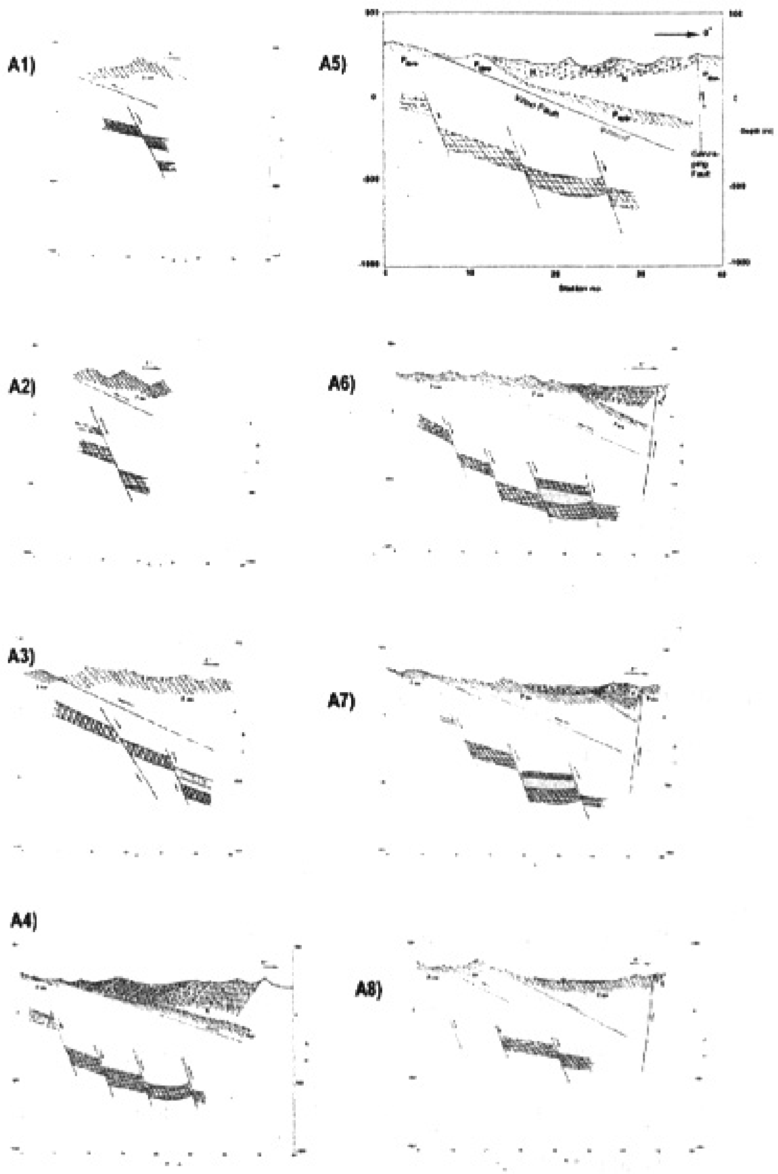


Figure 5-5. Interpreted geological cross sections of lines A1-A8.Main normal faults of the area are shown. Role of normal faults in cutting and displacing of the ore veins are evidently shown.

3. Four main mineralized zones were detected that two of them were explored and mapped based on new geological model at stage C in the north and south of the surveyed area.
4. Biased by the Woxi fault, the mineralized zone displaced towards the north. The area covers from west to east and gradually turns to the northeast and extends to north of Ganziping Fault
5. Main structures of the area were mapped by CSAMT including some newly recognized F3, F4, and F5. Moreover, some buried faults like F5 and F6 were traced and mapped.
6. Based on MT, CSAMT and geological evidences, a concealed Fault was detected as the main extension fault in the area and was named Wuhe (武赫). This fault along with other main faults in the area such as Woxi and Ganziping are all were classified as normal faults in an extensional basin.
7. Based on the strong anomalies detected by implemented geophysical methods on the north and south east of the study area, rich mineralized zones on both places are expected.
8. In the Proterozoic strata, there are some interlayer large mineralized slate strata. Some parallel normal faults, with dip angles larger than the layers' slope, cut the veins and dislocate them meters to hundred meters.
9. There is still a bright prospective in lines A1 and A2 in western Woxi.
10. Comparing to the known geological materials, all the drill holes including minerals are in the anomalous range of CSAMT .In the anomaly center, the minerals are rich such as in 24/CK 9 , but in the anomaly boundaries, the minerals are scarce and thin, such as 28/CK 7, 34/CK 3. There are many normal faults, which cut the mineral deposit in drilling holes, and all of these are verified the geophysical anomaly range.

11. Based on comprehensive geophysical, geological, and borehole data, the western part of A1 line is prosperous zone for more detailed exploration works.
12. Assuming Woxi fault as a postmineralization fault, the mineralized zone shrinking from east to west, there is still mineralized zone in the western Woxi.
13. Boreholes data such as 241 CK6, 241CK7, 241 CK8 and 241 CK9 verifies that economic ore bodies exist in mineral veins and the ore bodies have a certain scale and a good prospect.
14. Tanghuping Gold prospect is located at the south end of the mining area is in the southeast of Xian'ebaodan anticline, and at the south end of reverse "S" structures. In Northwest of north-east-east Tanghuping Fault, there are favorable structures for mineralization. In 300~500-m-long altered breccia zones on the strike on the surface, native gold grains and scheelite are found in silicified breccia rocks and thin reticulated quartz veins. In the depth, there is a mining prospect of looking for altered breccia gold.

III. Conclusions

Based on the mentioned results along with geological fieldworks and boreholes data, the following conclusions were achieved:

1. Woxi deposit is a deposit of strata-bound middle-low-temperature metamorphic solution quartz vein type. It is a large-scale gold deposit with antimony and tungsten as byproducts. The ore bodies are strata-bound, stable and successively extended along the layer to great depth. Towards the depth, thick-grain gold increases and grades of gold and antimony become higher while the grade of tungsten relatively decreases. In the depth of west of Shiliupenggong ore block, because of the F2 and F7 faults, the axis of ore-bearing synclines turns from northeast to east, and ore bodies also turn from northeast to northeast east.

2. Based on the offered model, majority of the mapped faults in the deposit are normal faults and therefore instead of east and west, north of the deposit considered as the most prosperous mineralized zone and first mining target in the area.

3. Northwest of Xintianwan Fault is the second important large reserve explored in this research work and is also classified as high priority mining target.

4. IP profiling and MIP sounding strongly support employed CSAMT technique results in the study area. Therefore, it proved implemented methods reliability, efficiency and productivity. CSAMT results, specially 2-D models, uncovered and located some new faults in the area. Moreover, CSAMT results hint at a concealed fault on the south of Woxi fault. Supporting CSAMT, MT results reveal faults' high-angle listric extensional characteristics.

5. Tensor CSAMT was very successful in its first mission by detecting and mapping the mineralized zones and geological structures clearly Tensor CSAMT supported the scalar CSAMT results in the study area. Moreover, Tensor results are more reliable and unambiguous than the scalar.

6. Tanghuping Gold prospect is located at the south end of the mining area and in southeast of Xian'ebaodan anticline, and at the south end of reverse "S" structures. In Northwest of north-east-east Tanghuping Fault, there are favorable structures for mineralization.

7. Bearing in mind limitation of the ordinary IP techniques ability for exploration of deep-buried orebodies, near-ore IP Method proved his competence and dependability in the surveyed area. While CSAMT outcome can direct us to the low resistivity zones caused by various sources, the near-ore induced polarization method can advise us the character of anomaly source and reveal the mineralized zones. Thus, employing of IP method as interpretation reference for CSAMT anomalies is recommended.

8. RRI inversion was found as an effective method for producing a resistivity image from CSAMT measurements made along surface traverses.

9. Pseudo-random multi-frequency IP sounding shows high conformity with the produced 2-D CSAMT models. The drilled boreholes based on the achieved results demonstrate the effectiveness of the combined use of both methods.

10. By rock sample testing and numerical calculation, a geophysical model was built to be employed in the Tanghuping prospect. Employing dual-frequency IP, nine gold-bearing quartz veins were found in the surveyed area and their mineralization and exploitation value were evaluated. This study shows while there are no economically feasible ore bodies in the west of the line 3, there are slightly sloped prosperous veins along the layer, in the east. They are similar to the Woxi deposits' veins and more exploration work need to be done for their exploitation.

11. In conclusion, North of Ganziping fault, northwest of Xintianwan Fault and Tanghuping in southwest of the deposit were found as the most important mining targets in the area.

III. Suggestion

Based on the achieved results it is suggested that:

1. There are some prosperous mineralized zones sparsely spread through the west of the surveyed area that need more detailed exploration work to be undertaken. Applying CSAMT survey in denser grid can provide more accurate information about the mineralized zones.
2. CSAMT locates the low resistive mineralized zones but in order to do drilling and other exploitation works, IP surveying should be carried out to locate the exact position of target veins.

3. Tensor CSAMT can be helpful to study ore controlling structures in the area in detail. Sounding with high density grids are strongly recommended for detailed structural research works and mineralization mapping.
4. Detailed exploration work and Mining in the north of Ganziping and northwest of Xintianwan faults are emphasized. Meanwhile, prospecting in the east and west should also be paid close attention to, i.e. prospecting in Yu'ershan and Hongyanxi to its west.
5. Deep seismic reflection profiles provide a means at testing structural models of the study area. Reflection seismic profiles may show only certain of the geologically necessary elements and important steeper dipping faults with small vertical offsets can be missed unless the basin development is examined regional. The analysis of the basin dynamics necessitates merging and reconciling apparently conflicting data from theoretical and observational branches of tectonics, stratigraphy, and geophysics. Ultimately, the model must be predictive and be capable of being tested. The implications of such basin model are important in that they begin to show how region's stratigraphy can evolve in response to structure and how the mineralization may distribute with time.
6. MT sounding with small dipole spacing in at lest two profiles can help getting more information about the regions strata and major structures.
7. Because of very close relation between ores and structures, detailed structural studies including faults and folds are recommended
8. Comprehensive Studies of main gold deposits in the gold belt in the region using geochemical, geophysical methods and employing GIS/RS techniques are recommended. The results can ease exploration of the similar deposits on the mineralization belt.

BIBLIOGRAPHY

Adam, A., Kaikkonen, P., Hjelt, S.E., Pajunpaa, K., Szarka, L., and Wallner, A. (1982) Magnetotelluric and audiomagnetotelluric measurments in finland. Tectonophysics, 90: 77-90.

Adam, A. (1998) Tensor decomposition with static shift correction of the deep magnetotelluric(MT) soundings to improve the asthenospheric depth values in the Great Hungarian Plain. Acta Geodaetica et Geophysica Hungarica. 33: 187-213.

Anderson, W.L., (1974) Electromagnetic fields about a finite electric wire source. US Geol. Surv. Rep., USGS-6D-74-04.

Arcon,S.A. (1984) Field observations of electromagnetic pulse propagation in dielectric slabs. Geophysics, 49(10): 1763-1773.

Bahr, K., (1991) Geological noise in magnetotelluric data: A classification of distortion types: Phys, Earth and planet. int., 66: 24-38.

Bahr, K. (1988) Interpretation of the magnetotelluric impedence tensor. regional induction and local telluric distortion. J. Geophys. 62:119-127.

Bartel, L.C. (1991) Results from a controlled-source audiofrequency magnetotelluric survey to characterize an aquifer. In: ward, S.H. (Ed.), Geotechnical and enviromental geophysics, Vol. 5 of investigations in Geophysics, SEG, pp. 219-233.

Bartel, L.C., and Ranganayaki, R.P. (1989) Acquisition of controlled-source audiofrequency magnetotelluric data at an active steam-drive site: 59th Ann. Internat. Mtg., Soc. Expl. Geophys., Expanded Abstracts, 213-215.

Bartel, L.C. (1982) Modeling and analysis of the CSAMT geophysical technique results to map oil recovery processes: Soc. Petr. Eng. paper SPE 11192.

Beamish, D. and Travassos, J.M. (1992) A study of static shift removal from magnetotelluric data, J. appl. Geophys., 29:1-19.

Benderitter, Y., Lach, H.N.t., and Jolivet, A. (1973) Magnetotelluric method for mining exploration. geophysics, 49: 925-933. (Wrong Volume No)

Berdichevskiy, M.N., and Zhadanov, M.S. (1984) Advanced theory of deep geomagnetic sounding. Amesterdam, Elsevier, 407 pp.

Bernard, J., Vachette, C., Valle, P. (1990) Deep groundwater survey with audiomagnetotelluric soundings. In: SEG, 60th Annu. Int. meeting. 60: 528-531.

Boerner, D.E., and West, G.F. (1989) A spatial and spectral analysis of the electromagnetic sensivity in a layered earth. Geophys. j., 98: 11-21.

Boerner, D.E. nad West, G.F. (1989) A generalized representation of the electromagnetic fields in a layered earth: Geophysics, j., 97: 529-548.

Bostick, F.X., Jr. (1997) A simple and almost exact method of MT analysis. Workshop on electrical methods in geothermal exploration. Utah.

Barnett, C.T. (1984) Simple inversion of time-domain electromagnetic data. Geophysics, 49: 925-933.

Bartel, L.C., and Jacobson, R.D. (1987) Results of a controlled-source audiofrequency magnetotelluric survey at the Puhimau thermal area, Kilauea Volcano, Hawaii. Geophysics, 52(5):665-677.

Basokur, A.T., Rasmussen, T.M., Kaya, C., Altun, Y., and Aktas, K. (1997)

Comparison of induced-polarization and controlled-source audio-magnetotellurics methods for massive chalcopyrite exploration in a volcanic area: Geophysics (62): 1087-1096.

Bhattacharya, B.B., Gupta, D., Banerjee, B., and Shalivahan (2001) *Mise-a-la-masse survey for an auriferous sulfide deposit. Geophysics, 66(1): 70-77.*

Boerner, D.E., Kurtz, R.D., and Jones, A.G. (1993) *Orthogonality in CSAMT and MT measurements. Geophysics, 58(7): 924-934.*

Boerner, D.E., Wright, J.A., Thurlow, J.G., and Reed, L.E. (1993) *Tensor CSAMT studies at the Buchans Mine in central Newfoundland. Geophysics, 58(1): 12-19.*

Boschetto, N.B., and Hohmann, G.W. (1991) *Controlled-source audiofrequency magnetotelluric responses of three-dimensional bodies. Geophysics, 56(2): 255-264.*

Boyd, G.W., and Wiles, C.J. (1984) *The Newmont drill-hole EMP system-Examples from eastern Australia. Geophysics, 49(7): 949-956.*

Cagniard, L. (1953) *Basic theory of the MT method of geophysical prospecting. geophysics, 18: 605-635.*

Calderon-Macias, C., and Sen, M.K., and Stoffa, P.L. (2000) *Artificial neural networks for parameter estimation in geophysics. Geophysical Prospecting, 48: 21-47.*

Chew, W.C., and Kong, J.A. (1981) *Electromagnetic field of a dipole on a two layer earth. Geophysics, 46: 309-315.*

Christopherson K.R. (1991) *Applications of magnetotellurics to petroleum exploration in Papua New Guinea; a model for frontier area. The Leading Edge, 10: 21-27.*

Claerbout, J.F. (1976) *Fundamentals of Geophysical Data Processing. New York: McGraw-Hill.*

Constable, S.C., parker, R.L, and Constable, C.G. (1987) *Occam's inversion: A practical algorithm for generating smooth models from electromagnetic data: Geophysics, 52:289-300.*

Cox, S.F., Sun, S.S., Etheridge, M.A., Wall, V.J., and Potter, T.F. (1995) *structural and geochemical controls on the development of turbidite-hosted gold quartz vein deposits, Wattle Gully Mine, Central Victoria. Economic Geology, 90:1722-1746.*

Cox, S.F., Wall, V.J., and Eatheridge, M.A. (1991) *Deformational and metamorphic processes in the formation of mesothermal vein-hosted gold deposits-examples from the Lachlan fold belt in central Victoria, Australia. Ore Geology review 6: 391-423.*

Craw, D., and McKeag S.A. (1995) *Structural control of tertiary Au-Ag-bearing breccias in an extensional environment, Nelson area, southern Nevada, USA. Mineral Deposita, 30: 1-10.*

Davies, B., Macnae, J., Green, A., Raiche, A., Hunter, D., and Spies, B. (1999) *Advances in AEM software: ASEG Preview, 82: 24-28.*

DeGroot-hedin, C., and Constable, S. (1990) *Occam's inversion to generate smooth, two-dimensional models from magnetotelluric data. Geophysics, 55(12): 1613-1624.*

De Ronde, C.E.J., Spooner, E.T.C., De Wit, M. J., and Bray, C.J. (1992) *Shear zone-related, Au quartz vein deposits in the Barberton Greenstone Belt, South Africa: field and petrographic characteristics, fluid properties, and light stable isotope geochemistry. Economic Geology, 87:366-402.*

Drews, C., Furch, N., Maurer, H.M., Husmann, G., nad Weidelt, P. (1989) *Active audio-magnetotellurics on Milos(Greece) for determination of electrical conductivity distribution and its correction with geothermal anomalies. Geothermics, 18:507-519.*

Dupis, A., Marie, P., and Petian, G. (1980) *Magnetotelluric prospecting of the Mont*

Dore area, in Advances in European Geoth. res.: D. Reidel, 935-943.

Dyck, A.V., and West, G.F. (1984) The role of simple computer models in interpretations of wide-band drillhole electromagnetic surveys in mineral exploration. geophysics, 49:957-980.

Egbert, G. (1997) Robust multiple-station magnetotelluric data processing, Geophys. J. Int., 130:475-496.

Egbert, G. and Livelybrooks, D., (1996) Single station magnetotelluric impedence estimation: coherency weighting and the regression M-estimate, Geophysics, 61: 964-970.

Eggers, D.E. (1982) An eigen state formulation of the magnetotelluric impedence tensor. Geophysics, 47: 1204-1214.

Fischer, G. (1982) Magnetotelluric observational techniques on land, Geophys. Surv., 4: 373-393.

Fischer, G and Le Quang, B.V. (1981) Topography and minimization of standard deviation in one-dimensional magnetotelluric modeling, Geophys, J. R. astr. Soc., 67: 279-292.

Fischer, G., Schnegg, P.A., Peguiron, M. and Le Quang, B.V. (1981) An analytic one-dimensional magnetotelluric inversion scheme, Geophys, J.R.astr.Soc., 67:257-278.

Fischer, G. and Le Quang, B.V. (1982) Parameter trade-off in one-dimensional magnetotelluric modeling, J. Geophys., 51:206-215.

Fitterman, D.V. and Stewart, M.T. (1986) Transient Electromagnetic Sounding for groundwater. Geophysics, 51(4):995-1005.

Gamble, T.D., Goubau, W.M., Miracky, R., and Clarke, J. (1982) Magnetotelluric regional strike. Geophysics, 47: 932-937.

Gamble, T.D., Goubau, W.M. and Clarke, J. (1979) Magnetotellurics with a remote reference, Geophysics, 44:53-68.

Garcia, X., D. Boerner and L.B. Pedersen 2001. Full galvanic distortion decomposition of tensor CSAMT data. Application to data from the Buchans Mine (Newfoundland). Geophysical Journal International, accepted.

Gasperikova, E. and Morrison, H.F. (2001) Mapping of induced polarization using natural fields. Geophysics, 66(1): 137-147.

Giroux, B., and Chouteau, M, Descloitres, M., and Ritz, M. (1997) Use of the magnetotelluric method in the study of the deep Maestrichtian aquifer in Senegal. Journal of Applied geophysics, 38: 77-96.

Goldberg, S. and Rostein, Y. (1982) A simple form of presentation of magnetotelluric data using the Bostick transform, Geophys. Prospecting, 30: 211-216.

Goldman, M.,du Plooy,A. and Eckard, M. (1994) On Reducing Ambigiutyin the Interpretation of Transient Electromagnetic Sounding Data. Geophysical Prospecting, 42(1): 3-25.

Goldstein, N.E., Mozley, E, E., and Wilt, M. (1982) Interpretation of shallow electrical features from electromagnetic and magnetotelluric surveys at mount Hood, oregon: J. Geophys. Res., 87, 2815-2828.

Goldstein, N.E., and Mozley, E. (1978) a telluric-magnetotelluric survey at Mount Hood, Oregon: Univ. california, lawrence Berkeley Lab. Rep. LBL-7050.

Goldstein, M.A., and Stangway, D.W., (1975) Audio-frequency magnetotellurics with a grounded electrodipole source. Geophysics, 40: 669-683.

Goubau, W.M., Gamble, T.D. and Clarke, J. (1978) Magnetotelluric data analysis: removal of bias. Geophysics, 43: 1157-1166.

Grant, F.S., and West, G.F. (1965) Interpretation theory in applied geophysics. McGraw-Hill Book Co., Inc.

Groom, R.W. and K. Bahr, 1992. Corrections for near-surface effects: decompositions of the magnetotelluric impedance tensor and scaling corrections for regional sensitivities: A tutorial. Surveys in Geophysics, 13: 341-379.

Guha, J., Archambault, G. and Leroy, J. (1983) A correlation between the evolution of mineralizing fluids and the geomechanical development of shear zone as illustrated by the Henderson #2 Mine, Quebec: Economic Geology, 78:1605-1618.

Gupta, D., Shyam, R., Agrawal, A.K., Bhattacharya, B.B., and Banerjee, B. (1999) Principles of mise-a-la-masse method and its application to polymetallic sulfide deposits of rajasthan. J. Geol. Soc. India, 53: 49-57.

Haber, E., Ascher, U.M., and Oldenburg, D. (1999) Inversion of 3D electromagnetic data, in Oristaglio, M., and Spies, B., Eds, Three-dimensional electromagnetics (3DEM-2). university of utah, 162-165.

HASTIE, L.M. & CEVALLOS, C. Hydrocarbon exploration using the CSAMT method of electromagnetic prospecting. ASEG 9th Geophysical Conference and Exhibition, 39. Abstract. Gold Coast, 1992.

Hathertone, T., Macdonald, W.J.P., and Thomson, G.E.K. (1966) Geophysical methods in geothermal prospecting in New Zealand: Bull. Volcan., 48-497.

Hayakawa, M., Itoh, T., Hattori, A., Yumoto, K. (2000) VLF electromagnetic precursors for an earthquake at Biak, Indonesia on february 17, 1996. Geophysical Research Letter, 27(10): 1531-1534.

He, J. (1993) Dual-frequency IP. Transactions of Nf Soc, 3(4): 1-10.

Hermance, J.F., Thayer, R.E., Bjornsson, A. (1976) The telluric-magnetotelluric method in the regional assessment of geothermal potential, in Proc. 2nd U.N. Sympos. on the Development and Use of Geoth. Res., San Francisco, 2: 1037-1048.

Hermance, J.F., and Thayer, R.E. (1975) The telluric-magnetotelluric method. Geophysics, 40: 664-668.

Hohmann, G.W., and Ting, S.C. (1978) Three-dimensional magnetotelluric modeling. ESL Final Rep. 77-15, DOE contract EY-76-s-07-1601, july, 48 p.

Hohmann, G.W. (1975) Three-dimensional Induced-polarization and Electromagnetic Modeling. Geophysics, 40(2): 309-324 (and discussion in Geophysics, 50(11): 2279).

Hoover, D.B., Long, C.L., and Senterfit, R.M. (1978) Some results from audiomagnetotelluric investigations in geothermal areas: Geophysics, 43: 1501-1514.

Hoover, D.B., and Loy, C.L. (1976) Audio-magnetotelluric methods in reconnaissance geothermal exploration. Proc. 2nd UN symp. dev. geothermal resources, 1059-1064.

Hoover, D.B., FrischKnecht, F.C., and Tippens, C. (1976) Audiomagnetotelluric soundings as a reconnaissance exploration technique in Long Valley, Calif., J. Geophys. res. 81: 801-809.

Hughes, L.J., and Carlson, N.R. (1987) Structure mapping at Trap spring oilfield, Nevada, using controlled-source magnetotellurics. First Break, 5: 403-418.

Hughes, L.J., Carlson, N.R., and Ostrander, A.G., 1984, Applications of CSAMT in mapping structure and alteration associated with petroleum: 54th SEG Meeting, Atlanta, GA, Expanded Abstracts, pp. 102-104.

Ingerov, O.I., Yamashita, Y., and Balint, G. (1994) Controlled source electromagnetic soundings in frequency domain: Phoenix Geophysics Ltd.

Jiracek, G.R., Baldridge, W.S., and Hix, L.J. (1991) SAGE-A unique geophysical experience. The Leading Edge, 10: 45-49.

Jiracek, G. (1990) Near-surface and topographic distortions in electromagnetic induction, survey in geophysics, 11: 163-203.

Jones, A.G., and Groom, R.G. (1993) Strike angle determination from the magnetotelluric impedence tensor in the presence of noise and *local distortion: Rotate at your peril!. Geophys. J. Internat., 113: 524-534.*

Jones, A.G. (1988) Static shift of magnetotelluric data and its removal in a sedimentary basin environment. Geophysics, 53: 968-978.

Jones, A.G. (1983) On the equivalence of the 'Niblett" and "Bostick" transformations in the magnetoteluric method. J. Gophys., 53: 72-73.

Jones, F.W., and Vozoff, K. (1978) The calculation of magnetotelluric quantities for three-dimensional conductivity inhomogeneities. Geophysics, 43: 1167-1175.

Junge, A. (1996) Characterization of and correction for cultural noise, Surv. Geophys. 17:361-391.

Jupp, D.L.B., and Vozoff, K. (1975) Stable interactive methods for the inversion of geophysical data. Geophys. J. Roy. Astr. Soc., 42: 957-976.

Jupp, D., and Vozoff, K. (1978) Two-dimensional magnetotelluric inversion. Geophys. J. Roy. Astr. Soc., 50: 333-352.

Kellett, R., Bishop, J., and Van Reed, E. (1993) The effects of source polarization in CSAMT data over two massive sulfide deposits in Australia. Geophysics, 58: 1764-1772.

Kunetz, G. (1972) Processing and interpretation of magnetotelluric soundings. Geophysics, 37:1005-1021.

Lakanen, E. (1986) Scalar audiomagnetotellurics applied to base-metal exploration in Finland. Geophysics, 51(8): 1628-1646.

Larsen, J.C., Mackie, R.L., Manzella, A., Fiordelisi, A., and Rieven, S. (1996) Robust smooth magnetotelluric transfer functions, Geophys. J. Int., 124: 801-819.

Li, X., Oskooi, B., and Pedersen, L.B. (2000) Inversion of controlled-source tensor magnetotelluric data for a layered earth with azimuthal anisotropy. Geophysics, 65(2):452-464.

Li, X., and pedersen, L.B. (1991) Controlled-source tensor magnetotellurics. Geophysics, 56: 1456-1461.

Lilley, F.E.M. (1998) Magnetotelluric tensor decomposition, 1.Theory for a basic procedure, geophysics, 63:1885-1897.

Lilley, F.E.M. (1998) Magnetotelluric tensor decomposition , 2. Exams of a basic procedure Geophysics, 63: 1898-1907.

Lilley, F.E.M. (1976) Diagrams for magnetotelluric data, Geophysics, 41: 766-770.

Lines, L.R., Schultz, A., and Treitel, S. (1988) Cooperative inversion of geophysical data: Geophysics, 53: 8-20.

Lines, L.R., and treitel, S., 1984, tutorial: A review of least-squares inversion and its application to geophysical problems: Geophys. prosp., 32, 159-186.

Lister, G.S. (1984) S-C Mylonites. J. of struct. Geol., 6(6): 617-638.

Luo, Y. Wan, L. and He, Z. (1996) A new near-field correction method for CSAMT data - The full frequency domain apparent resistivity method, Contributions to The Exploration Geophysics and Geochemistry, Vol. 20, (Collected Works of Electrical Method), Geological Publishing House, (ISBN 7-116-02167-1/P.1626), 69-77.

Luo, Y. Wan, L. and Yang, L. (1996) Equivalent dipolar source approximate calculation for One-dimension direct problem of bipolar source CSAMT method, Contributions to The Exploration Geophysics and Geochemistry, Vol. 20, (Collected Works of Electrical Method), Geological Publishing House, (ISBN 7-116-02167-1/P.1626), 78-90.

Macnae, J., Lay, L., and weston, L. (1998) Measurements of static shift in MT and CSAMT surveys. Exploration geophysics, 29: 494-498.

Macnae, J.C., Lamontagne, Y., and West, G.F. (1984) Noise processing techniques for time-domain EM systems. Geophysics, 49(7): 934-948.

Madden, T., and Swift, C.W.Jr. (1969) Magnetotelluric studies of the electrical conductivity structure of the crust and upper mantel. AGU Monograph 13, The Earth's Crust and upper mantle, p. 469.

Madden, T.R., and Mackie, R.L. (1989) Three-dimensional magnetotelluric modeling and inversion. Proc. IEEE, 77: 318-333.

Madden, T.R., and Park, S.K. (1982) Magnetotelluric modeling for a crustal environment, Final report, USGS contract no. 14-08-0001-G-643, 74 p.

Maurer, H.M. (1988) Discussion on :" Results of a controlled-source audiofrequncy magnetotelluric survey at the Puhimau Thermal area, Kilauea volcano, Hawaii".Bartel, L.C., and Jacobson, R.D., authors. Geophysics, 53: 724-725.

McCracken, K.G., Oristaglio, M.L., and Hohmann, G.W. (1986) The minimization of noise in electromagnetic exploration systems: Geophysics, 51: 819-832.

McNeice G.W. and Jones A.G. (2001) Multisite, multifrequency tensor decomposition of magnetotelluric data. Geophysics, 66(1):158-173.

Morrison, H.F., Goldstein, N.E., Hoversten, M., Oppliger, G, and Riveros, C. (1978) Description, field test, and data analysis of a controlled-source EM system (EM-60). Univ. of California Lawrence Berkeley Laboratory, Rep. LBL-7088.

Masero, W., Schnegg, P.-A. and Fontes, S.L. (1994) A Magneto-telluric investigation of the Araguainha impact structure in Mato Grosso-Goias, central Brazil. Geophysical J. Int., 116(2): 366-376.

Matthai, S.K., Henley, R.W., and Heinrich, C.A. (1995) Gold precipitation by fluid mixing in bedding-parallel fractures near carbonaceous slates at the Cosmopolitan Howley gold deposit, Northern Australia. Economic Geology, 90: 2123-2142.

Mills, T., Hoekstra, P., Blohm, M. and Evans, L. (1988) Time-domain electromagnetic sounding for mapping seawater intrusion in Monterey County, CA. Ground Water, 26: 771-782.

Mitsuhata, Y., Uchida T., and Amano, H. (2002) 2.5-D inversion of frequency-domain electromagnetic data generated by a grounded-wire source. Geophysics, vol. 67, No. 6, p. 1753-1768

Mitsuhata, Y., Matsuo, k., and Minegishi, M. (1999) Magnetotelluric survey for exploration of a volcanic-rock reservoir in the Yurihara oil and gas field, Japan. Geophysical Prospecting, 47: 195-218.

Muller, A. (2000) A new method to compensate for bias in magnetotellurics. Geophys. J. Int., 142: 257-269.

Nagy, Z. (1996) Advances in the combined interpretation of seismics with magnetotellurics. Geophysical Prospecting, 44: 1041-1083.

Newman, G.A. (1994) A study of down hole electromagnetic sources for mapping enhanced oil recovery processes. Geophysics, 59(4): 534-545.

Neves, A.S. (1957) The Magnetotelluric method in two-dimensional structure. dept. of Geology and Geophysics, MIT, Ph.D. thesis.

Newman, G.A., 3D magnetotelluric modeling and inversion, in Oristaglio, M., and Spies, B., Eds, Three-dimensional electromagnetics (3DEM-2): University of Utah, 113-118.

Newman, G.A., Wannamaker, P.E., and Hohmann, G.W. (1985) On the detectability of crustal magma chambers using magma chambers using the magnetotelluric method. Geophysics, 50:1136-1143.

Nichols, E.A., Morrison, H.F., Lee, S. (1994) Controlled-source magnetotellurics for groundwater. In: SEG, 64th Annu. Int. Meeting, pp. 553-554.

Noakes, J. (1951) An electromagnetic method of geophysical prospecting for application to drill holes: Ph.D. Thesis, Univ. of Toronto.

Oldenburg, D.W. (1994) Practical strategies for the solution of large scale electromagnetic inverse problems: Radio Science, 26: 1081-1099.

Oldenberg, D.W., and Li, Y. (1994) Inversion of induced polarization data. Geophysics, 59: 1327-1341.

Orange, A.S. (1989) Magnetotelluric exploration for hydrocarbons, Proc. IEEE, 77: 287-317.

Oristaglio, M., and Spies, B., Eds., (1999) Three-dimensional electromagnetics: Soc. of Expl. Geophys.

Ostrander, A.G., Carlson, N.R., and Zonge, K.L., 1983, Further evidence of electrical anomalies over hydrocarbon accumulations using CSAMT: 53rd SEG Meeting, Dallas, TX, Expanded Abstracts, pp. 60-63

Padilha, A.L., Vitorello, I. (2000) Magnetotelluric and geomagnetic depth soundings around the Torres Syncline hinge, Southeast Parana Basin, Brazil. Geophysical Research Letters, 27(22): 3655-3658.

Park, S.K., Orange, A.S., and Madden, T.R. (1983) Effects of three-dimensional structure on magnetotelluric sounding curves. Geophysics, 48(10): 1402-1405.

Park, S.K., Livelybrooks, D.W. (1989) Quantitative interpretation of rotationally invariant parameters in magnetotellurics. Geophysics, 54: 1483-1490.

Patrick, F.W., and Bostick, F.X. (1969) Magnetotelluric modeling techniqes. EERL, Univ. Texas, Tech. rep. No. 59.

Pederson, L.B., Rasmussen, T.M. (1989) Inversion of magnetotelluric data: A nonlinear least-squares approach. Geophys. prosp. 37: 669-695.

Pederson, L.B., and Gharibi, M. (2000) Automatic 1-D inversion of magnetotelluric data: Finding the simplest possible model that fits the data. Geophysics, 65(3): 773-782.

Pederson, L.B. (1982) The magnetotelluric impedence tensor- its random and bias error. Geophys. Prospect., 30:188-210.

Peng E.(Ensheng) (1994) Metallogenic characteristics of Haopinggou_Tieluping silver-gold polymetallic deposits. J. Cent. S. Inst. Mining and Metall., 25(4):430-434.

Portniaguine, O. Watts, D. and M.S Zhdanov, Interpretation of CSAMT data at Sellafield radioactive waste site (UK) by 3-D electromagnetic modeling and parameter estimation method, 1998 Consortium for Electromagnetic Modeling and Inversion final progress report, 403-420

Poulton, M.M., Sternberg, B.K., and Glass, C.E. (1992) Location of subsurface targets in geophysical data using neural networks: Geophysics, 57(12): 1534-1544.

Plouff, D. (1966) Magnetotelluric soundings in the southwestern United States. Geophysics, 31(6): 1145-1152.

Purucker, M.E., and Dyment, J. (2000) Satellite magnetic anomalies related to seafloor spreading in the Soth Atlantic Ocean. Geophysical research letters, 27:17: 2765-2768.

Qian, W. (1994) On small-scale near-surface distortion in controlled-source tensor electromagnetics: Geophys. Prosp., 42: 501-520.

Ranganayaki, R.P. (1984) an interpretive analysis of magnetotelluric data. Geophysics, 49(10): 1730-1748.

Rass, H.P., Nielson, D.L., Smith, C., Glenna, W.E., and Moore, J.N. (1981) An integrated case study of the Roosvelt Hot Springs Geothermal System, Utah: Submitted to AAPG Bull.

Ranganayaki, R.P. (1984) An interactive analysis of magnetotelluric data. Geophysics, 49: 1730-1748.

Reddig, R.P., and Jiracek, G.R. (1984) Topographic modeling and correction in magnetotellurics. Presented at the 54th Ann. Internat. Mtg. Soc. Explor. Geophys.; abstracts and biographies, 44-47.

Reddy, I.K., Rankin, D., and Phillips, R.J. (1977) Three-dimensional modeling in magnetotelluric and magnetic variational sounding. Geophys, J. Roy. Astr. Soc. (51): 313-325.

Reid, J.E., and Macnae. J.C. (1999) Doubling the effective skin depth with a local source. Geophysics, 64(3): 732-738.

Reid, J.E., and Macnae, J.C. (1998) Comments on the electromagnetic "smoking ring" concept. Geophysics, 63(6): 1908-1913.

Rijo, L. (1977) Modeling of electric and electromagnetic data: Ph.D. thesis, Univ. of Utah.

Ritter, O, Junge, A. and Dawes, G.J.K. (1998) New equipment and processing for magnetotelluric remote reference observations, Geophys. J. int., 132: 535-548.

Robert, F., and Kelly, W.C. (1987) Ore-forming fluids in Archean gold-bearing quartz veins at the Sigma Mine, Abitibi greenstone belt, Quebec, Canada. Economic Geology, 82:56-74.

Routh, P.S., and Oldenburg, D.W. (1999) Inversion of controlled source audio-frequency magnetotellurics data for a horizontally layered earth. Geophysics, 64(6): 1689-1697.

Routh, P.S., and Oldenburg, D.W.(1998) Advantages of inverting field components in CSAMT survey, The UBC geophysical inversion facility.

Roux, A.T. (1967) The application of geophysics to gold exploration in South Africa. Union Corporation, South Africa.

Sandberg, S.K., and Hohmann G.W. (1982) Controlled-source audiomagnetotellurics in geothermal exploration. Geophysics, 47(1): 100-116.

Sasaki, Y., Yoneda, Y., and Matsuo, K. (1992) Resistivity imaging of controlled-source audiofrequency magnetotelluric data. Geophysics, 57(7):952-955.

Scales, J.A., and Snieder, R. (1997) To Bayes or not to Bayes. Geophysics, 62: 1045-1046.

Shao J., Wang, P., and Chen D. (1996) Mineralogical studies on pyrites in Woxi Au-Sb-W deposit, western Hunan. Hunan Geology (in Chinese), 15(3): 151-158.

Sasaki, Y. (1989) Two-dimensional joint inversion of magnetotelluric and dipole-dipole resistivity data. Geophysics, 54: 254-262.

Sims, W.E., Bostick, F.X. and Smith, H.W. (1971) The estimation of magnetotelluric impedence tensor elements from measured data. geophysics, 36:938-942.

Sims, W.E., and Bostick, F.X. (1969) Methods of magnetotelluric analysis. Res. Lab. Tech. Rpt. 58, Univ. of Texas-Austin.

Singer, B.Sh. (1995) Method for solution of maxwell's equations in non-uniform media. Geophys. j. int., 120: 590-598.

Singer, B.Sh. (1992) Correction for distortions of magnetotelluric fields: Limits of validity of the static approach. Surv. Geophys., 13: 309-340.

Slankis, J.A., Telford, W.M., and Becker, A. (1972) 8 Hz telluric and magnetotelluric prospecting. Geophysics, 37: 862-878.

Smith, Chris. (1997) Theory and the Art of Communications Design. State of the University Press.

Smith, J.T. (1995) Understanding telluric distortion matrices, Geophys. j. int., 122: 219-226.

Smith, J.T., and Booker, J.R. (1991) Rapid inversion of two- and three-dimensional magnetotelluric data. J. Geophys, Res., 96: 3905-3922.

Smith, J.T., and Booker, J.R. (1988) Magnetotelluric inversion for minimum structure. Geophysics, 53(12):1565-1576.

Sowerbutts, W.T.C. and Mason, R.W.I. (1984) A microcomputer-based system for small-scale geophysical surveys. Geophysics, 49(2): 189-193.

Spichak, V., and Popova, I. (2000) Artificial neural network inversion of manetotelluric data in terms of three-dimensional earth macroparameters. Geophys. J. Int., 142: 15-26.

Spies, B.R. (1989) Depth of investigation in electromagnetics methods. Geophysics, 54: 872-888.

Spies, B.R., and Habashy, T.M. (1992) Numerical modeling of surface-to-borehole electromagnetic surveys sor monitoring thermal enhanced oil recovery. Geoexpl., 28:293-311.

Spies, B.R., and Eggers, D. E. (1986) The use and misuse of apparent resistivity in electromagnetic methods. Geophysics, 51(7): 1462-1471.

Strangway, D.W., Swift, C.M., Jr., and Holmer, R.C. (1973) The application of audio-frequency magnetotellurics(AMT) to mineral exploration. Geophysics, 38: 1159-1175.

Srivastava, S.P. (1967) Magnetotelluric two and three-layer master curves. Dom. Obs. Publ. 35(7), Canada Dept. of energy, Ottawa.

Srnka, L.J., and Crutchfield, W.Y. (1987) Riccati inversion of magnetotelluric data. Geophys. J. Roy. Astr. Soc., 91: 211-228.

Stanley, W.D. (1982) Magnetotelluric soundings on the Idaho National Engineering Laboratory facility, Idaho. J. Geophys. Res., 87: 2683-2691.

Stark, M., Wilt, M., Haught, J.R., and Goldstein, N.E. (1980) Controlled source electromagnetic survey at Soda Lakes Geothermal Area, Nevada. Lawrence Berkeley Laboratory, Berkeley, LBL-11221.

Stodt, J.A., Hohmann, G.W., and Ting, S.C. (1981) The telluric-magnetotelluric method in two- and three-dimensional environments. Geophysics, 46: 1137-1147.

Szarka, L. (1988) Geophysical aspects of man-made magnetotelluric noise in the earth- a review, Surv. Geophys., 9:287-318.

Szarka, L. (1988) Discusiion on: "Results of a controlled-source audiomagnetotellurics in geothermal exploration". Geophysics, 47: 100-116.

Szarka L., Menvielle M., Spichak V.V. (1998): 3D imaging properties of the magnetotelluric impedance tensor. Geophysical Journal International (under revision)

Szarka L. (1997): Illustration of some 3-D magnetotelluric parameters on a festive occasion. Publ. Univ. of Miskolc, Series A Mining, 52, 149-161.

Szarka L., Menvielle M. (1997): Analysis of rotational invariants of the magnetotelluric impedance tensor. Geophysical Journal International, 129, 133-142

Szarka L., Nagy Z., Szalai S. (1994): 3-D CSAMT analogue modeling studies. Paper P121, In: Extended Abstracts of 56th EAEG Meeting, Vienna

Szarka, L. (1988) Discusiion on: "Results of a controlled-source audiomagnetotellurics in geothermal exploration. Geophysics, 47: 100-116.

Szarka L. (1986): Geophysical mapping by using stationary electric and magnetic field components. Thesis for the title "Candidate of Earth Science" (Ph.D.), Sopron, (in Hungarian)

Tarantola, A. (1987) Inverse problem theory. Elsevier.

Teagle, R.J., Norris and dave C. (1990) structural controls on gold-bearing quartz mineralization in a duplex thrust system, Hyde-Macrase shear zone, Otago Schist, *New Zealand. Economic geology, 85:1711-1719.*

Tinlin, R.M., Hughes, L.J., Anzzolin, A.R. (1988) The use of controlled-source audio magnetotellurics (CSAMT) to delineate zones of groundwater contamination; a case history. In: Collins, A.G., Johnson, A.i. (Eds.), Groundwater contamination field methods, vol. 963 of ASTM Spec. Tech. Publ., American Society for Testing and Materials, pp. 101-118.

Ting, S.C., and Hohmann, G.W. (1981) Integral equation modeling of three-dimensional magnetotelluric response, Geophysics, 46: 182-197.

Torres-verdin, C. and Bostick, F.X. (1992) Implications of the Born approximation for the magnetotelluric problem in threee-dimensional environments. Geophysics, 57: 587-602.

Torres-verdin, C. and Bostick, F.X. (1992) Principles of spatial surface electric field filtering in magnetotellurics: Electromagnetic array profiling (EMAP): Geophysics, 57: 603-622.

Ting, S.C., Hohmann, G.W. (1981) Integral equation modeling of three-dimensional magnetotelluric response. Geophysics, 46(2): 182-197.

Tournerie, B., Chouteau, M., etal. ( ) three dimensional magnetotelluric survey in Hungar. Phoenix Geophys., ltd.

Treitel, S., and Lines, L. (2001) Past, present, and future of geophysical inversion- A new millennium analysis. Geophysics, 66(1): 21-24.

Uchida, T. (1993) Smooth 2D inversion for magnetotelluric data based on statistical criterion ABIC. Journal of Geomagnetism and Geoelectricity, 45: 841-858.

Uchida, T., and Ogawa, Y. (1993) Development of FORTRAN code of two-dimensional magnetotelluric inversion with smoothness constraint. Geological Survey of Japan, Open-File Report, No. 205, 115pp.

Vozoff, K., and Jupp, D.L.B. (1975) Joint inversion of geophysical data. Geophys. J. Roy. astr. Soc., 42: 977-991.

Vozoff, K and Ellis, R.M. (1966) Magnetotelluric measurements in southern Alberta. Geophysics, 31: p.1153-1157.

Vozoff, K. (1969) Discussion on "The magnetotelluric method in the exploration of sedimentary basins" by Keeva Vozoff (Geophysics, February 1972, p. 98-114), Geophysics : 325-328.

Vozoff, K. (1972) The magnetotelluric method in the exploration of sedimentary basins. Geophysics, 37(1): 98-141.

Wait, J.R. (1962) theory of Magnetotelluric fields. J.res.N.B.S., v. 66D, p.509.

Wait, J.R. (1982) Geoelectromagnetism, Academic precc Inc.

Wan, Le (1992) 1-D generalized inversion for CSAMT data, Annual of The Chinese Geophysical Society, Seismological Press (ISBN 7-5028-0704-7/P.472), 132.

Wan, Le. Luo Y. and, Ma R., (1993) Fast approximate calculation for CSAMT 1-D forward, Annual of The Chinese Geophysical Society, Seismological Press, (ISBN 7-5028-0967-8/P.597), 198.

Wan, Le, Y. Luo, (1996) Data processing and interpretation in CSAMT, The Advanced Development of Applied Geophysics, (ISBN 7-5625-1075-X) China University of Geosciences Press, 20-29.

Wan, Le, and Luo Y. (1996), One-dimensional inverse algorithm of bipolar source CSAMT method, Contributions to The Exploration Geophysics and Geochemistry, Vol. 20, (Collected Works of Electrical Method), Geological Publishing House, (ISBN 7-116-02167-1/P.1626), 91-102.

Wang Jingcao (1999) Modes of Extending and Sliding Ore-bearing Structure in Xiangxi Gold Deposit, not published manuscript

Wang, Z. (Zhiqiang) (1986) Paleotectonics and paleogeography of south China in late Proterozoic Era. The history of continental *margin tectonics in south China (in Chinese). Wuhan Institute of geology press.*

Ward, S.H., and Wannamaker, P.E. (1983) The MT/AMT electromagnetic method in geothermal exploration. U.N. Univ. Geothermal training Program, Iceland. Rep. 1983-5.

Ward, S.H., Parry, W.T., Nash, W.P., Sill, W.R., Cook, K.L., Smith, R.B., Chapman, D.S., Brown, F.H., Whelan, J.A., and Bowman, J.R. (1978) A summary of the geology, geochemistry, and geophysics of the Roosvelt Hot Springs thermal area, Utah. Geophysics, 43: 1515-1542.

Ward, S.H., and Harvey, H.A. (1954) Electromagnetic surveying of diamond drill holes. Can. Min. manual, 19-30.

Ward, S.H. (1983) Controlled-source electrical methods for deep exploration: Geophys. surv. 6, 137-152.

West, R.C., and Ward, S.H. (1988) The borehole controlled-source audiomanetotelluric response of a three-dimensional fracture zone. Geophysics, 53: 215-230.

Wilds, S.R., and MacInnes, S. (1991) High resolution CSAMT applications in gold exploration- a case history, in Raines, G.L., Lisle, R.E., Schafer, R.W.., and Wilkinson, W.H., Eds, Geology and ore deposits of the Great Basin: Geol. Soc. Nevada, II, 1231-1236.

Word, D.R., Smith,.... (1971) Crustal investigations by the magnetotelluric tensor impedance method, Wu, francis T. (1968) the inverse problem of Magnetotelluric tensor sounding. Geophysics, v.33, p.972.

Ward, S.H., Sternberg, B.K., LaBrecque, D.J., and poulton, M.M. (1995) Recommendations for IP research, The Leading Edge, 14: 243-247.

Weaver, J.T., agarwal, A.K., Lilley, F.E.M. (2000) Characterization of the magnetotelluric tensor in terms of its invariants. Geophys. J. int., 141: 321-336.

West, R.C., and Ward, S.H. (1988) The borehole controlled-source audiomagnetotelluric response of a three-dimensional fracture zone. Geophysics, 53(2): 215-230.

White, B.S., and Kohler, W. E., and Srnka, L. J. (2001) Random scattering in magnetotellrics. Geophysics, 66(1): 188-204.

Wilt, M., and Stark, M. (1982) A simple method for calculating apparent resistivity from electromagnetic data. Geophysics. 47: 1100-1105.

Windh, J. (1995) Saddle reef and related gold mineralization, Hill End gold field, Australia: Evolution of an auriferousb vein system during progressive deformation. Economic Geology, 90:1764-1775.

Wright, P. M., Ward, S.H., and Ross, H.P. (1985) State-of-the-art geophysical exploration for geothermal resources. Geophysics, 50(12): 2666-2696.

Wu, Y. (1996) Metallogenic characteristics and ore-prospecting future of pre-cambrian gold deposits in Mafu-Jiuling district. J. cent. South univ. Tech., 3(1): 9-11.

Xu A. (1992) Turbidite in the gold ore district of west Hunan and its geological significance. J. Cent. S. Inst. Min. and Mwtall (in Chinese), 23(2):124-129.

Yamashita, M. and Hallof, P.G. (1985) CSAMT case histories with a multi-channel CSAMT system and discussion of near-field data correction. 55th SEG annual convention , Washington, D.C.

Yutaka, S., Yoshihiro, Y., and Koichi M. (1992) resistivity imaging of controlled-source audiofrequency magnettelluric data. Geophysics, 57(7):952-955.

Zhang Ligang (1989) Theories of Mineralization and Mining- Main Types of Deposit in China and Geology of Stable

Isotope of Rocks like Granite. Beijing Industrial University Press

Zhang, Z., and Oldenburg, D.W. (1997) Recovering magnetic susceptibility from electromagnetic data over 1D earth. Geophys. J. Internat., 130: 422-434.

Zhang, P., Roberts, R.G. and Pederson, L.B. (1987) Magnetotelluric strike rules. Geophysics, 52: 267-278.

Zhang, Z., Routh, P.S., Oldenburg, D.W., Alumbaugh, D.L., and Newman G.A. (2000) Reconstruction of 1-D conductivity from dual-loop EM data. Geophysics, 65(2): 492-501.

Zhang, Z., and Oldenburg, W. (1999) Simultaneous reconstruction of 1-D susceptibility and conductivity from electromagnetic data. Geophysics, 64(1): 33-47.

Zhdanov, M.S., and Keller, G.V. (1994) The geoelectrical methods in geophysical exploration. Elsevier.

Zonge, K., L., and hughes, L.J. (1985) The effect of electrode contact resistance on electric field measurments: 55th Ann. Internat. Mtg., Soc. Expl. Geophys., Expanded Abstracts, 231-234.

Zonge, K. L., Ostrander, A.G., and Emer, D.F. (1980) Controlled source audio-frequency magnetotelluric measurments: 50th Ann. Internat. Mtg., Soc. Expl. Geophys., expanded Abstracts, 5, 2491-2521. Reprinted in: Vozoff, K. (Ed.) (1987), Magnetotelluric methods: Soc. of Expl. Geophys., 749-763. Geophysics, 46: 460.

Zonge, K.L., Emer, D.F., and Ostrander, A.G. (1980) Controlled-source audiofrequency magnetotelluric measurments. Technical Papers of 50th S.E.G. Conventionv. v. V, pp. 2491-2521.

摘 要

研究区域一沃溪金锑钨矿位于湘西沅陵县沃溪镇内。矿区位于东经 110°54′北纬 28°32′，面积 12 平方公里，距常德市西南 114 公里，距沅陵县东 79 公里。矿区构造可分为弧形、WE 向和 NE 向隆起构造。矿区位于仙鹅抱蛋背斜的东北翼。出露地层主要为元古界的冷家溪群、板溪群震旦系以及上白垩统地层。在早五岭运动也既众所周知的东安运动的作用下，冷家溪群和板溪群以非完全倾斜角相接触。

为了研究矿区矿床地质、圈出地质构造以及寻找新的远景矿区，已经开展了包括 CSAMT、MT 和 IP 在内的地球物理方法。在沃溪矿区在不同的两个阶段（A 和 B）分别进行了标量可控源音频大地电磁法（CSAMT），同时在第三阶段（C 阶段）进行了国内首次张量可控源音频大地电磁法。可控源音频大地电磁法为矿区提供了矿化层和构造的重要信息。后来详细的地质成图以钻探结果证实这些地球物理方法结果的正确性。对 CSAMT 测量进行 RRI 反演，可以有效的生成二维电阻率图像。

在高密度钻探和地质研究结果验证下，沿三条测线近上的矿 IP 剖面结果证实了 CSAMT 法的有效性和可靠性。

伪随机多频激电测深与二维 CSAMT 模拟具有高度一致性。根据已知结果部署的钻孔更加证明了这两种方法结合应用的有效性。

为了研究区域的深部构造，以 2 公里的点距在 20 公里长的测线上

采集到了 11 个点的 MT 数据。结果表明在沃溪矿区下面有一个相对深的延伸隐伏断层。

基于已开展的综合地电方法结果、野外地质和钻探数据，在矿区追索并圈定了一些新的断层，勘探出了三个大规模的矿床。地球物理工作的结果、野外工作以及钻探数据使我们对于矿床地质有了一些新的认识。在提供的模型中，此区域隐伏的主断层（武赫断层）与其它一些主断层如沃溪断层以及新圈定的断层如 F2、F3、F4 和甘子坪断层都可以归纳为外延盆地的正常断层。

基于假设的模型，应该在甘子坪断层北部和新田湾断层西北部进行勘探工作。与此同时，东部和西部的探矿也应给予密切的重视。